AGROFLORESTAL NOVO HORIZONTE REDD AUD PROJECT

Project title	*Agroflorestal Novo Horizonte REDD AUD PROJECT*
Project ID	*5295*
Monitoring period	*10-October-2021 to 30-September-2024*
Crediting period	*10-October-2021 to 09-October 2061*
Original date of issue	*04-October-2024*
Most recent date of issue	*15-October-2024*
Version	*01*
VCS Standard Version	*VCS Standard 4.7*
Prepared by	*Agroflorestal Novo Horizonte ltda*

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1 PROJECT DETAILS

1.1 Summary Description of the Project

The Agroforestry Novo Horizonte REDD AUD Project is located in the Baixo Amazonas mesoregion, within the municipality of Prainha, State of Pará. Situated in the heart of the Amazon, Prainha lies within the "Arc of Deforestation," a crescent-shaped area along the southern and eastern edges of the Amazon rainforest where extensive deforestation has occurred over the past few decades. This deforestation has been primarily driven by cattle ranching, agricultural expansion, logging, infrastructure development, and land speculation.

The project is implemented and in the continuous monitoring phase, with key milestones including the establishment of monitoring systems in October 2021, the community training programs in place since 2022, and the ongoing community's support. The project is expected to ensure the continuous reduction of greenhouse gas emissions and enhancement of carbon sequestration over the project's lifespan.

The project employs a combination of satellite image monitoring, fire prevention strategies, and the implementation of low-carbon agricultural techniques for local community. These measures include agroforestry systems, rotational grazing, and sustainable logging practices, all aimed at reducing the pressure on forested areas. Additionally, community education and stakeholder engagement are critical components, focusing on the adoption of sustainable value chain production methods that minimize environmental impact.

The project's goal is to promote the conservation of the Amazon rainforest through sustainable actions and practices aimed at reducing greenhouse gas emissions from deforestation and degradation, primarily caused by economic activities such as cattle ranching and agriculture expansion, illegal logging, and a lack of adequate conservation incentives. The strategy for forest protection includes the establishment of sustainable land management practices, education, and training of communities and stakeholders in low-carbon agricultural techniques and other sustainable value chain productions. The implementation of satellite image monitoring, fire prevention measures, and biodiversity protection, the project aims to significantly reduce emissions from deforestation and land use change while also increasing carbon sequestration in natural ecosystems.

Prior to the implementation of the project, the area was subjected to ongoing unplanned deforestation, driven by the expansion of cattle ranching and agriculture, coupled with illegal logging activities. The lack of effective land management practices and incentives for conservation exacerbated the deforestation rates, leading to significant greenhouse gas emissions and the degradation of natural ecosystems.

The project is proposed by Agroflorestal Novo Horizonte, a company that has been operating sustainable and certified forest management in the region for years. With a strong relationship with the community, the Agroforestry Novo Horizonte REDD+ AUDD Project is expected to generate significant socio-environmental and climate benefits.

Over the first crediting period 2021 – 2061, based on historical averages and modeling, it is estimated that the project will achieve cumulative reductions of **2,101,885** tCO2e by curbing unplanned deforestation of Amazon rainforest. These estimates are based on conservative assumptions and will be verified through periodic monitoring and evaluation.

Through monitoring and evaluation, the project will track and report on total emissions reductions generated during the monitoring period, providing valuable insights into its overall impact and effectiveness in combating climate change. The total GHG emission reductions achieved during the monitoring period will be documented in accordance with the Verra standards, ensuring transparency and accountability.

During the first monitoring period (2021–2024), the project successfully achieved an average annual reduction of **94,143.18** tCO2e, leading to a cumulative reduction of **376,572.70** tCO2e. Notably, there were no recorded emissions from the project itself, nor was there any leakage during this period.

1.2 Audit History

Table 1. Audit history of the Agroflorestal Novo Horizonte REDD AUD project

.Audit type	Period	Program	Validation/verification body name	Number of years
Validation/ verification	*10-October-2021 to 30-september-2024*	*VCS*	*Ecolance*	*4*

1.3 Sectoral Scope and Project Type

Table 2. Sectoral scope and project type

Sectoral scope	Sectoral Scope 14 Agriculture, Forestry, Land Use
AFOLU project category[1]	Project Category: Avoided Unplanned Deforestation (AUD Project Activity)
Project activity type	Single project

1.4 Project Eligibility

The Agroforestry Novo Horizonte Project commits to maintaining compliance with all the requirements listed and detailed in this section and by the VM0015 Methodology. It is strictly based on the principles defined in section 2.2.1 for its design, implementation, and operation to uphold high levels of social and environmental responsibility. The project is founded on the premise of adopting conservation measures to ensure the credibility of the project's GHG quantification.

The project applies the latest version of VM0015 Methodology for Avoided Unplanned Deforestation, v1.2, including all associated tools and modules specified by the methodology, following the definition 3.14.2 (VCS 4.7). Any updates requested, such as baseline reassessment or crediting period renewal, will be implemented. The project has already met the pipeline listing deadline and conducted the opening meeting with the validation/verification body on [specific date]. Additionally, the project is on track to meet the validation deadline, ensuring its eligibility under the VCS Program.

The project applies the latest version of VM0015 Methodology for Avoided Unplanned Deforestation, v1.2, including all associated tools and modules specified by the methodology, following the definition 3.14.2 (VCS 4.7). Any updates requested, such as baseline reassessment or crediting period renewal, will be implemented. The project is on track to meet the validation deadline, ensuring its eligibility under the VCS Program.

The project contributes to reducing carbon emissions by preventing or reducing illegal deforestation, thus helping to mitigate climate change. Additionally, by maintaining forest cover intact, it promotes biodiversity conservation, protects ecosystems, and preserves vital ecosystem services, thereby aligning it with the scope of the REDD+ mechanism's development. All activities, actions, and operations related to the Agroforestry Novo Horizonte REDD AUD project are designed within the boundaries of all applicable laws, regardless of the level of enforcement.

[1] *See Appendix 1 of the* VCS Standard

Despite being comprehensive and robust, Brazilian environmental legislation often falls short in enforcement. The lack of oversight and accountability leads to criminal activities in areas where the State is absent, undermining both the enforcement of environmental laws and the sustainability of solutions for communities in forested regions. This state absence has repercussions beyond environmental issues, directly impacting the quality of life and natural resources for these populations.

In this context, the primary objective of the project is to conserve a native tropical forest in the Amazon. These efforts are aligned with the principles established in Law 6.938/81, dated August 31, 1981, which aims to preserve, improve, and recover Brazil's natural environments. This law provides the foundation for the project's existence, which seeks to equip local communities with training and resources to develop effective strategies for monitoring and combating environmental crimes.

The project has adhered to all capacity limits set out in the VCS Program Definitions, ensuring that it is not a fragmented part of a larger project or activity that would otherwise exceed such limits. The areas where the project properties for VCU generation are located, according to land cover and land use described in section 1.1, total 23.092,85 hectares and encompass the municipality of Prainha - PA.

1.4.1 AFOLU project eligibility

Within the AFOLU category, there are six types of projects, and this project specifically falls under the VCS Program’s Reduced Emissions from Deforestation and/or Degradation (REDD) category. Although projects within a jurisdiction with a jurisdictional REDD program are expected to follow the VCS jurisdictional program requirements, these do not apply to this project as it is not located within a regulated jurisdictional REDD area.

This project is eligible to participate in the VCS Program as a REDD project according to the AFOLU project requirements. The project meets all the criteria of the REDD category, including the avoidance of unplanned deforestation and/or degradation of forests. According to the internationally accepted definition of "forest," the project area qualifies as a forest and has been classified as such for at least 10 years prior to the project start date, in line with UNFCCC and FAO guidelines. It is important to emphasize that, to ensure compliance with the VCS Program requirements, it has been verified that the project does not convert, drain, or degrade native ecosystems to generate GHG credits. Documented evidence has been collected to demonstrate that the project preserves hydrological functions and maintains the integrity of native ecosystems.

Additionally, the project complies with the VM0015 methodology, which outlines specific guidelines for REDD projects that avoid planned or unplanned deforestation. The project prevents the conversion of forest lands with high carbon stocks into lands with lower carbon stocks, thereby reducing GHG emissions. These measures are crucial to ensuring the project's eligibility and its contribution to climate change mitigation.

This project does not involve additional partners, but all proposed actions have been thoroughly described in the project description, ensuring transparency and clarity regarding the objectives and methods used. With these justifications and evidence, the project clearly demonstrates its eligibility and compliance with VCS Program requirements, ensuring its contribution to reducing GHG emissions through the preservation and protection of forests in the Brazilian Amazon.

1.4.2 Transfer project eligibility

Not Applicable

1.5 Project Design

Indicate if the project has been designed as:

☒ Single location or installation

☐ Multiple locations or project activity instances (but not a grouped project)

☐ Grouped project

Grouped Project Design

Not applicable.

1.6 Project Proponent

Table 3. Project Proponent

Organization name	Agroflorestal Novo Horizonte Ltda.
Contact person	Osmar José Ruschel
Title	Landowner and Project Developer
Address	Rod PA 371 KM 50, S/N, PRX Sta Ma. Do Uruarara, Fazenda Lote 23, 68.130-000, Prainha - PA..

Telephone	+55 93 99225-8414
Email	agroflorestal@novohorizonte.eco.br

1.7 Other Entities Involved in the Project

Table 4. Other entities

Organization name	Harpia Soluções Sustentáveis Ltda
Role in the project	Consultant for Project Design development, Forest inventory, Socioeconomic diagnostic
Contact person	Guilherme Berweth Stucchi
Title	Consultant
Address	Av Monteiro Lobato, 1043 - Araraquara- SP - Cep 14801-220
Telephone	16 98160 2545
Email	guiflorestal@yahoo.com.br

1.8 Ownership

Agroflorestal Novo Horizonte holds the legal authority to manage and oversee the project activities in accordance with Brazilian laws. Ownership is supported by legally recognized documents, including land titles, CARs (Rural Environmental Registry), and CCIRs (Rural Property Federal Registration Certificate). These documents confirm the right to the land, vegetation, and management processes responsible for generating GHG emission reductions and/or removals. According to Article 41, § 4 of the Brazilian Environmental Law, the maintenance activities of Permanent Preservation Areas and Legal Reserves are eligible for payments or incentives, establishing additionality for GHG reduction. Therefore, Agroflorestal Novo Horizonte has the legal ownership and right to implement and benefit from the project activities, in line with the established legal requirements, including the utilization of the generated carbon credits. Documentary evidence will be made available to auditors during the validation and verification process.

1.9 Project Start Date

Table 5. Project start date

Project start date	*10-October-2021*
Justification	The project start date is defined as the date when the farm installed a radio communication tower at its port on 10-October-2021, one of the primary entry points. This installation significantly enhanced the farm's boundary surveillance system, enabling real-time communication with the farm headquarters about the presence of individuals along the river Curuatinga. Importantly, the radio communication tower plays a crucial role in reducing the advancement of illegal activities that lead to deforestation by providing immediate alerts and improving the farm's ability to respond to unauthorized activities. The tower installation followed consultations with local communities conducted on 01/10/21 to ensure their awareness of the project and to mark the initiation of the project activities. Concurrently, the project initiated periodic deforestation monitoring through satellite analysis. This integrated approach marked the official commencement of the project's activities.

1.10 Project Crediting Period

Table 6. Crediting period

Crediting period	☐ *Seven years, twice renewable* ☐ *Ten years, fixed* X Other: 40 years, as this a REDD project, in conformance with requirement 3.2.11 VCS v4.7 aligned with the AFOLU Non-Permanence Risk Tool (p.11), where “Projects shall have a minimum of a 40-year project longevity.”
Start and end date of first or fixed crediting period	*10-october-2021* to *09-October-2061*

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

The Agroflorestal Novo Horizonte project estimates an average annual net GHG emission reduction of 79,648.32 tCO2e/year over a 6-year baseline (Table 1.7). Considering the following ranges, the project enters in the first of them:

☒ < 300,000 tCO2e/year (project)

☐ ≥ 300,000 tCO2e/year (large project)

Table 7. First Baseline crediting period.

Calendar year of crediting period	*Estimated GHG emission reductions or removals (tCO_2e)*
10-October-2021	75,260
2022	77,029
2023	78,786
2024	80,539
2025	82,274
2026	84,002
2027	63,971
2028	65,475
2029	66,968
2030	68,458
2031	69,933
2032	71,402
2033	54,375

2034	55,654
2035	56,923
2036	58,189
2037	59,443
2038	60,691
2039	46,219
2040	47,306
2041	48,384
2042	49,461
2043	50,527
2044	51,588
2045	39,286
2046	40,210
2047	41,127
2048	42,042
2049	42,948
2050	43,849
2051	33,393
2052	34,178
2053	34,958

2054	35,736
2055	36,506
2056	37,272
2057	28,384
2058	29,052
2059	29,714
2060	30,375
Total estimated ERRs during the first or fixed crediting period	2,101,885
Total number of years	40
Average annual ERRs	52,547.12

1.12 Description of the Project Activity

The REDD+ Agroforestry Novo Horizonte project, situated in Prainha, Pará, plays a important role in conserving Amazon rainforest areas and reducing greenhouse gas emissions. The project's importance is underscored by its proximity to rural settlements and roads, key drivers of deforestation in the region. By addressing these underlying causes, the project seeks to mitigate deforestation while fostering sustainable development and environmental conservation.

The project employs a multifaceted approach with short, medium, and long-term strategies. These include:

- **Installation of a Communication Radio Tower**: On 10-October-2021, the project installed a communication radio tower at the farm port. This upgrade significantly enhances the farm's surveillance capabilities, providing real-time alerts about unauthorized activities and strengthening the response to illegal encroachments. This measure is instrumental in reducing the advancement of illegal activities that contribute to deforestation.

- **Education and Training Programs**: The project prioritizes the development of comprehensive education and training programs. These programs focus on environmental education, sustainable land management, and climate change mitigation. Local communities and farm personnel receive training to enhance their understanding of sustainable practices and the importance of forest conservation. By engaging with educational institutions and local experts, the project aims to build local capacity and foster a culture of environmental stewardship.
- **Community Development:** In the short term, these efforts will enhance proficiency in social organization and association-building. Over the medium term, they will contribute to increased income opportunities, better community representation, and improved livelihoods and employment prospects.
- **Fire Brigade Formation and Training**: Recognizing the risk of fires in the Amazon, the project has established a dedicated fire brigade. This brigade undergoes regular training to effectively manage and prevent fire incidents, which are a significant threat to forest areas. The fire brigade is equipped with the necessary tools and resources to respond promptly to fire outbreaks, thereby safeguarding the project area from destructive fires.
- **Fire Monitoring**: The project utilizes satellite data to detect and respond to fire hotspots swiftly. Fire brigades, established in partnership with local communities, manage and control fires, minimizing damage and preventing further deforestation.
- **Monitoring and Leakage Control**: The project integrates advanced monitoring techniques, including satellite imagery and field visits, to track deforestation and land use changes. Platforms like Mapbiomas Alert are used to enhance the accuracy and reliability of deforestation alerts. Additionally, a proactive leakage management plan is in place, involving continuous monitoring of the project's leakage belt and collaboration with local stakeholders to address potential risks and promote sustainable land use practices.
- **Species Monitoring:** Regular monitoring of endangered species and their habitats is conducted based on a detailed fauna inventory. This monitoring supports effective conservation strategies and allows for adjustments to ensure the protection of biodiversity.

By combining these measures with community engagement and education, the project not only aims to achieve significant greenhouse gas emission reductions but also strives to empower local communities through training and active participation in conservation efforts. The project, while not located within a jurisdictional REDD+ program, aligns with best practices for forest conservation and sustainable development, ensuring a comprehensive approach to environmental and social impact.

1.13 Project Location

The Agroflorestal Novo Horizonte REDD+ project is in the municipality of Prainha, Pará, Brazil. The project area spans 23.092,85 hectares across 22 distinct lots. Following the VCS Methodology VM0015 v1.1, the project area may only include areas composed of forest for a minimum of ten years prior to the project start date, a definition that also includes secondary forests. Therefore, satellite images between 2011 and 2021, were analyzed and classified and the areas within the property that were defined as forests were separated and utilized to compose the project area. In addition, some non-forest areas were also excluded, such as rivers, rocks, and non-forest vegetation.

The definition used for forest areas is the one given by the Food and Agriculture Organization of the United Nations: “land spanning more than 0.5 hectares with trees higher than 5 meters and a canopy cover of more than 10 percent, or trees able to reach these thresholds in situ. It does not include land that is predominantly under agricultural or urban land use”.

The project area is adjacent to the Renascer Extractive Reserve, a Sustainable Use Unit aimed at conserving biodiversity while allowing sustainable resource use. It is also located near a settlement encompassing three main communities: Santa Terezinha, Ramal do Meio, and Vila Nova. These communities are primarily made up of smallholders and are not classified as traditional communities. Although they do not directly use the project area and are not impacted stakeholders, they will still benefit from the project's social framework. Accessibility to the project area is facilitated by existing transportation infrastructure. The area can be reached via roads and rivers, with Prainha serving as the nearest major town. This accessibility is crucial for project management and implementation, providing necessary connections to markets, health centers, and other essential services. Table 8 contains the location of each property part of the Agroflorestal Novo Horizonte it also describes the size of total property area and eligible forest area.

Table 8. Agroflorestal Novo Horizonte properties

Properties	Rural Brazilian Registration # (CAR)	Area	Project Area / Eligible Forest (ha)	Latitude	Longitude
Lote de Terra Rural nº 23 Setor A	PA-1506005-6700918475AC448D85F1962F7F7A6C47	1.474,30	1.080,35	2°32'11,852"S	53°38'16,631"W
Lote de Terra Rural nº 24-Setor A	PA-1506005-C888CDB93F8C4A119F842FCC5F323119	1.444,80	1.357,28	2°31'29,11"S	53°39'20,736"W

Lote de Terra Rural n° 25 Setor A	PA-1506005-EF255C2F307E4467B8F716A223207E3C	1.224,04	850,47	2°33'24,977"S	53°42'4,022"W
Lote de Terra Rural nº 28 - Setor A	PA-1506005-2DBA25C7B5D24A879B676BF23B99EF19	1.479,51	1.471,45	2°38'33,372"S	53°43'32,839"W
Lote de Terra Rural nº 30 - Setor A	PA-1506005-D807BFBD0346495F8A6968920369E2F2	1.042,05	833,64	2°44'4,52"S	53°38'59,65"W
Lote de Terra Rural nº 31 - Setor A	PA-1506005-BDA7BF1E9B4E4195B66DBB652AF4D545	1.472,09	1.171,32	2°43'10,557"S	53°41'48,737"W
Lote de Terra Rural nº 32 - Setor A	PA-1506005-E2CBEEE4AFA64A4EA0B71E826AAF9399	1.480,13	1.188,47	2°41'44,572"S	53°41'6,818"W
Lote de Terra Rural nº 33	PA-1506005-AE6BD15D939643ED8F32AF96FB2976CA	1.463,37	1.168,64	2°41'11,058"S	53°42'18,432"W
Lote de Terra Rural nº 34 - Setor A	PA-1506005-68BC7F2DA07743E1B3FC3423FD2389CD	1.481,62	1.184,89	2°40'20,454"S	53°42'33,3"W
Lote de Terra Rural nº 35 - Setor A	PA-1506005-5854ECC6D0454A42BD57E1C3C687003C	1.455,60	1.020,63	2°40'58,901"S	53°44'41,777"W
Lote de Terra Rural nº 36 - Setor A	PA-1506005-F59538D4BA794680A7BE5594295F2A63	1.472,57	1.020,35	2°41'7,329"S	53°46'0,97"W
Fazenda Novo Horizonte	PA-1506005-C727428311974777A8573A26F24C134F	1.488,84	1.188,10	2°31'53,465"S	53°44'11,667"W
Lote de Terra Rural nº 38 - Setor A	PA-1506005-42B639F2F8704289B47961439D6621CD	1.476,20	656,45	2°44'26,461"S	53°45'59,968"W
Lote de Terra Rural nº 39 - Setor A	PA-1506005-87BD4DF16F4C4FEB9AD7C29505BE93A7	1.482,38	1.179,89	2°45'18,213"S	53°45'13,864"W

Lote de Terra Rural nº 40 - Setor A	PA-1506005-BDCBB4BB6A924A968D8A9C9A6853D3DF	1.459,12	1.454,84	2°48'43,309"S	53°46'33,95"W
Lote de Terra Rural nº 41 - Setor A	PA-1506005-B223D400B1054D8D8D2470F8937162AE	1.439,25	1.431,78	2°47'22,906"S	53°49'2,7"W
Santa Carmelita Agropecuária S.A.	PA-1506005-774C099202DF488DAE606D939CD7A6B4	1.027,08	1.155,07	2°48'33,697"S	53°50'31,462"W
Fazenda Agroflorestal Novo Horizonte I	PA-1506005-D4BA709C4BF34A22967662AFD9F2E410	1.419,10	1.018,01	2°45'46,751"S	53°44'16,15"W
Fazenda Agroflorestal Novo Horizonte I	PA-1506005-D4BA709C4BF34A22967662AFD9F2E410	1.098,50	692,96	2°46'21,746"S	53°43'18,45"W
Fazenda Ipê	PA-1506005-CA76D9C1A0C84F05AF36D37E2563DA08	1.483,79	1.186,36	2°43'50,999"S	53°40'58,152"W
Fazenda Curral Grande	PA-1506005-EF891A014A874EB0B1576C38B69E961A	1.127,53	781,95	2°31'28,545"S	53°42'16,589"W
Total		28.991,87	23.092,90	2°32'11,852"S	53°38'16,631"W

The spatial location of the project is shown below on figure 1.

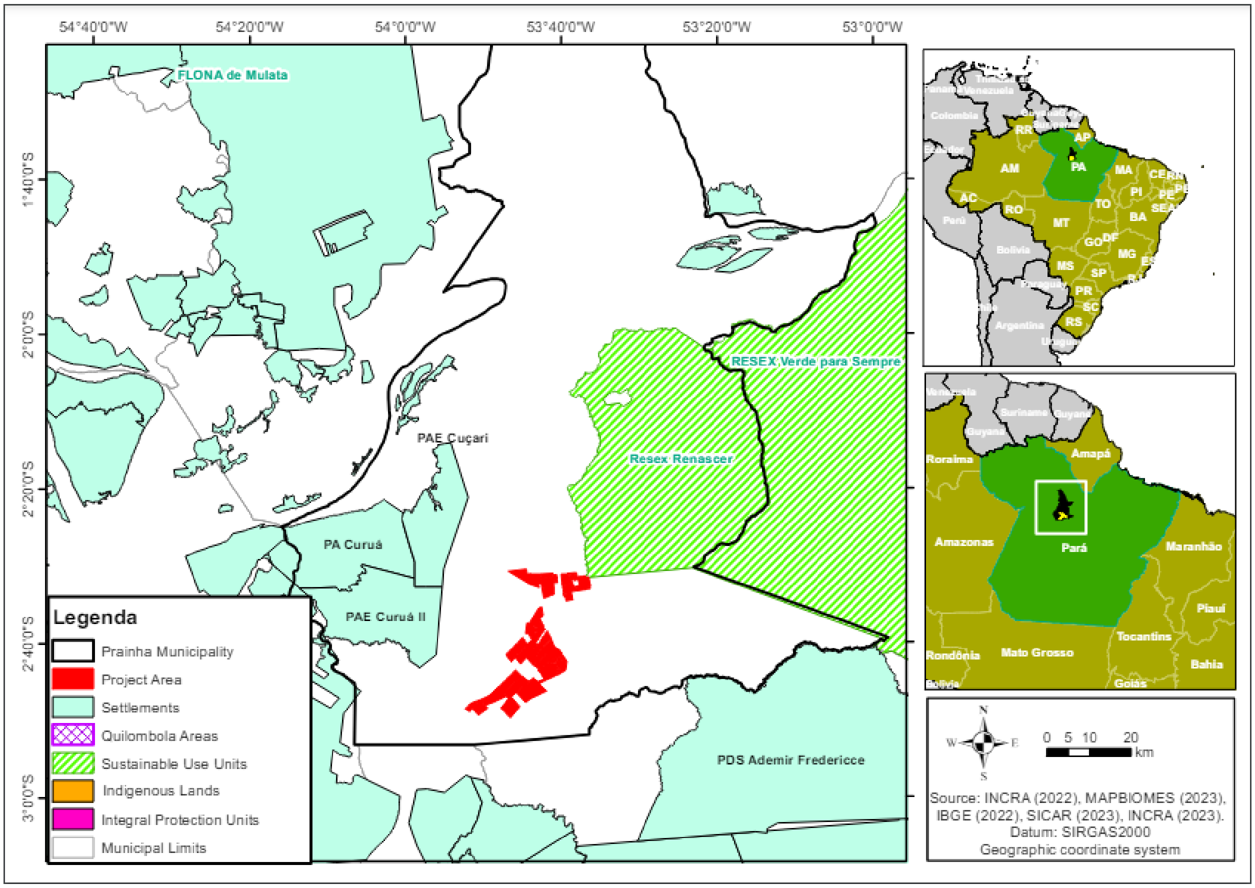


Figure 1. Spatial location of the project

1.14 Conditions Prior to Project Initiation

Ecosystem type: The project is located in a dense ombrophiles forest, characterized by perennial vegetation and large trees, with predominance of Latosol and Argisol soils (figure 2).

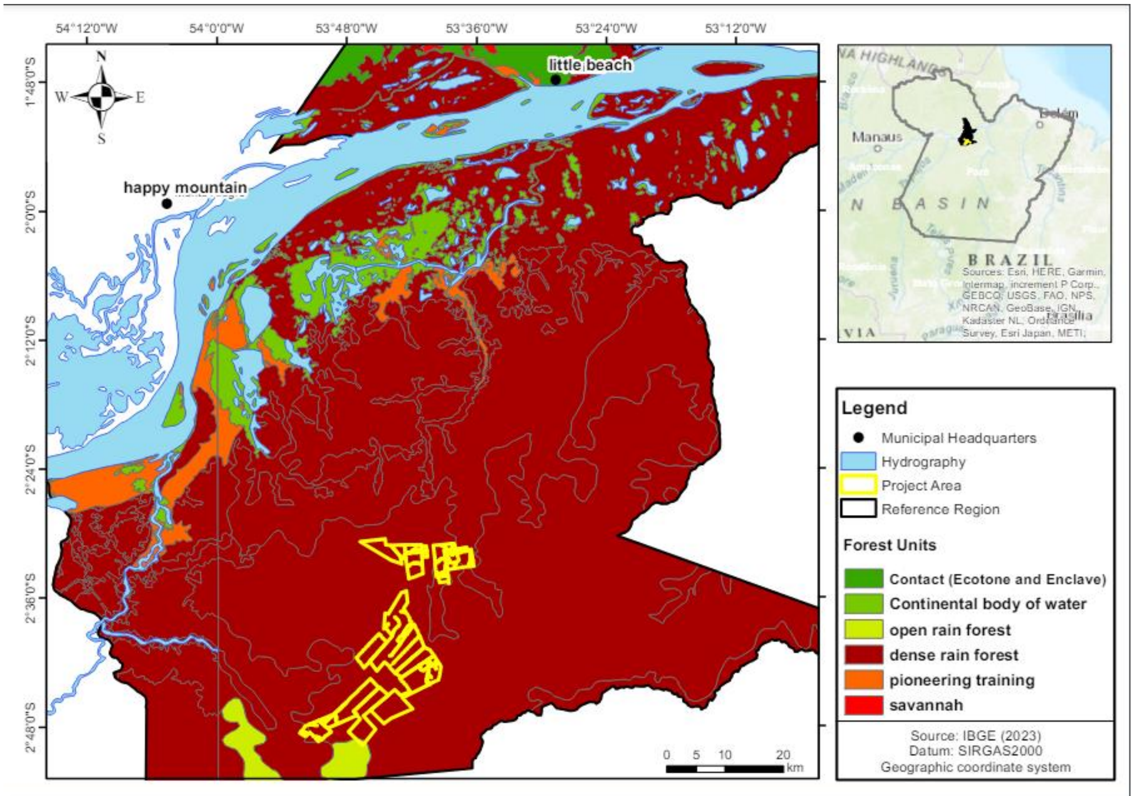


Figure 2. Ecosystem type

Current and historical land-use: Prainha is emblematic of the environmental and socio-economic challenges faced by municipalities in the Amazon. Its economy is primarily driven by activities such as logging, cattle ranching (figure 3), and agriculture, which have led to significant deforestation and environmental degradation. Over the past decade, Prainha has emerged as one of the leading municipalities in terms of deforestation, ranking alongside notorious areas like Altamira, São Félix do Xingu, Novo Progresso, and Itaituba.

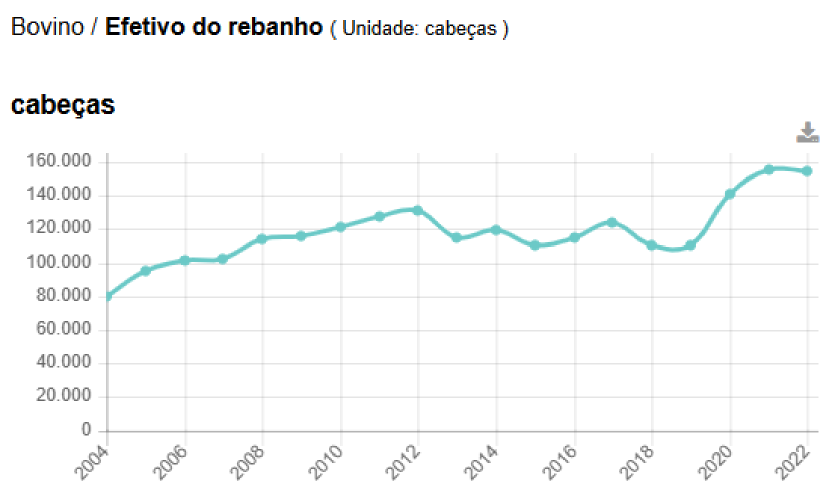


Figure 3. Growth of the cattle herd in the municipality of Prainha. Source: IBGE.

The municipality is part of the Pará State Green Municipalities Program but has not signed on to the program. According to the report available on the program's webpage[2], 78% of properties have registered Environmental Rural Cadastre (CAR). A pact against deforestation was signed in 2017, and the same year, a deforestation combat group was established. However, according to the program's information, the Local Group in the municipality is disbanded, leading to a lack of meetings and, consequently, no discussions or planning of actions aimed at combating deforestation. The municipality is classified as "Under Pressure" due to its high risk of deforestation, either from its deforestation rate or its proximity to major infrastructure projects. The municipality partially meets the goals set by the program, which likely exacerbates its classification. Unmet goals include:

- Having more than 80% of the municipal area in the CAR
- Conducting inspections of illegal deforestation hotspots
- Maintaining the annual deforestation rate below 40 km²

With strong potential for timber exploitation (figure 4), coupled with the expansion of livestock and agriculture production, intensified pressure on the region's forest cover. Infrastructure developments, including road expansions, have further facilitated access to previously remote areas, accelerating deforestation. In 2023, Prainha was reported among the top 10 municipalities contributing most to deforestation in the Amazon, marking a significant rise in forest loss driven by cattle expansion and infrastructure development.

[2] https://www.municipiosverdes.pa.gov.br/index.php/ficha-completa-municipios-participantes/?municipio=1506005&pesquisar=Pesquisar

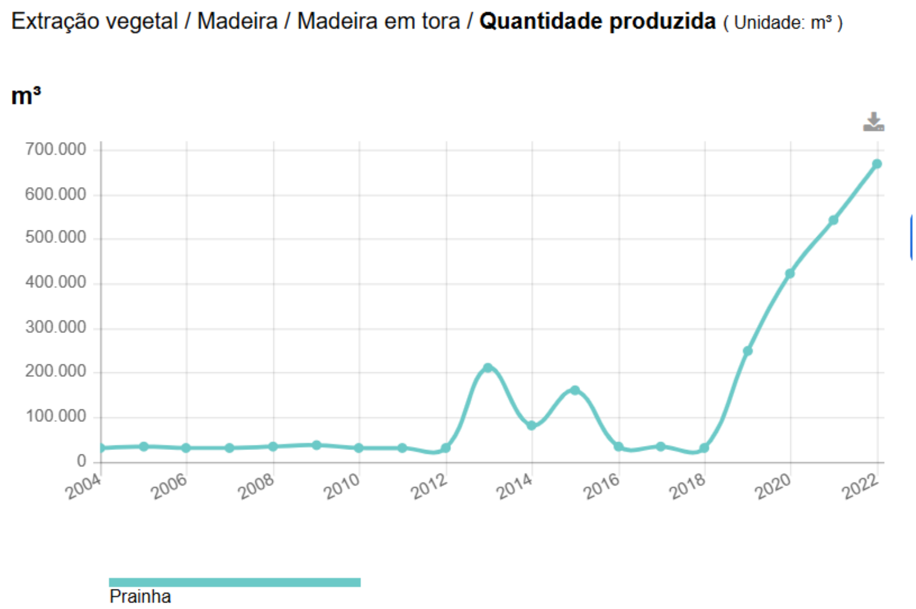


Figure 4. Growth of timber extraction in the municipality of Prainha. Source: IBGE.

The socio-economic context of Prainha complicates efforts to mitigate deforestation. The local economy's heavy reliance on land-intensive activities underscores the need for sustainable development strategies that can balance economic growth with environmental conservation. Moreover, Prainha's primary productive activities, such as artisanal fishing, cattle ranching, and cassava cultivation, are supported by local government initiatives aimed at improving the livelihoods of rural workers. However, these activities, particularly cattle ranching, contribute to the ongoing deforestation and environmental degradation in the region. MapBiomas data[3] indicates that in the municipality of Prainha, since the start of the reference period in 2011, there has been a 62% increase in the area allocated for pasture, rising from approximately 102,000 hectares to 164,000 hectares. Similarly, the area planted with soybeans more than doubled. In 2011, there were only 1.5 hectares planted, which increased to 1,930 hectares by 2023. This expansion occurred primarily over the last six years, coinciding with the financial incentives provided by the previous federal government.

The Agroflorestal Novo Horizonte property is intersected by PA 371, a road that plays a significant role in the process of deforestation. Numerous studies4,5,6 show that roads are a major driver of

[3] MapBiomas_Link for Prainha

[4] Lima, M., et al. (2021). "Roads, deforestation, and the role of protected areas in the Amazon rainforest." *Environmental Research Letters*, 16(6), 064026. doi:10.1088/1748-9326/ac0741.

[5] Soares-Filho, B.S., et al. (2020). "A global map of road networks and their impact on forest loss." *Nature Sustainability*, 3, 572-581. doi:10.1038/s41893-020-0520-2.

[6] Rufino, M.C., et al. (2022). "The influence of road construction on land-use change and deforestation in tropical forests: A review." *Journal of Environmental Management*, 305, 114375. doi:10.1016/j.jenvman.2021.114375.

deforestation, as they connect previously isolated areas, facilitate the opening of new trails, and promote the expansion of economic activities that can lead to forest degradation. These infrastructures not only increase access to remote regions but also encourage intensive land use and the conversion of forested areas into agricultural and pasture lands.

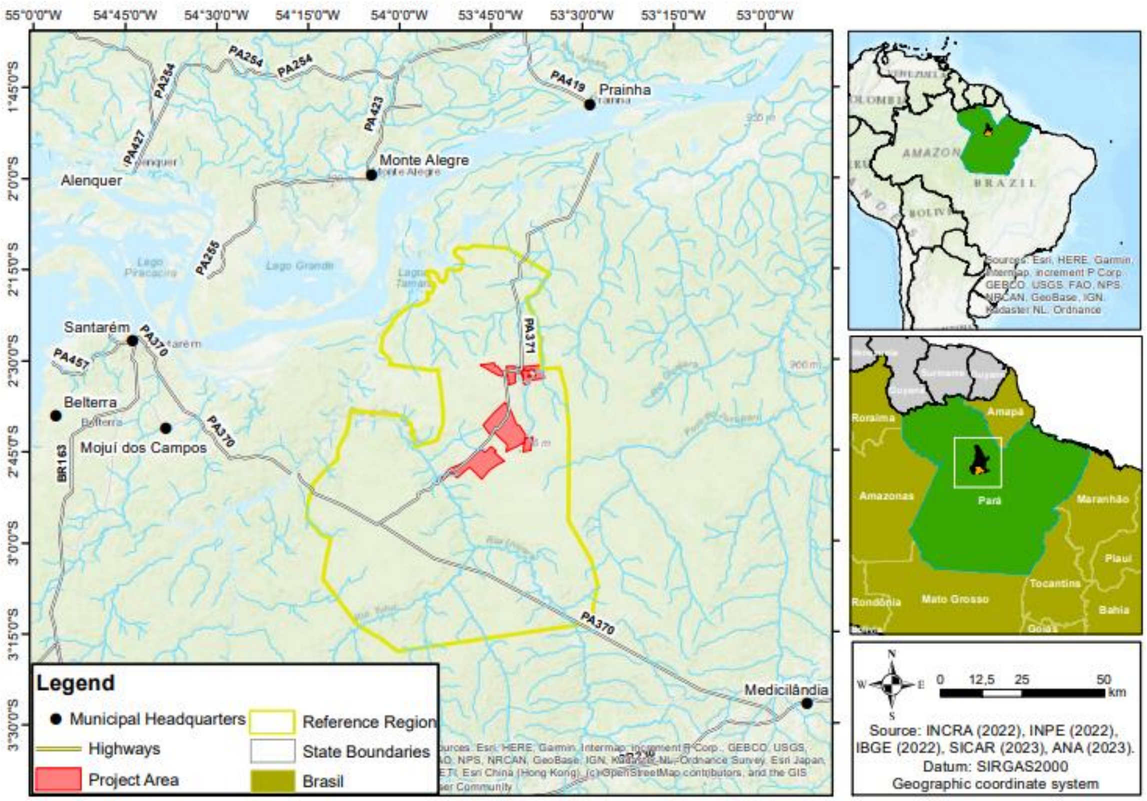


Figure 5. Project Area and the PA 371 Road

Climate: The project area is classified as Ami according to the Köppen-Geiger climate classification, characterized by high annual rainfall of approximately 1900 mm and temperatures ranging from 24ºC to 34ºC (Figure 5). The region experiences short dry periods, with lower precipitation from August to November and higher rainfall from February to MayHydrology

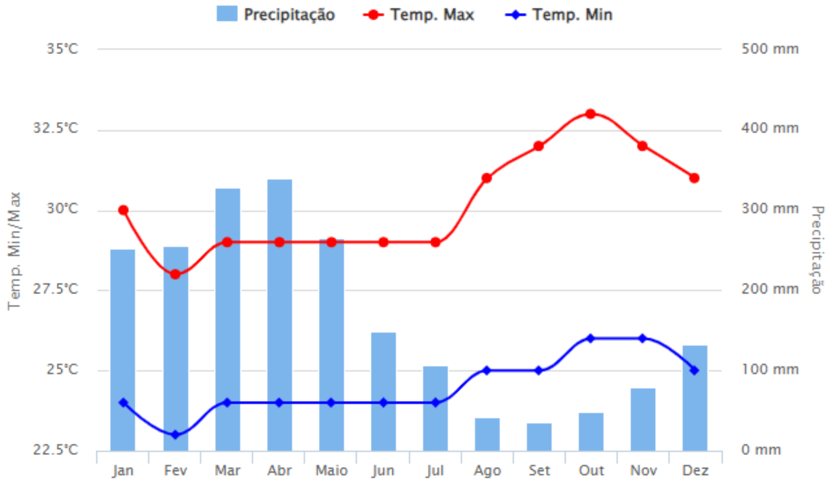


Figure 6. Annual climate variation

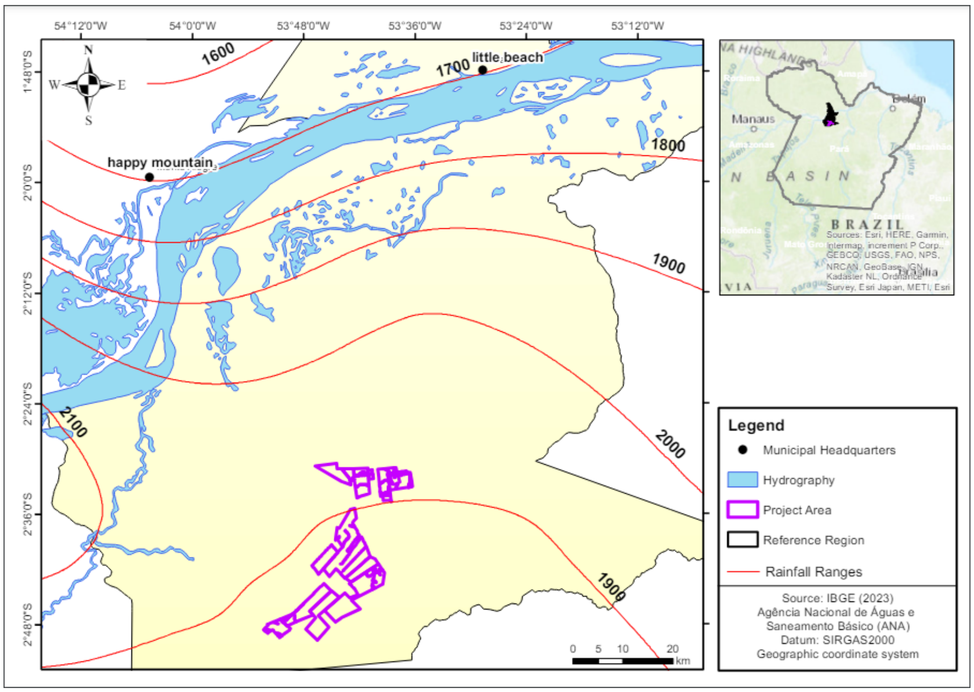


Figure 7. Annual climate variation in the Project Area

Hidrology: It is located northward and is bordered to the south by the Curuatinga River, a tributary of the Uruará River, which eventually flows into the Amazon River. The Curuatinga River is crucial for local communities, serving as a fishing and leisure area.

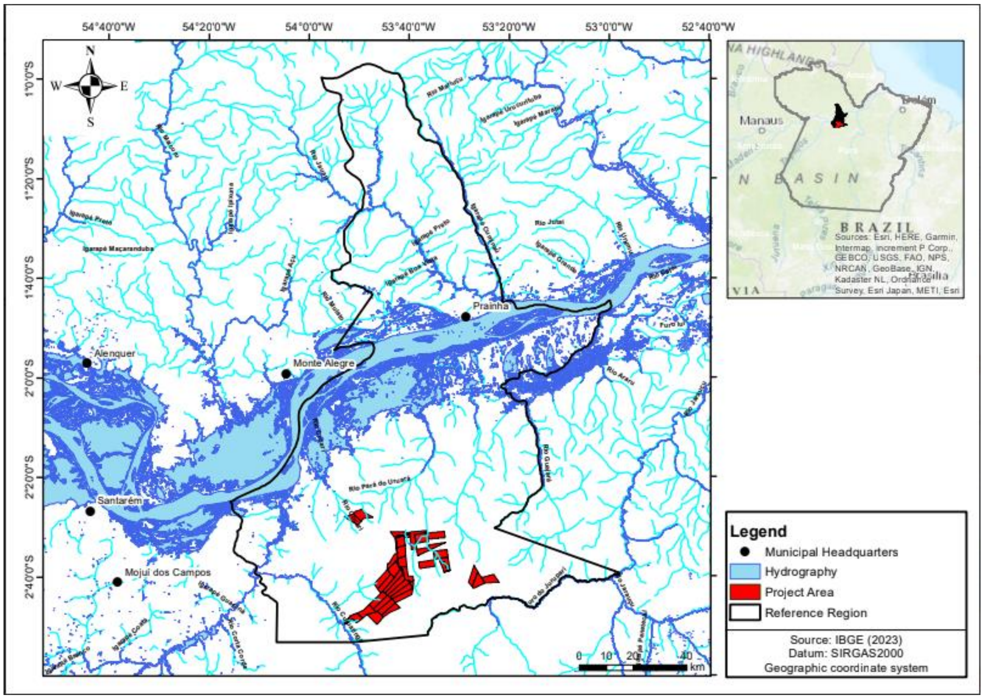


Figure 8. Hydrology in the project area

Topography: The topography of the project area is predominantly flat (0-3% slope) and gently undulating (3.1-20% slope). This terrain configuration influences the distribution of vegetation and land use in the region.

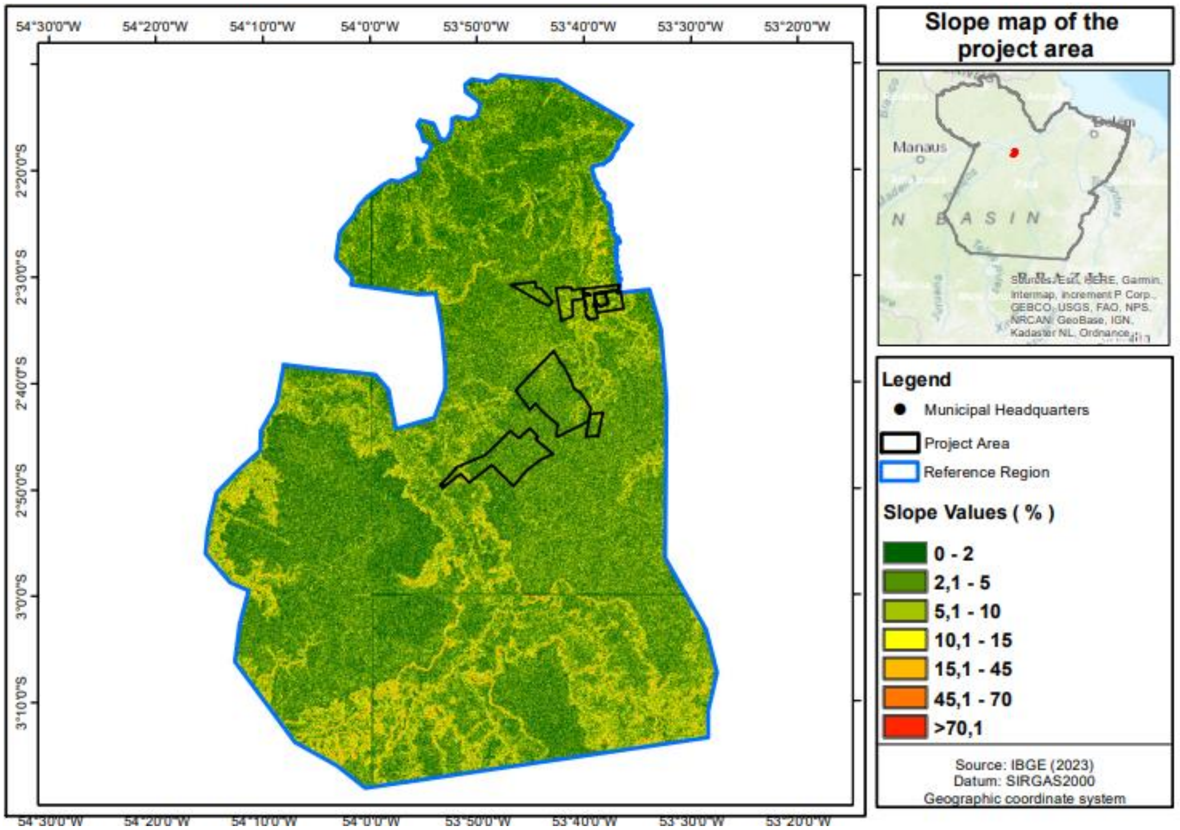


Figure 9. Topography of the project Area

Relevant historic conditions: Historically, Prainha was formed from small settlements that developed along the Amazon River in a region rich in natural resources but difficult to access. The local economy has always been based on the exploitation of natural resources, such as timber, fishing, and, more recently, livestock and agriculture. This dependence on natural resources has shaped the social and economic dynamics of the municipality, where a lack of economic diversification and pressure for land for cultivation and grazing have led to high rates of deforestation.

The historical formation of Prainha is intrinsically linked to the colonization process of the Amazon, where territorial occupation was driven by the search for new agricultural frontiers and the exploitation of natural resources. This context of land occupation and use resulted in a concentrated land structure and a development model based on agro-pastoral expansion, which, while generating economic growth for some sectors, has also intensified the environmental and social problems of the municipality.

Agroflorestal Novo Horizontes Ltda is a family-owned company that acquired its land in the region over 30 years ago through the purchase of lots from settlers and companies in the area. Since 2018, it has been conducting sustainable forest management activities for timber production on its 23,092.85 hectares with FSC-certified production (FSC-C156509).

Starting in 2026, the managed area will enter a fallow period to comply with the 25-year harvesting cycle, and the forested areas will remain without management activities until 2043. The REDD AUD project serves as an economic alternative for the maintenance and protection of the forest, which is under significant pressure from degradation and deforestation activities in its surroundings.

Soils: The main soil types in the project area are Dystrophic Yellow Latosol (LAd) and Dystrophic Red-Yellow Argisol (PVAd). LAd is characterized by good moisture retention and permeability, while PVAd has low fertility due to its acidity and high clay content.

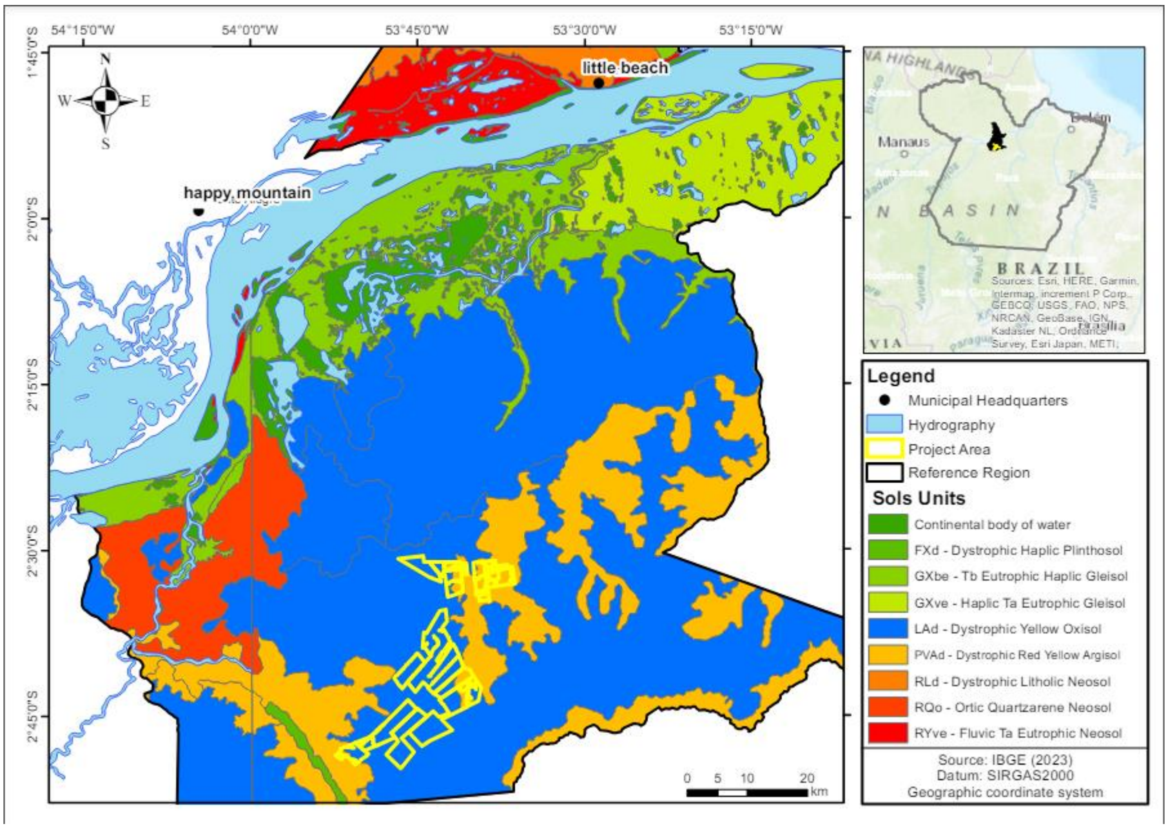


Figure 10. Soils in the project area

Vegetation and ecosystems of the project area: The predominant vegetation is dense ombrophilous forest, with large trees and rich biodiversity. The area features diverse flora, including species such as Bertholletia excelsa and Copaifera langsdorffii. The vegetation includes perennial plants and lianas, adapted to high humidity and frequent precipitation conditions.

Biodiversity

The Amazon Rainforest is home to an estimated 30 million animal species, harboring more than 73% of the mammal species and 80% of the bird species found in Brazil. Within its boundaries, the country hosts at least 311 mammals, 1,300 birds, 273 reptiles, 232 amphibians, and 1,800 freshwater fish species. Notable river-dwelling mammals include the pink river dolphin, tucuxi, Amazonian manatee, giant otter, and otter. Within the forest, species such as the endangered jaguar, anteaters, and a vast array of primates, including capuchin monkeys, spider monkeys, red howler monkeys, and woolly monkeys, can be found. According to ICMBio, the biome shelters 10% of the world's primate species.

As part of the Agroflorestal Novo Horizonte project, a wildlife survey was conducted with biologists. The sampling process was divided into two zones: the Direct Area of Interest (AID), corresponding to the areas where the REDD+ AUD project is executed, and the Indirect Area of Interest (AII). Sampling points were designated across 10 locations, each marked with indicator stakes, with F01 denoting Fauna Sample Point 01.

Three faunal groups were surveyed: ornithofauna (birds), herpetofauna (amphibians and reptiles), and mastofauna (mammals of small, medium, and large sizes). Each group was surveyed using distinct sampling methodologies, described below.

The ornithofauna survey was carried out using the Mackinnon list technique, which involves generating successive lists of 10 species recorded while traversing trails in the study areas (Ribon, 2010). Once 10 different species are observed, a new list begins. Each species is recorded only once per list, even if repeatedly sighted. It can only be recorded again if detected in a subsequent list. This technique, which produces many sample units (lists) in a short period, allows for a robust estimate of bird species richness and composition. It also provides general data on the relative abundance of species (% frequency in the lists). Additionally, the playback technique was employed. This method involves playing back the calls of species that were either not seen or are known to have unfamiliar vocalizations. The birds are stimulated to defend their territories, drawing closer to the observer, thus allowing for visual identification. This increases the likelihood of detecting rare or threatened species. Recordings can either be pre-recorded or made in the field using a digital recorder and directional microphone. Surveys were conducted during the morning (7:00 AM to 12:00 PM) and afternoon/evening (1:00 PM to 9:00 PM).

For the monitoring of herpetofauna, the active search technique was employed (Heyer et al., 1994). This method involves searching for individuals in environments such as burrows, tree hollows, leaf litter, under rocks, termite mounds, swamps, and temporary ponds. The active search was conducted at designated points, with a focus on covering wet areas, the interior, and the edges of forest fragments. During these searches, all species observed or identified by their vocalizations (in the case of frogs) were recorded. The activities were conducted in the morning (7:00 AM to 12:00 PM) and afternoon/evening (1:00 PM to 9:00 PM).

For the sampling of small, medium, and large mammals, camera traps and active searches for animals and traces were used. Camera traps are an efficient method to estimate the density of individuals, identify species, and determine activity patterns and habitat use, particularly for larger mammals. This technique allows for the sampling of a wide range of species with varying sizes and habits, regardless of visibility conditions affected by vegetation (Srbek-Araujo & Chiarello, 2005). Five camera traps were installed, prioritizing areas with low human traffic. Searches for mammal traces such as tracks, feces, fur, and burrows were carried out during the morning (7:00 AM to 12:00 PM) and afternoon/evening (1:00 PM to 9:00 PM).

Data analysis was categorized based on extinction threat levels according to global and national listings, endemism, migratory status for birds, environmental sensitivity, and habitat preference. To assess the efficiency of the method, a species rarefaction curve was generated for the recorded species at the sampling points using 100 randomizations. Subsequently, species richness estimators (Jackknife 1, Bootstrap, Chao 2) were used to predict the number of species. All calculations were performed using PAST4 software.

During the sampling period, the monitoring of birdlife recorded 69 bird species belonging to 17 orders and 30 families. The order with the greatest family representation was Passeriformes (n=12), while the families with the highest number of species were Psittacidae (n=11), Accipitridae (n=4), and Thamnophilidae (n=4) (Annex 1). Over the course of the survey campaign, 42 Mackinnon lists were compiled. The most frequently observed species were the Hyacinth Macaw (*Anodorhynchus hyacinthinus*, F=24), the Red-and-Green Macaw (*Ara chloropterus*, F=20), the Golden-winged Parakeet (*Brotogeris chrysoptera*, F=19), the White-throated Guan (*Penelope pileata*, F=13), and the Coraya Wren (*Pheugopedius coraya*, F=12).

In the case of herpetofauna, 3 orders, 16 families, and 35 species were sampled. Amphibians belonged to the order Anura, distributed across 5 families and 14 species. Reptiles were from the orders Sauria (5 families and 9 species), Ophidia (4 families and 9 species), and Testudines (2 families and 3 species). Anura (n=5) had the highest number of families, and the family Leptodactylidae (n=6) had the most species. The most frequently encountered species were *Dendropsophus cf microcephalus* (n=120 estimated in reproductive sites) and *Leptodactylus mystaceus* (n=27).

As for mammals, 16 species, 9 orders, and 14 families were recorded. The orders with the greatest family representation were Felidae (n=3) and Primates (n=3). The families with the highest species representation were Felidae (n=3).

Of the individuals sampled in both AID and AII, 15 species were listed as threatened, including 9 bird species, 2 herpetofauna species, and 4 mammal species. Among the threatened bird species were the Grey Tinamou (*Tinamus tao*), Harpy Eagle (*Harpia harpyja*), Red-necked Aracari (*Pteroglossus bitorquatus*), White-throated Guan (*Penelope pileata*), Blue-winged Macaw (*Primolius maracanã*), Black-spotted Bare-eye (*Phlegopsis nigromaculata*), Razor-billed Curassow (*Aburria cujubi*), Hyacinth Macaw (*Anodorhynchus hyacinthinus*), and the Helmeted Curassow (*Pauxi tuberosa*).

From the herpetofauna group, two reptiles were listed as threatened: the Scorpion Mud Turtle (*Kinosternon scorpioides*) and the Yellow-footed Tortoise (*Chelonoidis denticulata*). Among the mammals, the Jaguar (*Panthera onca*), Puma (*Puma concolor*), Brazilian Tapir (*Tapirus terrestris*), and Red-handed Howler Monkey (*Alouatta belzebul*) were listed as conservation threats.

The sampled area in AID and AII recorded 19 species endemic to the Amazon biome, 1 species endemic to the Brazilian Amazon basin, and 2 species endemic to Brazil. Of these species, birds had the highest number of endemisms, with 12 species endemic to the biome, 2 species endemic to Brazil, and 1 species endemic to the Brazilian Amazon basin. These species are, respectively: *Ara macao*, *Brotogeris chrysoptera*, *Cathartes melambrotus*, *Cercomacra cinerascens*, *Deroptyus accipitrinus*, *Microcerculus marginatus*, *Phlegopsis nigromaculata*, *Pheugopedius coraya*, *Ramphastos tucanus*, *Thraupis episcopus*, *Tinamus tao*, *Tyranneutes stolzmanni*, *Anodorhynchus hyacinthinus*, *Penelope pileata*, *Pteroglossus bitorquatus*.

Among the herpetofauna, five species were restricted to the Amazon biome: *Phyllomedusa tomopterna*, *Chatogekko amazonicus*, *Gonatodes humeralis*, *Plica plica*, and *Uracentron azureum*. Finally, mammal sampling revealed two species endemics to the Amazon biome: *Passalites nemorivaga* and *Bradypus tridactylus*.

The presence of both threatened and endemic species highlights the biological importance of conservation and ecosystem preservation projects in the area of the enterprise. These biodiversity protection measures play a crucial role in maintaining the ecosystem as a whole and contribute to the reduction of greenhouse gas emissions.

Social Conditions: With a population of 35,577 people according to the latest 2022 census and a per capita GDP of 15,000 reais, Prainha has a Municipal Human Development Index (IDHM) of 0.523, which is considered low. This value indicates significant challenges in the areas of health, education, and income within the municipality.

The Social Progress Index (IPS) Brazil is the most comprehensive measure of the socio-environmental reality across all 5,570 municipalities in the country. The index provides a multidimensional and accessible overview of how municipalities and states perform in meeting the basic needs of their citizens. The municipality of Prainha recorded a Social Progress Index (SPI) of 49.48[7], which is considered a low social progress score. This low score is primarily due to poor performance in areas such as access to higher education (17.99), individual rights (25.2), and access to information, water, and sanitation (39.12).

There are four local communities living near the project area (PA). These communities are primarily composed of residents from rural areas and local family farming producers. The project's social team conducted 40 interviews with the population living in the project areas. During the social analysis period, general presentations about the project were given, followed by individual interviews. Each issue

[7] https://ipsbrasil.org.br/explore/scorecard/PA/1506005

addressed by the social questionnaires will be presented, and future socioeconomic analyses will be conducted throughout the project's duration to monitor activities and impacts.

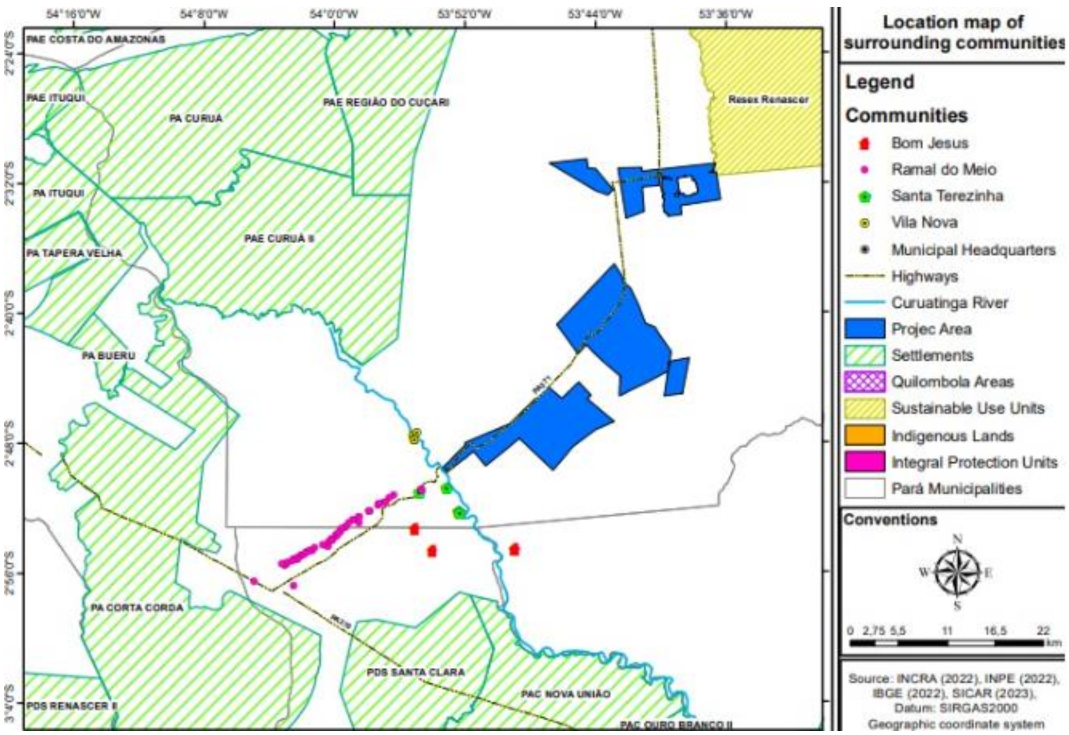


Figure 11. Location of communities, settlements, and conservation units near the project area.

The socio-environmental surveys conducted with communities near the project area revealed that 48% of the people are residents of these communities, 23% of the respondents reported using the land exclusively for agroforestry systems, 22% of the respondents, in addition to being residents, use the land for purposes such as agroforestry, agriculture, or other income-generating activities, and finally, 9% reported using the land solely for agriculture (3%), planting and investment (3%), and rural production (3%) (Figure 11).

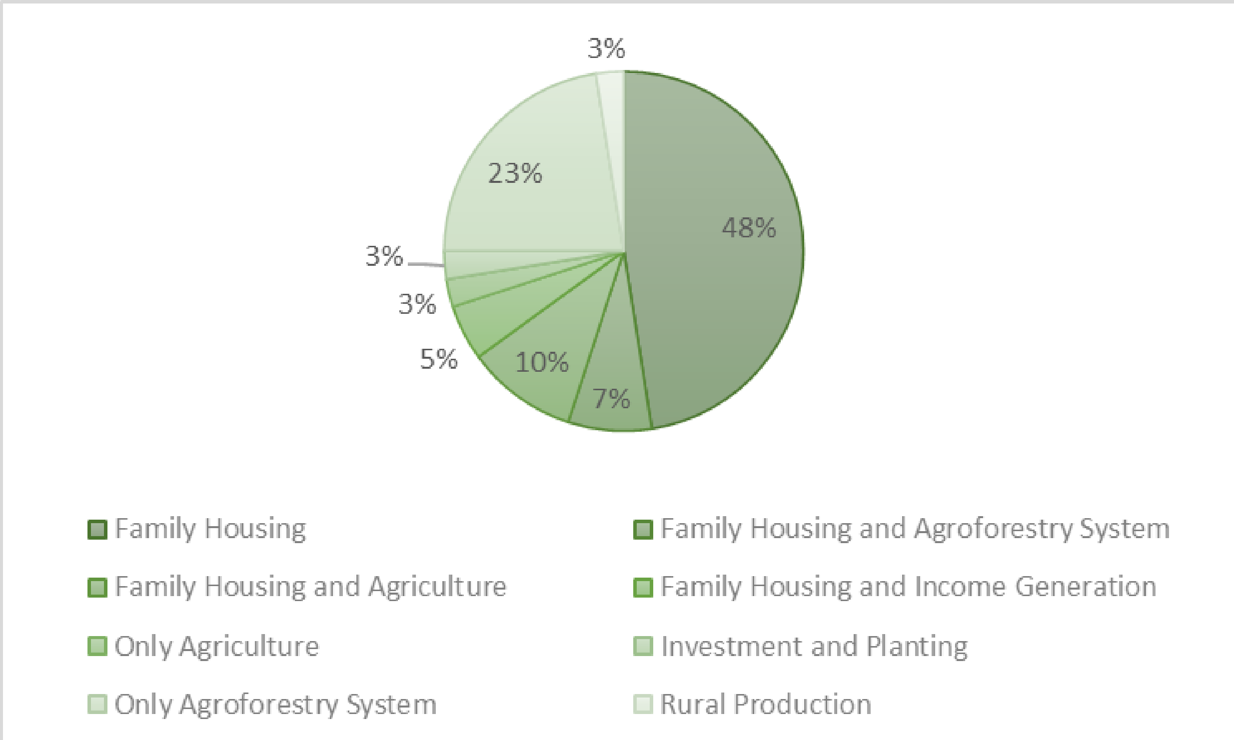


Figure 12. Community Land use

Of these landowners, 78% indicate that the land is owned and fully paid for, 15% say they are settlers, and 7% have land granted by relatives or the community. Of those who own land, out of the 40 people interviewed, 22 (55%) have public registration with the Rural Environmental Registry (Cadastro Ambiental Rural) and/or Georeferencing by the National Institute for Colonization and Agrarian Reform (INCRA). Additionally, 18% of residents do not have documentation but have started the process, 15% hold a Use Concession Contract, and only 8% have a title of ownership or are in the process with INCRA to obtain it. There were reports of people who were unaware of the land ownership documentation (2%) and people who purchased the property without documentation (2%).

Socioeconomic Characteristics

Among the local population interviewed, 31 people (78%) reported a family income of up to 1 Minimum Wage (MW) ($254.45), and 9 people (22%) reported having an income of 1 to 3 MW per month. The main source of family income identified was agriculture, mentioned by 83% of the respondents. Other sources were also cited but appeared less frequently and were often combined with agriculture. The figure 12 below proportionally indicates the primary sources of income for the respondents.

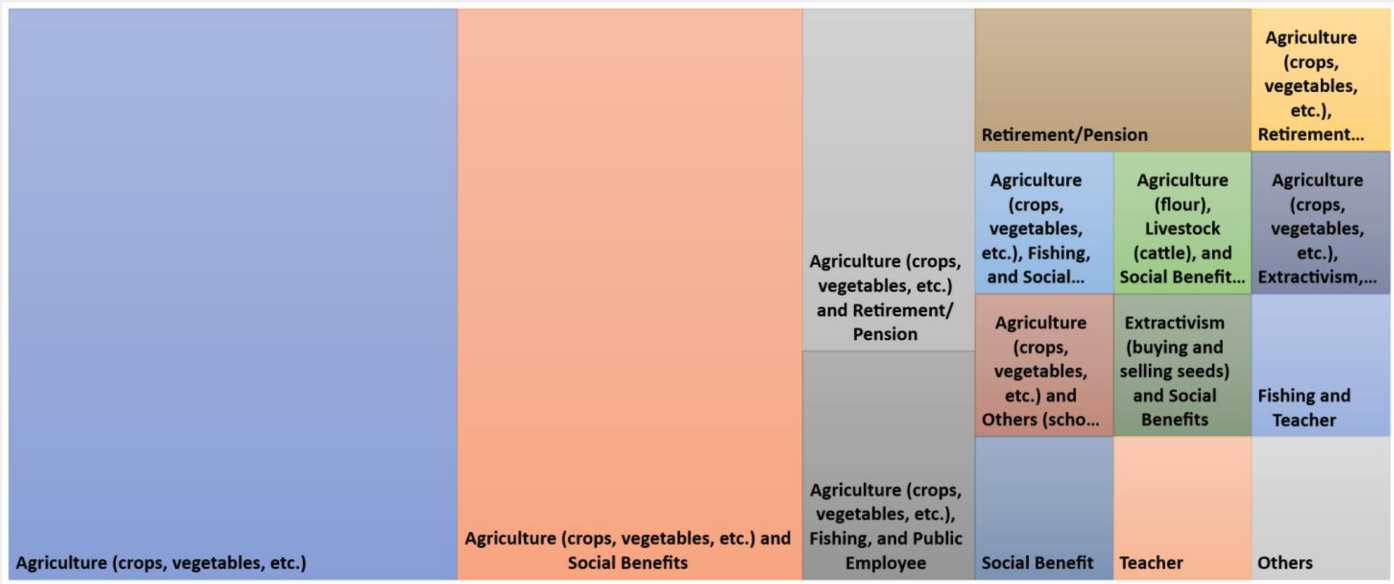


Figure 13. Map of income sources for families in communities near the project.

Infrastructure

a. Internet

The majority of the local population interviewed reported not having access to the internet. Quantitatively, 50% of the respondents said they did not have internet on their property. Of the total people who reported having internet access (45%), 8% stated they had access through a neighbor or via satellite, and 5% said they had occasional internet access.

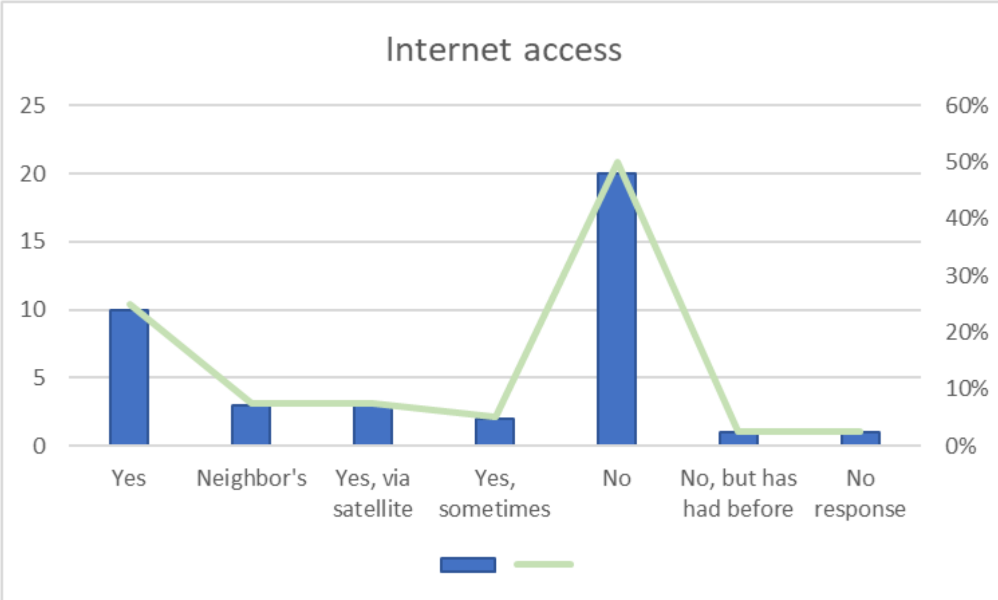


Figure 14. internet access

b. Sanitation and Potable Water

The tables below provide a comparative analysis of the communities adjacent to PA, focusing on their differences in basic sanitation (Table 9) and the quality of potable water (Table 10).

In the Bom Jesus community, nine representatives were interviewed. Regarding bathroom drainage systems, four individuals reported using rudimentary cesspools, four utilized septic tanks, and one indicated an alternative system. Concerning potable water supply, six respondents noted that their water source was located outside the property, two indicated it was sourced from within the property, and one mentioned that the water was obtained through micro-systems or a public distribution network.

In the Ramal do Meio community, thirteen individuals participated in the interview process. Among them, ten reported using rudimentary cesspools for bathroom drainage, two used septic tanks, and one specified an unspecified alternative system. Additionally, four individuals sourced their water from outside the property, two from within the property, and seven indicated various other supply methods, including a neighbor's well, third-party purchases, and/or rainwater collection.

The Santa Terezinha community was assessed with the involvement of five local residents. Of these, four indicated that rudimentary cesspools were the primary drainage system for their bathrooms, while one used a septic tank. The methods of water supply were varied: one individual sourced water from a well or spring outside the property, one from within the property, two reported having artesian wells, and one mentioned an alternative supply method involving direct collection from the river.

Finally, in the Vila Nova community, thirteen individuals were interviewed. Six reported using rudimentary cesspools, five utilized septic tanks, one declined to answer the question, and one indicated the use of another drainage system. Regarding water supply, three respondents collected water from artesian wells, one from a community well, five sourced it from springs or wells outside their properties, one used a cistern, and three mentioned alternative sources, including acquaintances or rainwater collection.

It is important to note that all interviewees across the four communities indicated the absence of a sewage collection service within their areas.

Table 9. Type of sanitation existing in nearby communities

Sanitation				
Community	**Interviewed Residents**	**Bathroom Drainage**		**Sewage Collection**
Bom Jesus	9 people	Primitive cesspool	4 people	None
		Septic tank	4 people	
		Other	1 people	
Ramal do Meio	13 people	Primitive cesspool	10 people	None
		Septic tank	2 people	
		Other	1 people	
Santa Terezinha	5 people	Primitive cesspool	4 people	None
		Septic tank	1 people	
Vila Nova	13 people	Primitive cesspool	6 people	None
		Septic tank	5 people	
		Don't know/No response	1 people	
		Other	1 people	

Table 10. Comparison between the type of water source and the water quality perceived by the communities.

Water source			Water Supply Quality	
Well or spring outside the property	6 people		Excellent	2 people
Well or spring on the property	2 people		Good/Regular	2 people
Public distribution network, microsystems	1 people		None	5 people
Well or spring outside the property	4 people		Good	4 people
Well or spring on the property	2 people		Poor/Very poor	4 people
Other	7 people		None	5 people
Well or spring outside the property	1 people		Excellent	1 people
Well or spring on the property	1 people		Good	2 people
Poço artesiano	2 people			
Other	1 people		None	2 people
Artesian well	3 people		Excellent	4 people
Community well	1 people		Good	4 people
Well or spring outside the property	5 people		Regular	2 people
Cistern	1 people		Poor	1 people
Other	3 people		Very poor	1 people
			None	people

c. Public Transportation and Roads

The perceived quality of public transportation varies across the four communities (Figure 14). The Ramal do Meio community provided the most favorable evaluations, with responses predominantly categorized as "excellent," "good," and "fair." In contrast, the Bom Jesus community reported the lowest ratings for public transportation quality, with responses primarily falling between "fair" and "very poor." Both the Vila Nova and Santa Terezinha communities received generally positive evaluations, with ratings ranging from "good" to "fair."

Regarding road quality, the Ramal do Meio and Santa Terezinha communities expressed favorable opinions, indicating that road conditions were "good" or "fair." Conversely, the Bom Jesus and Vila Nova communities reported significantly poor road conditions, with most responses categorized as "poor" to "very poor."

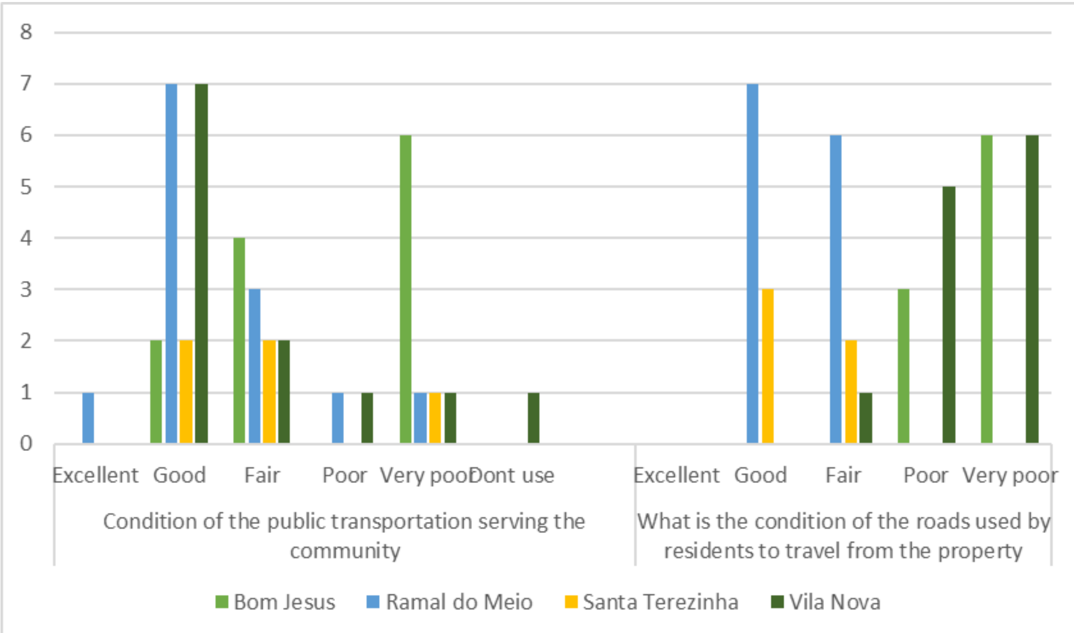


Figure 15. Communities' perception of the quality of transportation and roads

d. Health

A social analysis of access to healthcare, based on interviews with residents of communities near the project area, revealed that 11 residents from the Vila Nova community and one resident from the Ramal do Meio community have access to nearby health posts. The remaining residents indicated that they rely on healthcare services in the municipality of Santarém.

Of the 40 people interviewed, 25% rated the regional healthcare services as good, 20% reported no services available nearby, 18% rated the services as fair, 15% as poor, 10% as very poor, 10% did not respond, and 3% considered the healthcare services to be excellent.

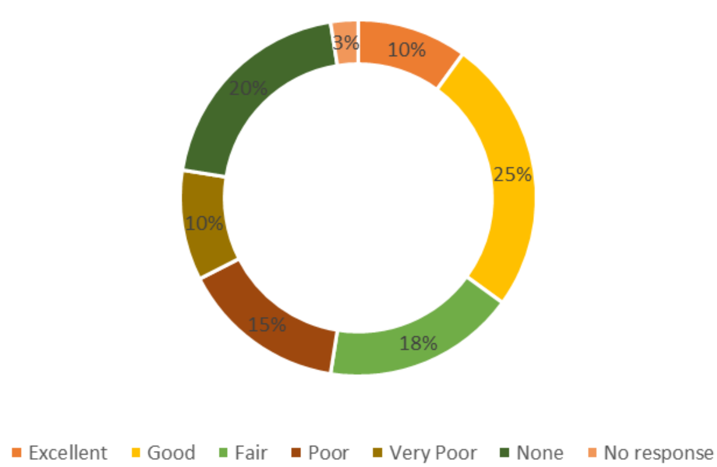


Figure 16. Access to health care.

e. Education

When asked about local education, residents of communities near the project area indicated that most have schools in close proximity to their homes. Of the 40 residents interviewed, 35 (88%) reported having access to a school near their residence, while 5 (12%) indicated no nearby school. The primary schools mentioned were Santa Terezinha School (serving the Santa Terezinha and Ramal do Meio communities), EMF Tereza de Almeida Costa School (Bom Jesus community), and Pequeno Polegar School (Vila Nova community).

Productive Aspects

a. Ramal do Meio Community

The cultivation of gardens and fruit trees in the backyards of properties for personal consumption is a common practice among all interviewed residents. The primary crop reported by residents for personal use was fruit trees. Notably, only one individual mentioned collecting forest resources for artisanal production, specifically wood for making kitchen utensils. Conversely, only one resident indicated that they do not collect forest resources for family food consumption. Among those who do, Brazil nuts were the most commonly gathered resource. The collection of forest resources extends beyond food and manufacturing; all residents of Ramal do Meio are knowledgeable about and utilize medicinal plants to promote well-being. The most frequently reported medicinal plant is “chá da preciosa” (precious tea).

Aniba canelilla, a plant from the Lauraceae family native to the Amazon, has various uses. The bark, wood, leaves, and seeds of this tree species are used to create teas with medicinal properties, primarily known for their beneficial effects on the digestive system, including antispasmodic properties and relief from dysentery.

b. Bom Jesus Community

Similar to the Ramal do Meio community, all interviewed property owners engage in the cultivation of fruit trees for personal consumption. Residents universally collect forest resources for both food and medicinal purposes. Among the various plants cited, Copaíba (Copaifera sp.) was the most frequently mentioned medicinal plant. Copaíba is a group of trees endemic to Brazil, found primarily in the Amazon and Cerrado biomes. These trees are renowned for their production of oils and resins that possess medicinal properties, particularly known for their anti-inflammatory and analgesic effects.

Fishing constitutes a significant cultural practice within the community, owing to its proximity to the Curuatinga River. All interviewed residents reported engaging in fishing for personal consumption, with some participating daily, others weekly, and a few monthly. Additionally, over half of the respondents utilize the river for transportation via motorized boats (rabeta).

c. Santa Terezinha Community

All interviewed residents from the Santa Terezinha community reported cultivating fruit trees on their properties for personal consumption. Most residents also gather forest resources for food, including Brazil nuts and uxi. Furthermore, all respondents are knowledgeable about or actively collect medicinal plants, with "chá da preciosa" being the most frequently mentioned, as noted in the Ramal do Meio community.

d. Vila Nova Community

In line with the traditions of neighboring communities, the interviewed residents of Vila Nova also cultivate fruit trees on their properties for personal consumption. Only one resident reported not collecting forest resources for food, while the majority primarily gather uxi, pequiá, and Brazil nuts. Additionally, all interviewed residents are either familiar with or actively collect medicinal plants.

Similar to the Bom Jesus community, Vila Nova is situated along the banks of the Curuatinga River. This proximity makes fishing a common practice among residents, who engage in it with varying frequency for family consumption.

Community Engagement

Community residents exhibit limited knowledge regarding the origins of their respective communities. However, some individuals shared insights about their community's beginnings. The Ramal do Meio and Santa Terezinha communities originated from an area formerly associated with a logging company known locally as Madesa. The Bom Jesus community was previously referred to as the Anta Community, although residents had little additional information about its history. In contrast, the Vila Nova community was established in 2007 through the efforts of two long-time residents, "Zezinho" and "Dona Conce."

When asked about their participation in associations or unions, residents of Ramal do Meio displayed a notable level of collective engagement. Among those interviewed, 54% are members of various organizations, while 46% are not (Figure 16). The most prominent organizations among those affiliated include the Corta Corda Association, Terra Nova Association, Community Union, Corta Vento Association, and the Boa Esperança Cooperative.

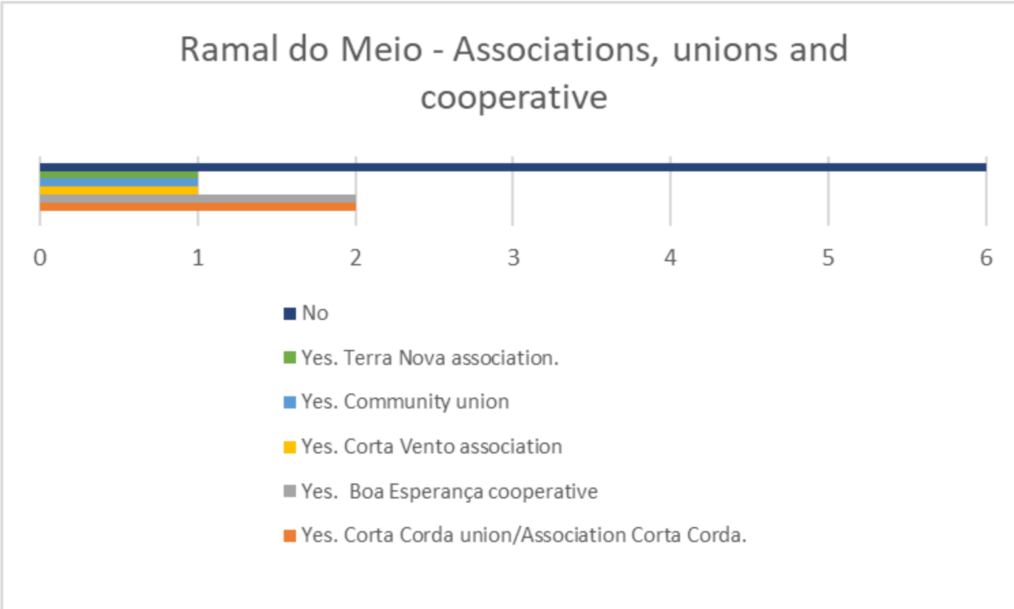


Figure 17. Ramal do meio communiy - Main social organizations participation

In the Santa Terezinha community, unlike Ramal do Meio, the majority of interviewees (80%) are not affiliated with any association, union, or cooperative, as illustrated below (Figure 17).

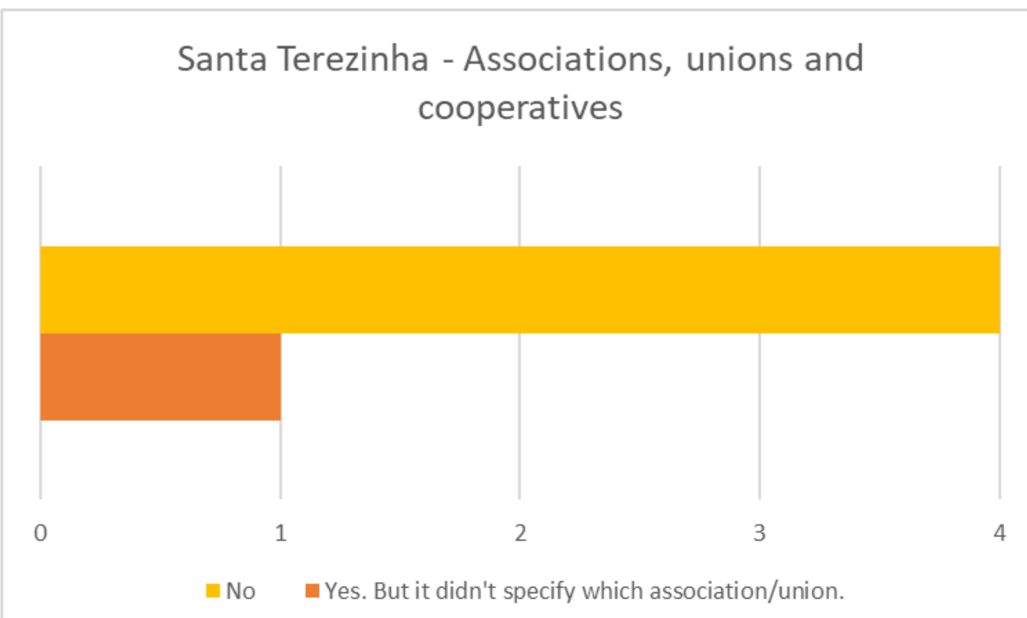


Figure 18. Santa Terezinha communiy - Main social organizations participation

The Bom Jesus community had the highest number of residents (89%) affiliated with associations, unions, or cooperatives. Three associations were identified: the "community association," APROSTINGA, STTR, and AASFLOR (agro-extractive association). A smaller proportion (11%) of residents reported not being affiliated with any organization (Figure 18).

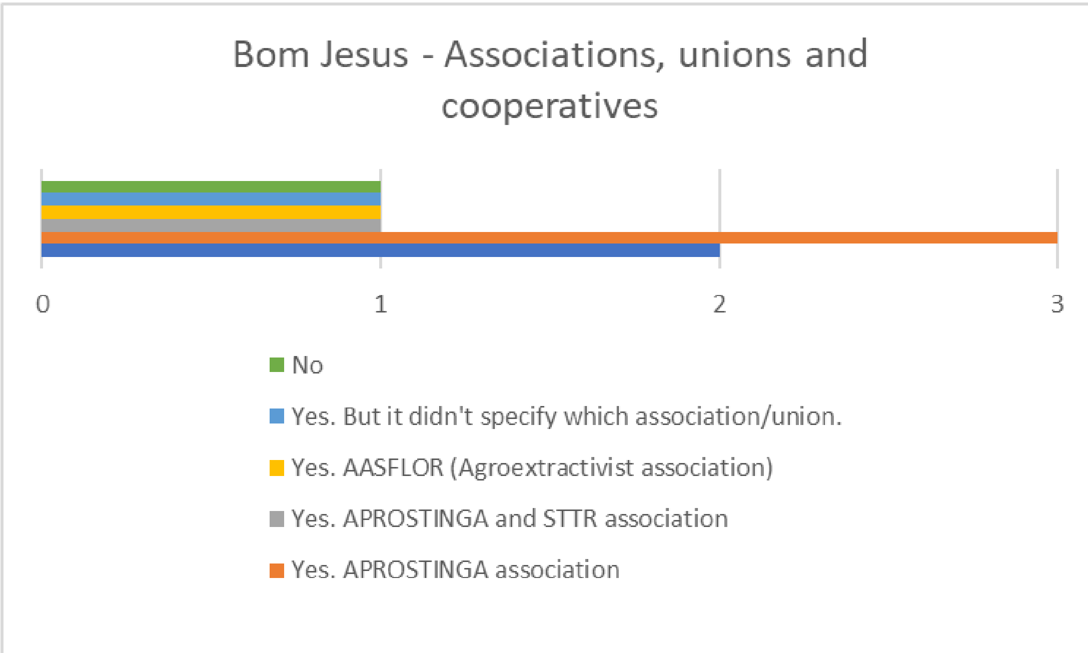


Figure 19. Bom Jesus communiy - Main social organizations participation

Lastly, in the Vila Nova community, 84% of the respondents reported being affiliated with the SEPROVIM association, ASCOMOVIN association, or another type of unspecified association, union, or cooperative, as shown in Figure 19 below.

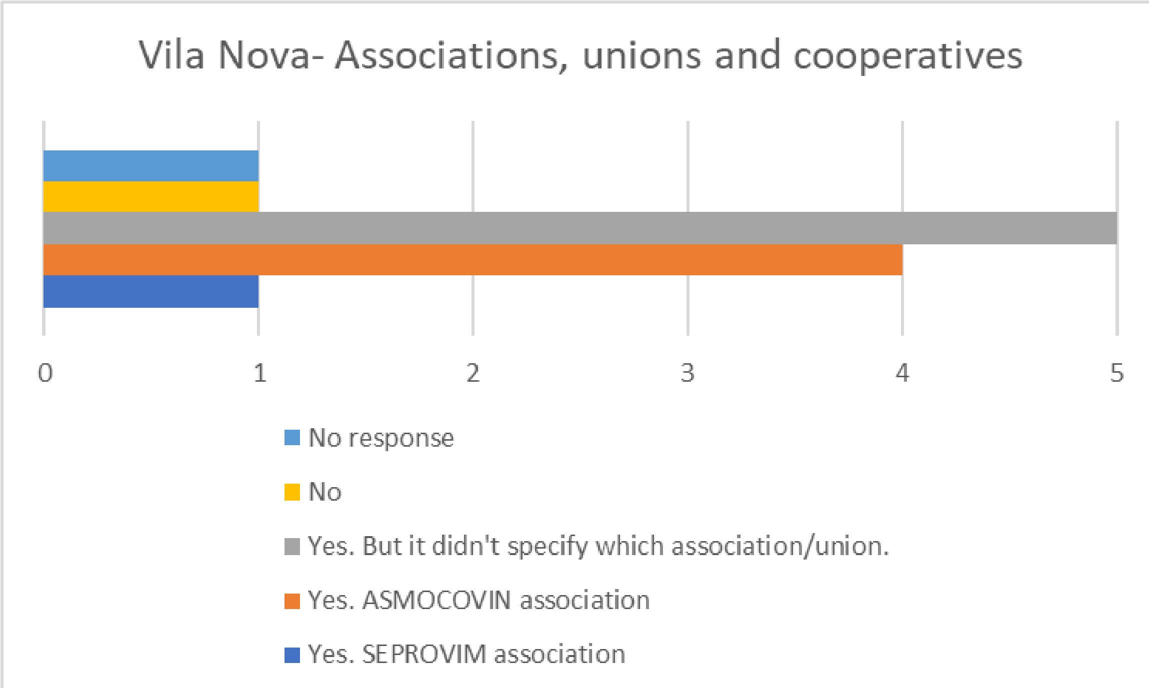


Figure 20. Vila Nova communiy - Main social organizations participation

Community and the Agroflorestal Novo Horizonte Company

The local population was asked about the relationship between the neighboring communities and Agroflorestal Novo Horizonte. Of the 40 people interviewed, 5 reported having participated in activities organized by Agroflorestal Novo Horizonte. All of these participants attended lectures on the Carbon Project, and 2 of them also participated in oral health seminars.

Residents mentioned several positive impacts of the company on their communities. The most frequently cited positive impact was the improvement and maintenance of roads. Other, less frequently mentioned positive aspects included job creation, the provision of potable and non-potable water during the summer, assistance in schools, and forest conservation.

Most respondents indicated that there were no negative points in the relationship between the company and the communities.

Resex Renascer traditional communities

The Resex Renascer currently hosts more than 600 families in 16 tradition riparian communities organized into three sectors distributed along three main rivers: Uruará, Tamuataí, and Guajará. These families primarily rely on agriculture, small-scale animal husbandry, and fishing for income and food security.

Although a small portion of RESEX Renascer territory borders the project area, the villages of traditional communities are distant and do not use the area for resource extraction or cultural practices. It was found that RESEX inhabitants are located far from the project areas and do not enter the project area for resource use. Therefore, the implementation of socio-environmental actions in RESEX is not a priority for the Agroflorestal Novo Horizonte REDD+ project. However, future socio-environmental actions may be proposed within the programmatic axes of the conservation unit management plan, in alignment with managers and community leaders

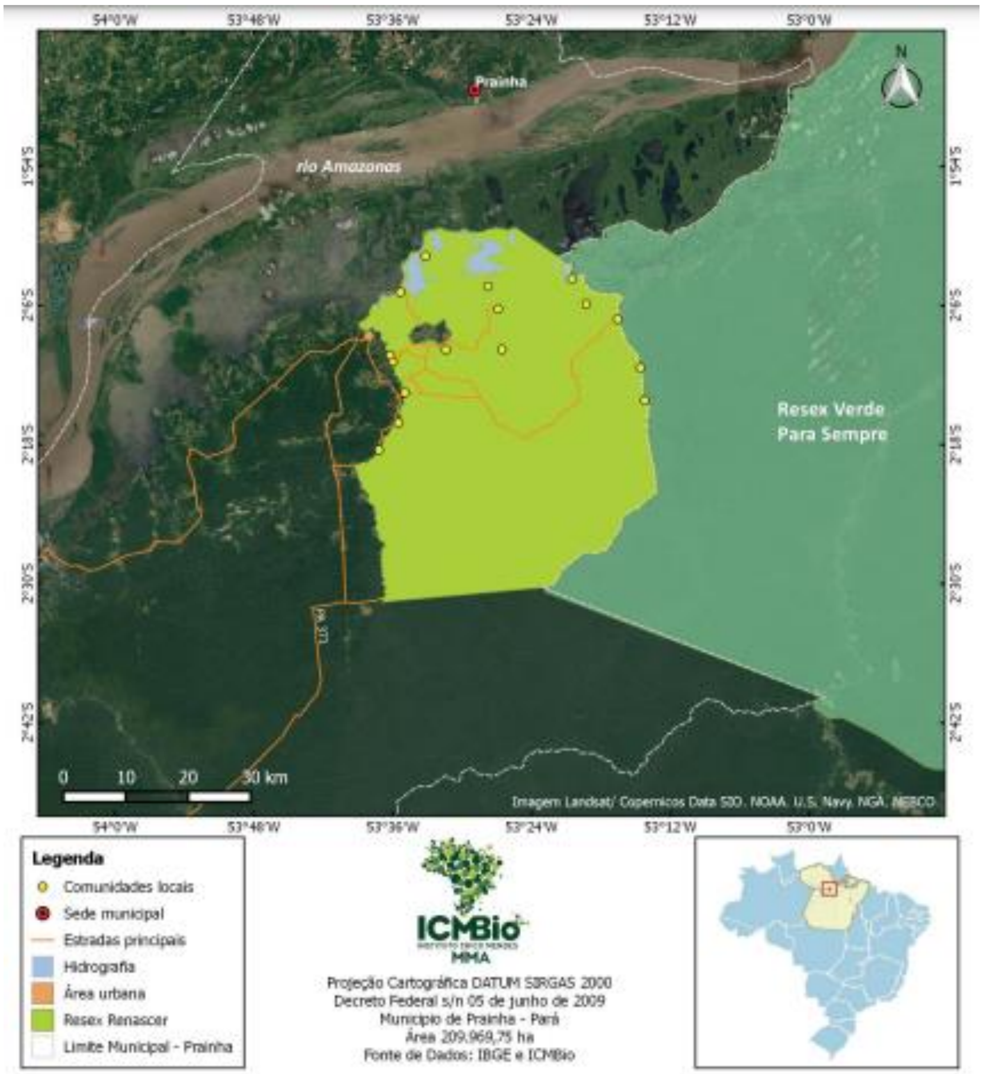


Figure 21. Location of Communities within the RESEX Renascer

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

Agroflorestal Novo Horizonte ensures that all relevant Brazilian laws and regulations are strictly followed in the activities carried out in the project area. The laws, statutes, and regulatory frameworks highlighted below are particularly relevant to the project action plan:

Federal Law No. 5,197, dated January 3, 1967: This law regulates the protection of Brazilian fauna, prohibiting the use, pursuit, destruction, hunting, or capturing of animals of any species, except under special conditions. Fauna protection is essential for maintaining biodiversity in forests. The Agroflorestal Novo Horizonte project ensures fauna protection and conducts environmental education activities related to local animals and the maintenance of ecological balance in forest areas.

Federal Law No. 5,889, dated June 8, 1973: Establishes regulatory norms for rural labor in Brazil, including working conditions and rights of rural workers. All workers involved in forest management activities and in the Agroflorestal Novo Horizonte REDD+ project area, whether they are employees or third parties, have their rights ensured, with decent and safe conditions. This is a premise of the company also ensured by third-party audits on FSC socio-environmental standards.

National Environmental Policy (Law No. 6,938, dated August 31, 1981): Defines the National Environmental Policy, its objectives, instruments, and mechanisms of application, including the protection of ecosystems and areas threatened by degradation. Since this policy establishes the legal framework for environmental protection in Brazil, it is fundamental for the implementation of projects aimed at preserving and restoring forest areas.

Law No. 7,804, dated July 18, 1989: Amends Law No. 6,938, from 1981, improving mechanisms for the formulation and application of the National Environmental Policy.

Federal Law No. 9,605, dated February 12, 1998: Provides for criminal and administrative sanctions for conduct and activities harmful to the environment. By defining administrative sanctions, this law provides a regulatory environment that discourages destructive practices, favoring the viability of conservation projects.

Federal Law No. 9,985, dated June 18, 2000: Establishes the National System of Conservation Units of Nature (SNUC), establishing criteria and norms for the creation, implementation, and management of conservation units. Considering the proximity of the Agroflorestal Novo Horizonte project to the Renascer Extractive Reserve (RESEX), this legislation strengthens actions that can be developed to protect and, where appropriate, support productive chains of sociobiodiversity in these areas.

Federal Law No. 12,187, dated December 29, 2009: Institutes the National Policy on Climate Change, defining guidelines for the reduction of greenhouse gas emissions.

Federal Law No. 12,305, dated August 2, 2010: Institutes the National Solid Waste Policy, promoting the adoption of clean technologies to minimize environmental impacts. Any activity, whether productive or conservation-related, to be carried out within the project boundaries and associated with proposed activities, must consider waste reduction and proper management.

Forest Code (Law No. 12,651, dated May 25, 2012): Defines general norms on the protection of native vegetation, including the recovery of degraded areas. The Forest Code is essential for the implementation of conservation projects, as it offers clear guidelines for the protection and restoration of forests and can guide the conservation actions proposed by the Agroflorestal Novo Horizonte project.

Law No. 12,727, dated October 17, 2012: Establishes norms for the protection of vegetation, permanent preservation areas, and legal reserves. This law strengthens forest protection legislation, essential for the execution of projects focusing on the conservation and restoration of native forest areas.

Federal Law No. 14,119, dated January 13, 2021: Considers carbon sequestration an ecosystem service, incentivizing forest preservation for ecological balance control. This legislation directly supports the activities planned within the Agroflorestal Novo Horizonte project, as it recognizes the benefits of stored carbon as a climate change mitigation strategy.

Municipal Law No. 120, dated December 22, 2021: Provides for environmental fees and the exercise of environmental policing powers in the municipality of Prainha. Similarly to the federal law that defines sanctions for environmental crimes, this municipal law facilitates local implementation of conservation projects, ensuring the necessary environmental regulation and oversight for forest conservation.

Decree No. 11,548, dated June 5, 2023: Institutes the National Commission for the Reduction of Greenhouse Gas Emissions from Deforestation and Forest Degradation - REDD+. Establishes the institutional and regulatory framework for the implementation of REDD+ projects, creating a favorable environment for the development of these initiatives.

Federal Constitution (Article 225): Establishes everyone's right to an ecologically balanced environment and imposes on the government and the community the duty to defend and preserve it. It provides the constitutional basis for environmental protection in Brazil, essential for the legitimacy and legal support of conservation projects such as Agroflorestal Novo Horizonte.

State Law No. 8.633/2018: Establishes, within the scope of this agency, the technical guidelines for the preparation of Sustainable Forest Management Plans (SFMP) mentioned in Article 19 of Law No. 4,771, of September 15, 1965.

State Normative Instruction No. 05/2011: Regulates Forest Management in the State of Pará and other provisions.

State Normative Instruction No. 37/2010-02/2023: Establishes the technical parameters and procedures for the automatic analysis of the Rural Environmental Registry (CAR).

State Law No. 5.887/1995: Provides for the State Environmental Policy and other provisions.

Conventions of the International Labour Organization (ILO): The Agroflorestal Novo Horizonte's REDD+ AUD project not only aims at forest conservation and climate change mitigation but also has a significant impact on local communities and workers' rights. In this context, several conventions of the International Labour Organization (ILO) are highly relevant to the project, as one of their premises is to ensure that activities are implemented ethically and fairly, respecting human rights and promoting decent working conditions. The main applicable ILO conventions are:

- Convention No. 2 - Abolition of Forced Labour, 1930
- Convention No. 87 - Freedom of Association and Protection of the Right to Organise, 1948
- Convention No. 97 - Migration for Employment Convention, 1949
- Convention No. 98 - Right to Organise and Collective Bargaining Convention, 1949
- Convention No. 100 - Equal Remuneration Convention, 1951
- Convention No. 105 - Abolition of Forced Labour Convention, 1957
- Convention No. 111 - Discrimination (Employment and Occupation) Convention, 1958
- Convention No. 131 - Minimum Wage Fixing Convention, 1970
- Convention No. 138 - Minimum Age Convention, 1973
- Convention No. 142 - Human Resources Development Convention, 1975
- Convention No. 155 - Occupational Safety and Health Convention, 1981
- Convention No. 169 - Indigenous and Tribal Peoples Convention, 1989
- Convention No. 182 - Worst Forms of Child Labour Convention, 1999

United Nations Framework Convention on Climate Change (UNFCCC): The Paris Agreement, ratified by Brazil, is a milestone in the global effort to combat climate change. It aims to limit global warming to well below 2°C above pre-industrial levels, preferably to 1.5°C. Reduced-impact forest management practices align with the objectives of the UNFCCC. These practices include forest conservation, restoration of degraded areas, and sustainable management, which help minimize greenhouse gas emissions without altering land use. Thus, Agroflorestal Novo Horizonte's REDD+ project directly contributes to Brazil's climate commitments by promoting emissions reduction through forest conservation and project actions.

Convention on Biological Diversity (CBD): The Convention on Biological Diversity, adopted during the United Nations Conference on Environment and Development (Rio 92), aims at conserving biological diversity, the sustainable use of its components, and the fair and equitable sharing of benefits arising from the use of genetic resources. Although policies resulting from the CBD do not directly apply to business activities, REDD+ projects indirectly promote the objectives of the CBD. By conserving and restoring forests, these projects help protect biodiversity, ensuring habitats for numerous species, and contributing to the maintenance of ecosystem services.

1.16 Double Counting and Participation under Other GHG Programs

This project has not been registered and is not seeking registration under any other GHG Programs.

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

If yes, provide all required evidence of no double claiming as outlined by the VCS Standard.

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

If yes, provide all required evidence of no double claiming as outlined by the VCS Standard.

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."

☐ Yes ☒ No

1.18 Sustainable Development Contributions

1.18.1 Sustainable Development Contributions Activity Description

The Agroflorestal Novo Horizonte REDD+ Project is an initiative designed to mitigate greenhouse gas (GHG) emissions from deforestation and forest degradation in the Brazilian Amazon. The project adopts sustainable practices and integrates local communities and workers, promoting socioeconomic development and environmental conservation.

1. **Education and Capacity Building (SDG 4 - Quality Education):**
    - **Training and Capacity Building:** Provides environmental and technical training to workers and local communities, focusing on sustainable practices, human rights, and gender equality.
    - **Support to Local Schools:** Supplies educational materials, equipment, and internet access to local schools, enhancing education quality and integrating sustainability into the curriculum.
    - **Capacity Development:** Conducts ongoing education sessions to raise awareness about sustainable development, climate change, and environmental conservation.

2. **Decent Work and Economic Growth (SDG 8):**
    - **Income Generation:** Stimulates the local economy by purchasing inputs and services from local communities, creating job opportunities, and reducing poverty.
    - **Local Employment:** Promotes social inclusion and ensures safe and dignified working conditions.
    - **Development of Local Production Chains:** Supports associations, cooperatives, and rural technical assistance, strengthening sustainable production practices.
3. **Climate Action (SDG 13):**
    - **Education and Awareness:** Conducts awareness-raising campaigns on climate change mitigation and adaptation.
    - **Institutional Capacity Building:** Strengthens local capacities to manage climate impacts through vulnerability mapping and the implementation of adaptation strategies.
4. **Life on Land (SDG 15):**
    - **Deforestation Reduction:** Encourages sustainable productive practices and continuously monitors forest integrity to prevent deforestation.
    - **Ecosystem Restoration:** Maps and implements reforestation and restoration projects in degraded areas.
    - **Biodiversity Protection:** Conducts educational actions and continuous monitoring to protect threatened species and their natural habitats.

Explanation of How Project Activities Result in Expected SD Contributions:

The project's activities are designed to generate significant impacts on the SDGs in various ways:

- **Education and Capacity Building:** Improve educational outcomes and environmental awareness, preparing communities to adopt sustainable practices.
- **Income Generation and Employment:** Create economic opportunities and improve working conditions, boosting local economic development.
- **Climate Action:** Campaigns and institutional capacity building increase local understanding and capacity to address climate challenges.
- ***Environmental* Conservation:** Practices to reduce deforestation, restore ecosystems, and protect biodiversity help conserve natural resources and local biodiversity.

1.18.2 Sustainable Development Contributions Activity Monitoring

During the monitoring period, Agroflorestal Novo Horizonte's REDD+ project carried out various impactful initiatives in local communities. Key activities included training programs, renovations of local schools with the provision of educational materials and equipment, and the adoption of sustainable production practices. Improving roads was also a crucial action, providing communities with better access to the city and ensuring that children can safely reach their schools.

These actions are directly connected to Sustainable Development Goals (SDGs) 4, 8, 13, and 15, which focus on quality education, economic growth, climate resilience, and environmental conservation. By addressing issues such as educational inequality, local job creation, and forest preservation, the project aligns with Brazil's national sustainability goals. To ensure the effectiveness of these initiatives, monitoring and reporting mechanisms are in place, with documented evidence available.

Below are the results of the actions established during the monitoring period.

Table 11. Sustainable development contributions

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
1)	4 QUALITY EDUCATION 4.3 4.5 4.7	**4.3** By 2030, ensure equal access for all women and men to affordable and quality technical, vocational and tertiary education, including university. **Indicator 4.3.1:** Participation rate of youth and adults in formal and non-formal education and training. **4.5** By 2030, eliminate gender disparities in education and ensure equal access to all levels of education and vocational training for the vulnerable, including persons with disabilities, indigenous peoples and children in vulnerable situations. **Indicator 4.5.1:** Parity indices (female/male, rural/urban, and	*Activities Implemented:* *Conducted environmental and technical training for workers and communities.* *Supported local schools with educational materials and internet access.* *Provided continuous education on human rights, gender equality,*	Number of training sessions conducted: 7 Number of participants in the training (by gender, age, and social group): A total of 366 participants, including 51 children and 49 adults from the community (23 women and 27 men), along with 268 workers. Number of schools supported: 3 Quantity of educational materials and equipment provided: 6	Agroflorestal Novo Horizonte project have provided significant benefits to the local population by offering access to seven training sessions. These sessions reached a total of 366 participants, including 51 children, 49 adults (23 women and 27 men) from the community, and 268 workers. Additionally, three schools have been supported, with two of them gaining internet access, and six educational materials and equipment have been provided. Through partnerships with local educational institutions, these initiatives have contributed to professional qualification and environmental education, enhancing both skill development

		bottom/top wealth quintile) for all education indicators. **4.7** By 2030, ensure that all learners acquire the knowledge and skills needed to promote sustainable development, including, among others, through education for sustainable development and sustainable lifestyles, human rights, gender equality, promotion of a culture of peace and non-violence, global citizenship and appreciation of cultural diversity and of culture's contribution to sustainable development. **Indicator 4.7.1:** Extent to which (i) global citizenship education and (ii) education for sustainable development are mainstreamed into national education policies, curricula, teacher education, and student assessment.	*and sustainable practices.*	Number of students with internet access: 2 schools	and awareness. As the project progresses, these benefits are expected to expand even further, reaching more people and strengthening community engagement.

2)	8.8.	**8.8** Protect labour rights and promote safe and secure working environments for all workers, including migrant workers, in particular women migrants, and those in precarious employment **Indicator 8.8.1:** Frequency rates of fatal and non-fatal occupational injuries, by sex and migrant status.	*Stimulated local economy through the purchase of local inputs and services.* *Created job opportunities and promoted social inclusion.* *Ensured safe and dignified working conditions.* *Supported local production chains and provided rural technical assistance.*	Total Expenditure on Local Inputs and Services (2021 – 2024): $226,942.27 USD Number of Jobs Created: 274 Number of Supported Associations and Cooperatives: 3 Amount of Agricultural Technical Assistance Provided: [to be specified]	Between 2021 and 2024, the Agroflorestal Novo Horizonte project invested 226 thousand dollars in social actions and opportunity development. Throughout the project's development, the implementation of these actions will be expanded so that all associations, cooperatives, and producers have access to technical assistance, and all mapped cooperatives will benefit.
3)	13.3	**13.3** Improve education, awareness-raising and human and institutional capacity on climate change mitigation, adaptation, impact reduction and early warning	*Conducted awareness-raising campaigns on climate change.* *Strengthened local capacities*	Number of awareness campaigns conducted: 5 Number of participants in the campaigns: 183	Annual awareness campaigns will be conducted in the communities and within the RESEX Renascer to reduce emissions from deforestation and degradation in the local communities within the leakage belt and surrounding

			for climate change mitigation and adaptation.	Number of local institutions trained: 3 Amount of greenhouse gas emissions avoided or removed: 486892 tCO_2e	areas. The aim is to maintain 100% forest cover within the project area.
4)	15.1 15.2 15.5	**15.1** By 2020, ensure the conservation, restoration and sustainable use of terrestrial and inland freshwater ecosystems and their services, in particular forests, wetlands, mountains and drylands, in line with obligations under international agreement. **Indicator 15.1.1:** Forest area as a proportion of total land area. **15.2** By 2020, promote the implementation of sustainable management of all types of forests, halt deforestation, restore degraded forests and substantially increase	*Encouraged sustainable productive practices and forest conservation.* *Mapped and implemented reforestation and restoration projects.* *Protected threatened species and their habitats.*	Conserved forest area (hectares): 23,092.90 Number of monitored and protected species: 120 species of fauna, including birds, reptiles, and small, medium, and large mammals. Flora: 108 tree species Number of sustainable practices implemented: 1 – Reduced Impact Logging Number of reforestation and restoration projects	Throughout the project duration, forest cover in the PA will be maintained at 100%, ensuring the protection of wildlife species. Periodic wildlife surveys will be conducted, and strategies for the preservation of endemic species will be implemented over the course of the project. Communities with areas designated for restoration will receive technical assistance for agroforestry planting, focusing on both income generation and forest preservation.

		afforestation and reforestation globally. **Indicator 15.2.1:** Progress towards sustainable forest management.		completed: to be implemented	
		15.5 Take urgent and significant action to reduce the degradation of natural habitats, halt the loss of biodiversity and, by 2020, protect and prevent the extinction of threatened species. **Indicator 15.3.1:** Proportion of land that is degraded over total land area.			

1.19 Additional Information Relevant to the Project

Leakage Management

During the monitoring period, the Agroflorestal Novo Horizonte REDD+ Project implemented a comprehensive leakage management plan, as outlined by Verra requirements and aligned with the Climate, Community, and Biodiversity (CCB) Standards. This plan includes identifying leakage management zones and implementing mitigation measures to reduce the risk of shifting deforestation activities outside the project area.

- **Leakage Management Zones:** Zones surrounding the project area were identified based on mobility and vulnerability analyses of nearby non-forest areas. These zones help minimize the risk of deforestation activities being displaced outside the project area. The CCB methodology was applied to analyze the social, environmental, and economic conditions of communities affected by the project, ensuring that the leakage management zones are effective and beneficial in the long term.

- **Sustainable Practices:** The project promotes sustainable practices such as sustainable cattle ranching and low-impact cultivation of crops (citrus, cocoa, and coffee) within the leakage management zones. These practices are carried out in non-forest land areas and aim to reduce pressure on forested areas and improve productivity per area. The CCB methodology ensures that these practices not only minimize leakage risk but also provide additional social and environmental benefits to local communities.

- **Mapping and Analysis:** A buffer was established around the project area to identify vulnerable areas. The leakage management plan is periodically adjusted and verified based on mobility analyses and monitoring results. The CCB methodology contributes to the ongoing evaluation and documentation of the project's social and environmental benefits.

The leakage management plan and the mapping of leakage management zones are included in Section 3.3 of the project documents. The effectiveness of the implemented measures will be continuously monitored and adjusted as needed to ensure that project goals are achieved and leakage risks are minimized. The CCB methodology will be used to ensure that the project’s social and environmental benefits are effectively documented, and that project performance is continuously assessed.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the Project Description. Through the entire section 1 of the Project Description document, all relevant and necessary information regarding the project proponent and the conditions prior to the project implementation has been made available. The same affirmation is valid for the entire PD.

Further Information

No further information to disclose. Include any additional relevant legislative, technical, economic, sectoral, social, environmental, geographic, site-specific and/or temporal information that may have a bearing on the eligibility of the project, the GHG emission reductions or carbon dioxide removals, or the quantification of the project's reductions or removals

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Table 12. Stakeholder Identification

Stakeholder Identification	To identify the project stakeholders, the proponent conducted socio-environmental assessments since 2021. Consequently, beyond the employees of Agroflorestal Novo Horizonte, the residents of a rural settlement were also considered stakeholders. These local communities are divided into three main villages: Vila Nova, Santa Terezinha, and Ramal do Meio. These residents are small non-traditional farmers and do not use the project area for their subsistence. For the REDD+ project, a new consultation with local communities was conducted, involving a deeper investigation of previously obtained information. Employees were also consulted and received specific training on climate change and the REDD+ project.

	To the north of the project area is the RESEX Renascer, where traditional riverside communities reside. To evaluate potential customary use overlaps and identify socio-environmental program opportunities, the Agroflorestal Novo Horizonte team met with ICMBio conservation unit managers. The objectives of RESEX Renascer include the maintenance and conservation of forest areas, commercial forest management, agriculture as a supplementary income activity for communities, conservation of water bodies and their fauna, fishing activities as a source of income, and protection of the local community's cultural expressions. It was found that RESEX inhabitants are located far from the project areas and do not enter the project area for resource use. Therefore, the implementation of socio-environmental actions in RESEX is not a priority for the Agroflorestal Novo Horizonte REDD+ project. However, future socio-environmental actions may be proposed within the programmatic axes of the conservation unit management plan, in alignment with managers and community leaders. Additionally, a comprehensive consultation was conducted with the local government and neighboring landowners.
Legal or customary tenure/access rights	There is no overlap of land or customary use by local communities in the project area. There are no indigenous or quilombola populations near the project. Although a small portion of RESEX Renascer territory borders the project area, the villages of traditional communities are distant and do not use the area for resource extraction or cultural practices.
Stakeholder diversity and changes over time	<u>Local Villages</u> The residents and landowners of the local communities are small rural farmers who rely on agriculture and agroforestry systems as their primary source of income. These farmers are familiar with the Novo Horizonte Agroforestry Project area and do not use any

forest resources within it. Therefore, the residents are not directly dependent on the project or the resources within the project area. However, due to their proximity, they will be considered stakeholders and will receive benefits from the emissions reductions. Similarly, these communities are located within the leakage zone, and project activities will be conducted to prevent potential leakages. The interviews show that since the start of the project, there have been no significant changes in the local community in terms of migration.

Other Farmers

There are other large private properties neighboring the project area. These properties are used for soybean cultivation and/or sustainable forest management. The boundaries between these properties are marked by rivers or firebreaks to mitigate any fires that could impact the project area. Additionally, this project will encourage owners of large neighboring areas to develop REDD projects, thereby creating a larger protected area.

Resex Renascer traditional communities

The Resex Renascer currently hosts more than 600 families in 16 tradition riparian communities organized into three sectors distributed along three main rivers: Uruará, Tamuataí, and Guajará. These families primarily rely on agriculture, small-scale animal husbandry, and fishing for income and food security.

The community organization of Resex Renascer is represented by the Association of Residents of the Communities of Resex Renascer — GUATAMURU, which takes its name from the initials of the Guajará, Tamuataí, and Uruará rivers, where the communities are distributed. The association is managed by a board composed of three coordinators, one from each river, ensuring the representativeness of the communities throughout the entire reserve.

Agriculture is a significant activity for the Resex Renascer communities and is guided by traditional low-impact environmental practices, utilizing fallow areas in a rotational planting system to maintain soil fertility over time. This activity meets both consumption and trade demands and is notable for the production of cassava, beans, corn, yam, rice, pumpkin, fruits, and vegetables. Along with small animal husbandry, it contributes to nutrition and income generation.

The waters in Resex Renascer, found in rivers, lakes, streams, and other floodable surfaces, constitute approximately 15% of the UC territory and are crucial for the 16 communities living along the banks of the Guajará, Uruará, and Tamuataí rivers. These waters serve as a source of consumption, food, and transportation. A central element in Resex culture, water plays a significant role in community life and the identity of the residents, who experience a dynamic closely tied to the seasonal flood and drought cycles.

The forests are a primary source of sustenance for the families and are used for craftwork (such as bark, vines, and leaves), alternative medicine (including cumaru, resins, oils, honey, amapá, sucuuba, verônica, saratudo, cat's claw, envirataia, purple lapacho, ananim, and others), as well as non-timber forest products (such as bacaba, açaí, Brazil nuts, taperebá, patauá, piquiá, uxi, buriti, and others), and food supplementation.

Although in the ecological zoning of the RESEX, the community use area has a small zone bordering the project area, ICMBio managers have reported that the communities are aware of and do not exceed the conservation unit's boundaries. The use areas are characterized by extractive activities without the clearing of fields.

<u>Agroflorestal Novo Horizonte workers</u>

The workers at Agroflorestal Novo Horizonte are key stakeholders who, while not directly dependent on the project, are crucial to its success. Their training in fire brigades and sustainable forest

	management techniques plays a significant role in reducing greenhouse gas emissions. The project implements technical development activities specifically for this group to enhance their skills and effectiveness. It is important to note that most of these workers do not live locally and do not rely on the forest for subsistence. They do not engage in hunting or extraction activities, which further emphasizes their role as contributors to the project's sustainability goals rather than users of forest resources.
Expected changes in well-being	Expansion of Pasture Areas: The growth of pastureland for cattle ranching results in significant deforestation, reducing the availability of vital forest resources for local communities and biodiversity. Illegal Logging: The illegal extraction of high-value timber disrupts local ecosystems and depletes resources that communities rely on for their livelihoods. Frequent Fires: Forest fires and intentional burnings contribute to environmental degradation, leading to poorer air quality and increased health risks for residents. Destruction of Water Springs: Clearing areas around water springs for pastureland reduces water availability, impacting both household consumption and agricultural needs. Stream Droughts: Deforestation and fires decrease water flow in streams, which affects water availability for domestic use and agriculture.
Location of stakeholders	1. Traditional Communities: Location: The traditional communities are situated at a considerable distance from the project area within Resex Renascer.

	Rights and Use: These communities do not engage in customary use of the project area and do not have traditional rights related to this area. Potential Impact: Although they are not directly involved with the project area, the project may have positive indirect impacts on the Resex Renascer. Enhanced monitoring and protection activities may help mitigate deforestation activities near the project area and the Resex, thereby indirectly benefiting these traditional communities and biodiversity by preserving surrounding forests. 2. Local Communities: Location: The local communities are situated within close proximity to the project area. Rights and Use: These communities do not possess traditional rights of use related to the project area and do not utilize resources from this area. Potential Impact: The project is expected to primarily benefit these local communities due to their proximity. They are closely related to areas of potential leakage, making them a key focus for the project's benefits. The project's benefits include improved environmental conditions and possible economic opportunities stemming from enhanced conservation efforts. 3. Private Neighboring Properties: Location: The neighboring private properties are adjacent to the project area. Rights and Use: These properties do not use the project area and do not hold rights related to the project area. Potential Impact: While these properties do not directly interact with the project area, the project's conservation efforts may indirectly influence them by improving surrounding environmental conditions. The reduction in deforestation and

	forest degradation in the project area can help stabilize the ecological balance in adjacent lands. 4. Areas Outside the Project Area: Impact Prediction: The project is expected to have an indirect positive impact on areas outside the project area. Through enhanced monitoring and protection, the project aims to mitigate increased deforestation pressures and reduce leakage impacts. This broader impact supports overall regional conservation efforts, benefiting both local and traditional communities by preserving ecological integrity and promoting sustainable practices.
Location of resources	1. Traditional Communities: Location of Territories and Resources: Traditional communities are situated at a considerable distance from the project area, within the Resex Renascer, a Public Conservation Unit established by the federal government in 2009. Their territory, delineated by extensive studies and documented in the Resex Renascer Management Plan[8], encompasses all the resources utilized by these communities. This plan outlines the boundaries and resource allocation within the Resex Renascer, ensuring that the needs and customary rights of the traditional communities are recognized and managed appropriately. Customary Access: As they do not engage in customary use of the project area, there are no specific territories or resources related to the project area that these traditional communities' access. 2. Local Communities: Location of Territories and Resources: Local communities are situated close to the project area and are associated with the surrounding agricultural settlement. They do not hold traditional

[8] https://www.gov.br/icmbio/pt-br/assuntos/biodiversidade/unidade-de-conservacao/unidades-de-biomas/amazonia/lista-de-ucs/resex-renascer/arquivos/plano_de_manejo___resex_renascer___versao_final.pdf

	rights related to the project area but are involved in the surrounding environment. Customary Access: These communities do not utilize resources directly from the project area, but they may access resources from nearby lands that are connected to the project area. Their key activities include agriculture, small-scale animal husbandry, and fishing, which rely on the surrounding natural environment. 3. Neighboring Private Properties: Location of Territories and Resources: The neighboring private properties are adjacent to the project area. These properties are involved in activities such as soybean cultivation and sustainable forest management. Customary Access: These properties do not use the project area nor have rights related to it. Their access to resources is limited to their private lands, which are managed separately from the project area.

2.1.2 Stakeholder Consultation and Ongoing Communication

Table 13. Stakeholder consultation

Date of stakeholder consultation	01-October-2021
Stakeholder engagement process	Agroflorestal Novo Horizonte already has a good relationship with the communities in the settlement along the access route to its area. For engagement and consultation, the company's team visited the residents, explained the project, and conducted the socio-environmental assessment. The information collected during these visits forms the basis for the implementation of socio-environmental projects and programs in the community. The invitation below (Figure 21) was delivered for each community: Santa Teresinha, Vila Nova and Bom Jesus.

Figure 22. Agroflorestal Novo Horizonte invitation for Santa Terezinha Community.

Informative folders were distributed, detailing the purpose and key features of the Agroflorestal Novo Horizonte project.

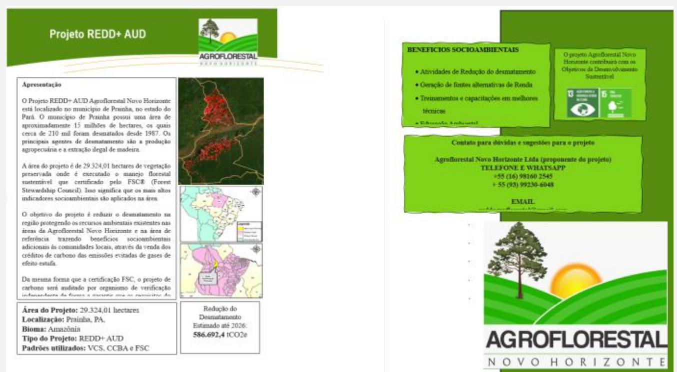


Figure 23. Folders about Agroflorestal Novo Horizonte project

	Figure 24. Local community members during the periodic consultation. Meetings were also held with the workers of Agroflorestal Novo Horizonte to introduce and train them on the project, as well as to gather information on communication methods. Additionally, consultations were conducted with NGOs, public agencies, and conservation unit managers. These interactions were aimed at ensuring that all relevant parties are informed about the project and to establish effective communication channels with stakeholders.
Consultation outcome	Since the REDD+ AUD Agroflorestal Novo Horizonte project does not overlap with areas used by traditional communities, indigenous populations, or local populations, it was not necessary to conduct a FPIC process, nor to discuss the design and implementation of the REDD+ project. The employees working in forest management activities participated in a large plenary session where training on climate change and conservation projects was presented. The company regularly conducts training on fire brigades, health and safety issues, and operational techniques to reduce environmental impacts. Local residents were invited to participate in a meeting where the project was presented in detail. During this meeting, the communities had the opportunity to clarify their doubts and provide valuable information, which was incorporated into the development of the project's socio-environmental impact activities, as previously described. In addition to these consultations, a socio-environmental diagnosis was conducted with the residents. To ensure ongoing and responsive management, the communities are periodically monitored through annual meetings. These meetings aim to assess any complaints, suggestions, or additional information

	that can be used to improve the project's activities and ensure that socio-environmental impacts are managed effectively. The labor laws outlined in item 1.15 are fully complied with by the company. Health and safety documents (PGRTR) are updated annually, and employees have their rights to rest and remuneration ensured. No risks or negative impacts of the project on stakeholders have been identified. On the contrary, the project has the potential to benefit the socio-economic and environmental development of the territory much more broadly than the company can achieve through the social programs of certified forest management.
Ongoing communication	During the initial consultation phase, a communication channel with local communities was established and validated. Following this initial setup, periodic assessments are conducted to maintain active and updated communication. WhatsApp has been identified as the primary means of communication with these communities, facilitating direct and immediate contact. Additionally, the project area manager maintains a regular presence within the communities to further strengthen this relationship. This communication process is reinforced through ongoing assessments, where information about the project's responsible parties is presented. This practice ensures transparency and provides stakeholders with direct access to relevant information. For workers, daily dialogues are conducted, and a suggestion box, along with WhatsApp, is used to ensure that their concerns and suggestions are effectively received and addressed. These mechanisms are designed to guarantee a continuous flow of information and opportunities for ongoing feedback. The communication strategy includes several key components. For the communities, dedicated WhatsApp groups are created for real-time updates, and bi-monthly community meetings are scheduled to discuss project progress and address concerns.

	Monthly information bulletins are distributed through local leaders and community centers to keep everyone informed about project developments and upcoming events. For workers, daily briefings are held at the start of each work shift to share important information and updates. A suggestion box is installed at strategic locations on the project site, allowing workers to provide anonymous feedback. WhatsApp is also used to quickly disseminate information and address immediate concerns. Feedback collection and management involve distributing feedback forms during community meetings and through local representatives. For workers, regular surveys assess satisfaction and safety concerns, while periodic one-on-one meetings provide a space for personal feedback. In response to the collected feedback, action plans are developed and implemented to address specific concerns raised by both communities and workers. Follow-up meetings are organized to review the effectiveness of the actions taken and to gather additional input. Issues raised by workers are resolved through a clear process, with timely responses and updates. Safety improvements are promptly acted upon, with changes communicated to the workers. The communication strategy is continuously monitored and evaluated. Periodic impact assessments are conducted to gauge the effectiveness of the communication methods and make necessary adjustments. Annual reports are prepared and shared with stakeholders, detailing how their feedback has been addressed and the outcomes of the changes implemented
Stakeholder input	The consultations and communications conducted with stakeholders up to this point have highlighted the need for improvements in communication channels and identified priorities for project activities. As a result, actions have been

	prioritized to include collaboration with the local school and support for the maintenance of local community infrastructure. Additionally, technical training in forest management has been offered through field days, addressing one of the key demands from the local community. These efforts reflect a commitment to enhancing community engagement and addressing the specific needs and priorities identified through ongoing stakeholder interactions.
Consultation Attending list	The full evidence of stakeholder consultations will be presented during the Validation and Verification process.

To expand awareness about the project, the following information sheet was shared with various types of stakeholders:

Figure 25. Information sheet for the general public regarding the Agroflorestal Novo Horizonte project

2.1.3 Free Prior and Informed Consent

There are no traditional communities, indigenous territories, or quilombola territories that customarily use the areas of the project. An FPIC process is not necessary.

2.1.4 Grievance Redress Procedure

The Agroflorestal Novo Horizonte REDD AUD project places significant emphasis on the development and implementation of an effective grievance redress mechanism to address any conflicts or concerns raised by stakeholders. Recognizing the importance of maintaining transparent and open communication, the project has established a dedicated grievance channel to ensure that all stakeholders, including local communities and external parties, have a clear and accessible process for submitting complaints or suggestions.

Table 14. Grievance procedure development process

Development process	The grievance procedure for the Agroflorestal Novo Horizonte REDD+ Project was effectively communicated to both the local settlement and project workers through a meeting. These meetings aimed to ensure that all stakeholders were informed about the process for receiving, addressing, and

	resolving grievances in a manner that is transparent, culturally sensitive, and aligned with the project's commitment to fair practices. 1. Meeting Preparation Prior to the meetings, comprehensive materials were prepared, including: Grievance Procedure Document: Detailed information on the grievance process, including channels for submission, steps for investigation, and resolution mechanisms. Visual Aids: Diagrams and flowcharts illustrating the grievance process to enhance understanding. 2. Presentation to the Local Settlement The grievance procedure was presented during a community meeting held in the local settlement. The key components of the presentation included: Introduction of the Procedure: An overview of the grievance mechanism, including the purpose and importance of having a structured process for addressing community concerns. Explanation of Channels: Detailed descriptions of the various channels available for submitting grievances, such as grievance boxes, a dedicated hotline, and in-person submissions. Steps of the Process: Clear explanation of each step, from receiving and acknowledging grievances to investigating and resolving them. Emphasis was placed on the inclusion of local customs and conflict resolution practices in the process. Cultural Sensitivity: Discussion on how the grievance mechanism respects and incorporates culturally appropriate methods for conflict resolution, involving local leaders or respected community members if needed. 3. Presentation to Workers The grievance procedure was also presented to project workers during a separate meeting. The presentation included:

Procedure Overview: A summary of the grievance process tailored to workers' concerns, including how they can raise issues related to working conditions, safety, and other employment-related matters.

Submission Channels: Information on how workers can use grievance boxes, the hotline, or direct communication with project management to raise their concerns.

Resolution Process: Description of how grievances will be investigated, including the involvement of worker representatives if applicable, and how workers will receive feedback on the resolution.

Feedback Mechanism: Emphasis on the importance of worker feedback and how it will be used to improve the grievance mechanism.

4. Interactive Q&A Sessions

Both meetings included interactive Q&A sessions where community members and workers could ask questions, express concerns, and seek clarification on the grievance procedure. This interactive component was crucial in ensuring that all stakeholders understood the process and felt comfortable using it.

5. Follow-Up Actions

After the meetings, follow-up actions included:

Distribution of Materials: Providing copies of the grievance procedure document.

Feedback Collection: Gathering feedback from attendees on the clarity and effectiveness of the grievance procedure presentation to make any necessary improvements.

Ongoing Communication: Establishing channels for ongoing communication to address any further questions or concerns about the grievance mechanism.

Grievance redress procedure	1. Receiving Grievances The project has set up multiple channels for community members to submit their grievances. These channels include: Grievance Boxes: Strategically placed in accessible locations within the community to allow anonymous submissions. Dedicated Grievance Hotline: A telephone number where community members can call to voice their concerns directly. In-Person Submissions: Community members can submit grievances during regular project meetings or through designated project staff. 2. Hearing Grievances Upon receiving a grievance, the project follows a structured approach to ensure it is heard and documented properly: Acknowledgement: Each grievance is formally acknowledged within five working days of receipt. A receipt or confirmation is provided to the complainant, detailing the next steps and expected timelines. Initial Assessment: An initial assessment is conducted to determine the nature and urgency of the grievance. This assessment considers the cultural context and ensures that the grievance is assigned to the appropriate team for further action. 3. Responding to Grievances Investigation: The grievance is investigated by the Agroflorestal Novo Horizonte team. The investigation process includes gathering relevant information, interviewing involved parties, and reviewing documentation. Resolution Meeting: A resolution meeting is held with the complainant and relevant stakeholders to discuss findings and possible solutions. This meeting is conducted in a manner that respects local customs and conflict resolution practices. Response Documentation: A formal response is documented and shared with the complainant, detailing the findings of the investigation and the proposed

resolution. This response is provided within 30 days of the grievance being received.

4. Attempting to Resolve Grievances

Efforts are made to resolve grievances in a manner that is acceptable to all parties involved:

Negotiation and Mediation: The project uses culturally sensitive conflict resolution methods, such as negotiation and mediation, to facilitate agreements. Local leaders, respected community members, and NGOs may be involved to support and guide the discussions towards a consensus. If these methods do not lead to a resolution, the grievance may be escalated to legal proceedings for court adjudication.

Action Plan: An action plan is developed to address the grievance and prevent future occurrences. This plan outlines specific actions to be taken, timelines, and responsibilities.

5. Monitoring and Follow-Up

To ensure the effectiveness of the grievance mechanism, the project includes:

Follow-Up Meetings: Regular follow-up meetings are scheduled with the complainant to review the implementation of the action plan and ensure that the grievance has been satisfactorily resolved.

Feedback Mechanism: A feedback mechanism is in place to gather input from the complainant on the resolution process and overall satisfaction.

6. Reporting and Transparency

The project maintains transparency in its grievance handling process by:

Public Reporting: Summarized reports of grievance trends and resolutions are made available to the community through public meetings or project newsletters.

Review and Improvement: The grievance mechanism is regularly reviewed and updated based on feedback and lessons learned to improve its effectiveness and responsiveness.

Figure 26. Grievance Procedure

Table 15. Grievances received reports

Grievances received	Resolution and outcome
NA	No grievances have been raised since the beginning of the project. The channels for addressing grievances and communication continue to be actively managed by Agroflorestal Novo Horizonte, as reflected in the project reports. A folder detailing the communication channels has been distributed to the communities and other stakeholders to ensure ongoing awareness and access.

2.1.5 Public Comments

Verra invites public comments on the compliance of projects with its program requirements, which are made available for public viewing on the Verra Registry. Projects currently open for comment can be accessed at https://verra.org/consultations/projects-open-for-public-comment/.

In addition to Verra's public comment period, the Agroflorestal Novo Horizonte Project has established internal channels to gather comments, feedback, and suggestions from stakeholders. For general inquiries or feedback, stakeholders can directly contact the Carbon department via email at reddagroflorestal@gmail.com. Compliance-related concerns can be addressed through a dedicated channel, available by phone at +55 93 99225 8414.

This multi-channel communication approach ensures that stakeholders have various means to engage with the project. All feedback received will be carefully reviewed, and relevant points will be incorporated into the Project Description, verification reports, or other appropriate communication formats. Agroflorestal Novo Horizonte is committed to keeping stakeholders informed about project developments and making necessary adjustments. By integrating internal feedback mechanisms with the Verra public comment process, we emphasize our commitment to transparency, accountability, and the successful implementation of the project.

Table 16. Public Comments received

Comments received	Actions taken
Periodic consultations with the communities have contributed to the development of actions implemented over the years. The main feedback received up until 2024 highlighted the need to focus on the following areas: • Regular road maintenance • Improvements in education and infrastructure • Technical assistance for productive activities • Employment and training opportunities	The actions taken to address these demands were incorporated into the project description and have been implemented from the outset. With the revenue from carbon credits, the goal is to expand these actions, always in dialogue with the community, ensuring an adaptive management approach to meet the evolving needs of these stakeholders.

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

Guilherme Berwerth Stucchi, Project Consultant - Forest Engineer from University of São Paulo – Superior School of Agriculture "Luiz de Queiroz". Has done his Msc.Forest Science Program: Forest Resources with Silviculture and Forest Management from University of São Paulo – Superior School of Agriculture

"Luiz de Queiroz". Mr. Guilherme has more than 15 years of experience in the field of Forest management activities, silviculture and forest carbon inventory. He has more than 15 years of Auditing experience in VCS REDD projects and FSC Forest certification audits. He is a qualified lead assessor in Environmental Management Systems, FS-COC and forest management certification. He is also a qualified auditor from Rainforest alliance standard for natural forest management (business and community management). He has been actively involved in validation/verification of forest carbon projects (REDD) and verification of legality since 2011. Participated in more than 80 forestry, chain of custody and carbon audits in Brazil. He is also involved in providing advisory activities for forest companies in the Amazon on issues related to forest management and suitability for FSC certification.

Alessandra Daniele de Sousa Brandão SIG Specialist Project Consultant Agronomist Engineer graduated from the Federal Rural University of the Amazon (2011), specializing in Georeferencing of Rural Properties at the Federal Institute of Pará - IFPA and Applied Geoprocessing at Faculdade Ideal (2016). She holds a Master's degree in Agronomy, specializing in Agricultural Entomology/Geoprocessing (2014), and a Ph.D. in Agronomy, focusing on Plant Production in Agricultural Systems from the Federal Rural University of the Amazon (2018). Her work primarily involves the following areas: Geotechnologies and applied geoprocessing. She has experience in Agronomy, with an emphasis on plant health protection, geostatistics, Global Positioning System (GPS), Geographic Information System (GIS), geoprocessing, remote sensing, and precision agriculture. Since 2014, she has worked at CKBV Florestal LTDA as a Geoprocessing Analyst and Environmental Coordinator.

Kaue Leopoldo Ferraz- Project Consultant - Biologist graduated from the Federal University of São Carlos (UFSCar). He has experience in ecotoxicological analysis and behavioral ecology of aquatic organisms, with a published scientific article (https://doi.org/10.1007/s00244-024-01052-2). He worked for five years in biodiversity conservation through the monitoring of benthic communities, with expertise in ecological data analysis, focusing on limnological analyses and taxonomy. He has experience in formal and non-formal education, as well as environmental education, having contributed as an author to environmental education and scientific dissemination books for high school and elementary education (ISBN 978-65-5889-275-5 and ISBN 978-65-5889-276-2). Currently, he works in the field of socio-environmental technical consulting, developing and applying socio-environmental protocols and certifications, as well as consulting for the development of REDD+ projects.

Rafael Nejar Badu Mahfud de Oliveira Project Consultant**:** Biologist graduated from the Federal University of Alfenas - Unifal. Experience in wildlife studies with an emphasis on herpetofauna. Involved in wildlife displacement and rescue activities in the MRN - Mineração Rio do Norte project. Biologist responsible for field data collection in the Belo Monte Hydroelectric Power Plant project. Biologist responsible for field data collection for the herpetofauna group in the Faunal Inventory of the Agroflorestal Novo Horizonte project.

Pedro Henrique Dos Santos Angeliti: Project Consultant Biologist with a Bachelor's degree in Biological Sciences from the Barão de Mauá University Center. With three years of experience, I have worked on several conservation projects such as the Pró-Muriqui Institute and the Pró-Carnivores Institute, focusing on the conservation of Southern Muriquis and the Pantanal Jaguars, respectively. I also spent my entire undergraduate program at the Ribeirão Preto Forest and Zoo, assisting in the care of injured animals that arrived there.

Matheus William Moschegni Baia: Project Consultant Biologist graduated from the Barão de Mauá University Center and postgraduate in environmental education. He has been working in the wildlife field for nine years, with experience in caring for orphaned animals and rehabilitating birds, as well as wildlife studies. He has worked as an animal caretaker at the Fábio Barreto Municipal Forest. Environmental consultant, working in wildlife rescue, displacement, inventory, and monitoring with an emphasis on ornithofauna. Biologist responsible for field data collection for the ornithofauna and mammal groups in the Faunal Inventory of the Agroflorestal Novo Horizonte project.

Eloá Fernanda Santos: Project Consultant Biologist graduated from the University of São Paulo (Ribeirão Preto School of Philosophy, Sciences, and Letters - FFCLRP-USP) with both a Bachelor's degree and a Teaching degree. She has experience in the fields of Education and Zoology, having spent two years conducting scientific research in the Laboratory of Bioecology and Systematics of Crustaceans (LBSC) and two years working as an intern in Environmental Education and the care of orphaned and wild animals at the Fábio Barreto Municipal Zoo in Ribeirão Preto. She is currently conducting field data collection on herpetofauna for the Faunal Inventory of the Agroflorestal Novo Horizonte project.

Erivelton Vinhote da Silva: Civil Construction and Building Technician. He has worked at concrete plants for MRN - Mineração Rio do Norte and at the concrete plant for Alcoa in Juruti, PA. He is currently working on the Agroflorestal Novo Horizonte project as an Operational Supervisor, also participating in the social aspects of FSC certifications.

Diego Monteiro Imbiriba: Graduated in Information Technology from Paulista University (UNIP). Currently works as an IT technician for the Agroflorestal Novo Horizonte project, also participating in the social and administrative aspects of FSC certifications and providing operational support for wildlife activities.

Mário Bruno Bizon: Forest Engineer graduated from the University of Contestado (UNC) in Santa Catarina, with a specialization in IFT. He has experience in forest consulting, forest management, suppression, environmental licensing, and the creation of environmental maps.

Yohana Cunha de Mello: Forest Engineer with a Master's in progress in Conservation of Biodiversity and Sustainable Development (ESCAS/IPÊ), specializing in geoprocessing and carbon project development. Over the past 5 years, has led carbon projects in the AFOLU sector, ensuring compliance with VCS and

CCB standards across all stages, from design to verification. Experienced in driving initiatives that contribute to climate change mitigation and biodiversity conservation

2.2.2 Risk Assessment

All activities carried out by Agroflorestal Novo Horizonte are executed following a thorough risk analysis, ensuring that appropriate mitigation measures have been planned and are ready for implementation. To achieve this, compliance with local, state, and national laws is strictly adhered to, and a comprehensive assessment of risks, impacts, and mitigation strategies is conducted.

Table 17. Risk Assessment

	Risks identified	Mitigation or preventative measure taken
Natural and human-induced risks to stakeholders' wellbeing	No risk identified	Given that the project's activities are designed to enhance forest conservation and biodiversity, as well as to provide net social benefits to local communities, there are no actions that could pose risks to the well-being of the communities. On the contrary, the activities are aimed at improving local well-being.
Risks to stakeholder participation	1) Risks related to the loss of the project's integrity and reputation, affecting the stakeholders – the implementation and enforcement of conservation measures must have sufficient resources and capabilities for an extended period of monitoring and reporting. Otherwise, the desired social and	Resource Allocation and Capacity Building: Agroflorestal Novo Horizonte ensures the project is well-funded and staffed to sustain effective monitoring and reporting systems throughout its lifespan. This includes investing in staff training and capacity-building to address long-term conservation and reporting requirements effectively. Transparent Reporting and Accountability: A transparent reporting system is established to

	environmental outcomes may not be achieved. 2) Generation of Misaligned Expectations in Local Communities Regarding the Socio-Environmental and Productive Benefits the Project May Deliver. 3) Impacts on Community Relationships Depending on the Implementation of Socio-Environmental Actions 4) Residents of local communities involved in illegal practices may face coercion.	regularly communicate the project's progress, challenges, and outcomes to stakeholders. The project will also undergo independent audits to verify compliance with conservation measures and promptly address any issues that arise. Clear Communication and Education: The project is committed to conducting thorough community engagement sessions to clearly explain its goals, expected outcomes, and limitations. Educational materials are provided to ensure that all community members have a realistic understanding of what the project can deliver. Feedback Mechanisms: Agroflorestal Novo Horizonte has set up several mechanisms for continuous feedback and dialogue with the community. This includes regularly updating the community on project progress and adjusting expectations based on the feedback received. Inclusive Planning and Participation: Community representatives are actively involved in the planning and implementation of socio-environmental actions to ensure all perspectives are considered. Collaborative efforts are emphasized to build consensus and strengthen community relationships.

		Support and Transition Programs: Support programs are provided to help residents transition away from illegal practices. This includes offering alternative livelihoods through training and resources for sustainable practices, thereby reducing dependence on illegal activities.
Working conditions	No risk identified	The project does not present any risk conditions for workers, either directly or indirectly, beyond those associated with conventional forest management activities. Any such risks are effectively mitigated through ongoing training, comprehensive guidance, and continuous monitoring.
Safety of women and girls	No risk identified	The project does not engage in activities that expose women and girls to any form of risk. Additionally, the proponent has established channels for reporting any issues, concerns, or complaints, ensuring that all stakeholders have access to mechanisms for raising and addressing their concerns.
Safety of minority and marginalized groups, including children	No risk identified	The project does not engage in activities that expose vulnerable groups, regardless of gender or age, to any form of risk. Additionally, the proponent has established channels for reporting any issues, concerns, or complaints, ensuring that all stakeholders have access to

		mechanisms for raising and addressing their concerns.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	No risk identified	The project does not engage in activities that create or worsen environmental issues related to pollutants. All implemented tools and methods are designed to avoid the use of fuel combustion, waste generation, or effluents, ensuring minimal environmental impact.

2.3 Respect for Human Rights and Equity

The project proponent upholds universally proclaimed human rights and is committed to not engaging in any form of violence or human rights abuse, as established in the Universal Declaration of Human Rights, the UN Declaration on the Rights of Indigenous Peoples, ILO Convention 169 on Indigenous and Tribal Peoples, and the ILO Declaration on Fundamental Principles and Rights at Work.

Furthermore, the project rejects any form of discrimination and ensures the participation and inclusion of all workers, local communities, riverine populations, and other stakeholders. It guarantees equal opportunities in gender and ensures the inclusion of women throughout the project's planning and implementation phases.

The project adheres to a Code of Conduct that prohibits discrimination and sexual harassment, reflecting the company's commitment to professional culture and ethics. It provides equal employment opportunities regardless of gender or race, ensuring inclusive participation during the project's execution.

2.3.1 Labor and Work

Table 18. Risks related to labor and work

	Risks identified[9]	Mitigation or preventative measure(s) taken
Discrimination	No risk identified	

[9] The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

Sexual harassment	No risk identified	According to the Code of Ethics of Agroflorestal Novo Horizonte, the company does not tolerate any form of discrimination, including based on gender, race, ethnicity, religion, sexual orientation, disability, or any other personal characteristic. All employees and members of local communities are encouraged to report any form of discrimination, sexual or moral harassment, forced labor, child labor, human trafficking, or any other abuse. We provide secure and confidential communication channels for these grievances. Workers and communities are encouraged to report any abuse, whether caused by Agroflorestal Novo Horizonte employees or by acquaintances or family members. Complaints will be taken seriously and investigated in accordance with company policies. Training sessions are conducted annually and cover the conduct code.
Equal pay for equal work	No risk identified	
Gender equity in labor and work	No risk identified	
Forced labor	No risk identified	
Child labor	No risk identified	
Human trafficking	No risk identified	

2.3.2 Human Rights

Table 19. Risks to human rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	During the monitoring period, no reports, complaints, or grievances regarding risks or human rights violations were received, whether related to

	the activities conducted by Agroflorestal Novo Horizonte or any other situations experienced by stakeholders. No communication channels were utilized for this purpose, and in meetings held since the project's inception, no issues related to such violations were raised.

2.3.3 Indigenous Peoples and Cultural Heritage

Table 20. Risks to indigenous people and cultural heritage

Risks identified	Mitigation(s) or preventative measure taken
No risk identified	There are no Indigenous populations near the project area. Traditional and local communities do not have customary use or engage in any activities within the project area. There are no rights of use or possession by traditional, Indigenous, or local communities. The project activities do not impact the livelihoods of local communities. Even though there is no customary or economic dependency on the area, the communication channel remains open for any potential complaints if necessary. During the monitoring period, no comments related to risks affecting the communities were received.

2.3.4 Property Rights

Table 21. Risks to the Property rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	There are no Indigenous populations near the project area. Traditional and local communities do not have customary use or engage in any activities within the project area. There are no rights of use or possession by traditional, Indigenous, or local communities. The project activities do not impact

	the livelihoods of local communities. Even though there is no customary or economic dependency on the area, the communication channel remains open for any potential complaints if necessary. During the monitoring period, no comments related to risks affecting the communities were received.

2.3.5 Benefit Sharing

Table 22. Information of Benefit Sharing

Process used to design the benefit sharing plan	Novo Horizonte Agroforestry, as the proponent and owner of the project area, retains all carbon credits generated during the verified period. To implement the project, the company hired consulting services and professionals to support the planned activities. Since the property is privately owned and there are no traditional communities or Indigenous territories utilizing or residing on the land, there is no direct benefit-sharing. However, the proponent continuously develops socio-environmental activities for local communities. Currently, project activities are funded with the company's own resources, which limits their implementation capacity. Over the three years of the project, the proponent held several meetings with the community to develop a collaborative and adaptive action plan, which will be fully implemented using the resources generated by the project. The communities prioritized for implementing actions are those closest to the Fazenda: Vila Nova, Bom Jesus, and Santa Terezinha. Although the implementation of activities in the Resex Renascer area may occur over time, it is not considered a priority.
Summary of the benefit sharing plan	Even though the property is private and does not house traditional communities or indigenous territories, the project

	adopts an inclusive approach to ensure that local communities benefit. It has been determined that a portion of the carbon credits will be allocated to social initiatives, defined in collaboration with stakeholders. Specifically, 5% of the carbon credits, after the BUFFER deduction, will be reserved for the development of social projects that promote the well-being of local communities and encourage sustainable practices. Over the three years of implementation, Agroflorestal Novo Horizonte has held several meetings with neighboring communities, including Vila Nova, Bom Jesus, and Santa Terezinha, which have been identified as priorities for project actions. During these meetings, a joint and adaptive action plan was developed to address the specific needs of these communities and strengthen cooperation between the project and local beneficiaries. This plan will be fully implemented with the resources obtained from the project, aligning with the commitment to transparency and collaboration. The project integrates continuous monitoring strategies to ensure the long-term permanence of benefits, while also utilizing advanced technologies to control potential leakages and ensure the effectiveness of conservation measures. The plan also includes the formation of fire brigades, environmental education programs, and species monitoring, reinforcing Agroflorestal Novo Horizonte's commitment to sustainability and the preservation of local biodiversity.
Approval and dissemination of benefit sharing plan	During the meetings held throughout the three years of project implementation, we employed a clear and accessible communication approach, ensuring that all parties fully understood the project. Although specific numerical values were not initially communicated, visits and public consultations were conducted where social benefits were transparently and directly presented to all groups involved. This approach aims to prevent

	the creation of unfounded expectations while maintaining clarity and honesty in our interactions with the communities.
Benefit sharing during the monitoring period	Regarding the benefit-sharing plan, it is expected that the implementation of the short, medium, and long-term social and environmental action plan, discussed with the local communities, will begin in full after the first verification period, following the initial negotiations of Verified Carbon Units (VCUs). However, during the monitoring period, Agroflorestal Novo Horizonte will continue to strengthen partnerships with representatives of local communities and carry out feasible actions as already established.

2.4 Ecosystem Health

The Agroflorestal Novo Horizonte REDD+ project primarily aims to mitigate climate change by preventing carbon emissions through deforestation control. It contributes positively to global climate change mitigation and supports biodiversity conservation and sustainable ecosystem services. By preserving native forests, the project sequesters carbon dioxide, fosters biodiversity, and supports sustainable land management practices, benefiting local communities through enhanced livelihoods and financial returns from forest management.

The project is committed to addressing potential leakage of illegal activities such as deforestation and hunting into nearby areas. To mitigate this risk, the project collaborates closely with local settlements to monitor and manage any adverse effects. Furthermore, community engagement is integral to the project’s strategy, with ongoing dialogue and collaboration to ensure that local needs and concerns are addressed effectively.

Overall, Agroflorestal Novo Horizonte's approach is centered on transparent communication, proactive engagement with local communities, and continual monitoring to ensure that the project’s positive impacts are maximized while minimizing potential negative effects.

Table 23. Risks identified for Ecosystem health

Risks identified	Mitigation or preventative measure(s) taken

Impacts on biodiversity and ecosystems	1) The project contributes positively to biodiversity conservation, particularly for species with large home ranges such as felines, tapirs, and other mammals. However, there is a risk of increased hunting pressure on these species due to heightened visibility and population increases. 2) Risks may arise from vegetation deforestation or degradation not directly caused by the project, such as wildfires, neighboring land fires, or environmental crimes by external agents.	To address potential risks, the project implements community-based training and communication campaigns focused on ecosystem understanding and the importance of conserving high-conservation-value species. Local communities are engaged in educational programs to raise awareness about sustainable practices and the value of biodiversity. Regular workshops and discussions help to reinforce these messages and promote responsible behaviors. The project has allocated Buffer credits to address unforeseen circumstances. This buffer serves as an insurance reserve for unexpected events like wildfires or external encroachments that are difficult to prevent. In the event of such incidents leading to carbon stock loss, credits from the buffer pool may be used to offset these losses. The buffer system ensures that credits can be reinstated when risks are effectively managed. The project engages in proactive educational efforts. Informative brochures using simple and accessible language explain climate issues, carbon definitions, REDD+, and project details. These materials are distributed during in-person meetings with farm workers and local communities. Additionally, project

		conversations and environmental awareness initiatives are conducted to strengthen engagement and understanding of climate issues and conservation principles.
Soil degradation and soil erosion	There is no significant risk of soil degradation or erosion directly related to the project activities.	Since no major risks are identified in this area, the project continues to monitor soil conditions and employs best practices for land management to maintain soil health.
Water consumption and stress	The project activities do not pose a substantial risk of increased water consumption or stress.	The project maintains monitoring efforts to ensure that water resources are used sustainably. Any potential impacts are addressed proactively through local consultations and adjustments in project activities as needed.

Our commitment to the future

With the revenue generated from the sale of carbon credits, the project will continue expanding its activities and collaborating with communities to improve the quality of life and ensure that development is both sustainable and inclusive. The project's management is adaptive, meaning we remain responsive to the needs of the communities and ready to adjust actions according to their demands.

2.4.1 Rare, Threatened, and Endangered species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☒ Yes ☐ No

If yes, list such species and habitats in the table below and provide evidence that the project has not adversely impacted these areas during the monitoring period.

Table 24. List of vulnerable species

Species and habitat	Threatened Bird Species:

	• Grey Tinamou (Tinamus tao) • Harpy Eagle (Harpia harpyja) • Red-necked Aracari (Pteroglossus bitorquatus) • White-throated Guan (Penelope pileata) • Blue-winged Macaw (Primolius maracanã) • Black-spotted Bare-eye (Phlegopsis nigromaculata) • Razor-billed Curassow (Aburria cujubi) • Hyacinth Macaw (Anodorhynchus hyacinthinus) • Helmeted Curassow (Pauxi tuberosa) Threatened Herpetofauna Species: • Scorpion Mud Turtle (Kinosternon scorpioides) • Yellow-footed Tortoise (Chelonoidis denticulata) Threatened Mammal Species: • Jaguar (Panthera onca) • Puma (Puma concolor) • Brazilian Tapir (Tapirus terrestris) • Red-handed Howler Monkey (Alouatta belzebul) By ensuring the maintenance of forest cover and reducing deforestation in leakage zones, the project promotes the protection of these species. The project area is monitored to prevent the entry of hunters, and there are fire brigades and initiatives to combat wildfires. These actions have a positive impact on wildlife, as they ensure that the area serves as a refuge for various species.
Areas needed for habitat connectivity	*Between 2021 and 2024, the vegetation in the project area has been conserved. There is no habitat fragmentation caused by deforestation, ensuring connectivity within the forest and providing suitable habitats for wildlife.*

The following table summarizes the identified risks related to habitats for rare, threatened, and endangered species, as well as areas for habitat connectivity, within the Agroflorestal Novo Horizonte REDD AUD project. Recognizing the importance of maintaining biodiversity and ecosystem integrity, this assessment aims to outline potential threats that could impact these critical habitats and the connectivity between them.

For each identified risk, we have detailed commensurate mitigation and preventative measures that are in place to effectively address and minimize these risks during the monitoring period. These measures are designed to enhance habitat protection, support species conservation, and ensure the long-term sustainability of the project. By implementing these strategies, Agroflorestal Novo Horizonte is committed to fostering an environment that supports biodiversity while achieving its carbon reduction goals.

Table 25. Risks for species and habitat connectivity

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	1) Illegal Deforestation 2) Illegal Hunting Activities	**1) Mitigation Measures:** Implement regular monitoring and patrols to detect illegal logging activities. Establish partnerships with local law enforcement and community groups to report and act against illegal activities. **2) Mitigation Measures:** Create awareness campaigns in local communities about the importance of protecting endangered species. Establish a community-based wildlife protection program that involves local residents in monitoring and reporting illegal hunting. **3) Mitigation Measures:** Implement fire prevention strategies, including controlled burns and firebreaks, to

	3) Wildfires 4) Lack of Monitoring and Enforcement	reduce the risk of uncontrolled wildfires. Train local communities in fire management techniques and establish firefighting brigades. **4) Mitigation Measures:** Establish a robust monitoring and reporting system that includes local communities and stakeholders. Allocate resources for regular inspections and enforce environmental regulations effectively.
Areas for habitat connectivity	1) Illegal Deforestation 2) Wildfires	**1) Mitigation Measures:** Implement regular monitoring and patrols to detect illegal logging activities. Establish partnerships with local law enforcement and community groups to report and act against illegal activities. **2) Mitigation Measures:** Implement fire management plans that include creating firebreaks and conducting controlled burns to maintain habitat integrity. Train local communities in fire management and prevention techniques.

2.4.2 Introduction of species

This section is N/A because this project does not have any planting or species introduction.

2.4.3 Ecosystem conversion

The project area contains only areas which were defined as “forest” in 2011 (more than 10 years prior to the project start date), as described in the forest cover benchmark maps in the section 1.13 (Project Location). That means, the project area’s ecosystem has been preserved as its natural state, with no conversion activities.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Table 26. Methodologies applied

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	VM0015	Methodology for Avoided Unplanned Deforestation	1.2
Tool	VT0001	Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities	3.0
Module	VCS	AFOLU Non-Permanence Risk	4.7
Tool	NPRT	AFOLU Non-Permanence Risk Tool	4.2

3.2 Applicability of Methodology

This document serves to demonstrate and justify how the activities of the Agroflorestal Novo Horizonte REDD AUD project meet each of the applicability conditions outlined in the relevant methodologies, tools, and modules utilized in the project. The adherence to these conditions is crucial for ensuring that the project aligns with established standards and effectively contributes to carbon reduction efforts.

Each applicability condition will be addressed separately, providing a clear explanation of how the project complies with the requirements. This thorough evaluation underscores our commitment to implementing best practices in forest conservation, enhancing biodiversity, and fostering sustainable development within the project area. By fulfilling these conditions, Agroflorestal Novo Horizonte aims to ensure the integrity and effectiveness of its REDD initiatives.

Table 27. Applicability

Methodology ID	Applicability condition	Justification of compliance
VM0015 – Methodology for Avoided Unplanned Deforestation, v1.2	a) Baseline activities may include planned or unplanned logging for timber, fuelwood collection, charcoal production, agricultural and grazing activities as long as the category is unplanned deforestation according to the most recent VCS AFOLU requirements.	The primary land uses in the baseline scenario are: cattle ranching, mainly for producing beef cattle; and timber harvest, acting both legally and illegally. These unplanned deforestation and degradation agents have been attracted due to infrastructure, such as waterways and roads.
VM0015 – Methodology for Avoided Unplanned Deforestation, v1.2	b) Project activities may include one or a combination of the eligible categories defined in the description of the scope of the methodology (table 1 and figure 2) ….	The project falls under category D - Avoided deforestation with logging in the baseline and project case.
VM0015 – Methodology for Avoided Unplanned Deforestation, v1.2	c) The project area can include different types of forest, such as, but not limited to, old growth forest, degraded forest, secondary forests, planted forests and agroforestry systems meeting the definition of "forest".	The area is considered forest as per the definition adopted definition of adopted by FAO3: Land spanning more than 0.5 hectares with trees higher than 5 meters and a canopy cover of more than 10%, or trees able to reach these thresholds in situ.
VM0015 – Methodology for Avoided Unplanned Deforestation, v1.2	d) At project commencement, the project area shall include only land qualifying as "forest" for a minimum of 10 years prior to the project start date.	The Project Area consisted of 100% tropical in 2014 This was ascertained using satellite images, as described in the section Baseline Scenario of the present VCS PD.

VM0015 – Methodology for Avoided Unplanned Deforestation, v1.2	e) The project area can include forested wetlands (such as bottomland forests, floodplain forests, mangrove forests) as long as they do not grow on peat. Peat shall be defined as organic soils with at least 65% organic matter and a minimum thickness of 50 cm. If the project includes project at a includes forested wetlands growing on peat (e.g. peat swamp forests), this methodology is not applicable.	The only soil type within the Project Area is the dystrophic Yellow Latosol Dystric. Therefore, no peat or peat swamp forests were found within the Project Area, satisfying this applicability criterion.
VT0001 – Tool for demonstration and assessment of additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities, v3.0	a) AFOLU activities the same or similar to the proposed project activity on the land within the proposed project boundary performed with or without being registered as the VCS AFOLU project shall not lead to violation of any applicable law even if the law is not enforced;	Sustainable Forest Management Plan is an authorized and endorsed activity in Brazil, and the project area have all environmental and legal authorizations necessary to conduct the activity
VT0001 – Tool for demonstration and assessment of additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities, v3.0	b) The use of this tool to determine additionality requires the baseline methodology to provide for a stepwise approach justifying the determination of the most plausible baseline scenario. Project proponent(s) proposing new baseline methodologies shall ensure consistency between the determination of a baseline scenario and the determination of additionality of a project activity.	The Methodology provides a stepwise approach to justify the determination of the most plausible baseline scenario, which is detailed at Section 3.4 – Baseline Scenario.
NPRT – AFOLU Non-Permanence Risk Tool, v4.2	a) AFOLU activities the same or similar to the proposed project activity on the land within the proposed project boundary performed with or without being registered as the VCS	Sustainable Forest Management Plan is an authorized and endorsed activity in Brazil, and Instances must have all environmental and legal

	AFOLU project shall not lead to violation of any applicable law even if the law is not enforced;	authorizations necessary to conduct the acti

3.3 Project Boundary

The project encompasses 21 properties, covering a total area of 23,092.90 hectares (ha), predominantly covered by native vegetation. To address potential leakage, a leakage belt of 128,115.48 ha has been delineated to the north of the project area, adjacent to major deforestation drivers such as roads and settlements.

The delineation of the project area, in compliance with the VERRA REDD+ methodologies, and the specifications of the leakage belt are detailed in Table 28.

Table 28. Agroflorestal Novo Horizonte Project Areas

Name	Area
Project Area	23,092.9
Leakage Belt	35,001.74

Agroflorestal is presented in the following figures (Figure 27), located in the municipality of Prainha - PA, an area currently under threat of deforestation. The total property area is 28,991.87 hectares; however, the project accounting area is 23,092.90 hectares, and it was 100% forested at the start of the project. The property boundaries were delineated through geospatial analyses and land titling records. The proponent carries out activities within and around the project area to mitigate deforestation pressures, adhering to the guidelines of the applied methodology and its respective modules and tools.

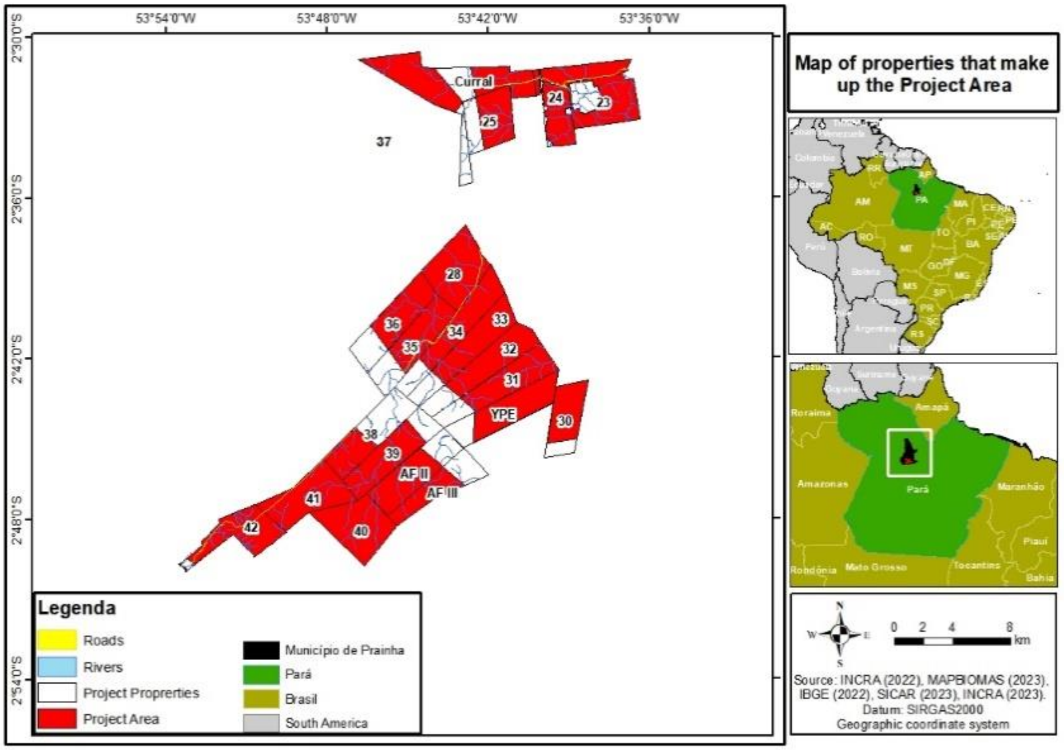


Figure 27. Project Area of the Agroflorestal Novo Horizonte

Reference Region

The RRD (Reference Region for Deforestation Rate Projection) is the only defined Reference Region, as the deforestation dynamics in the Project Area are transitional, and the BL-UP requirement is met.

To define the RRD, the proponent excluded the project area and the leakage belt, along with all non-forested areas up to the start of the historical reference period in 2011. Additionally, the reference region was defined with knowledge of the agents and drivers of unplanned deforestation in Legal Reserves and Permanent Preservation Areas within the region.

The main deforestation agents in the RRD are small-scale farmers, primarily aiming to establish or expand pastures for cattle ranching and, secondarily, agricultural crops through forest conversion. The same behavior observed among farmers and ranchers in the RRD is expected in the project area at the baseline. However, landscape factors (i.e., soil type, vegetation type, and slope) do not guide small farmers' decision-making processes. It can be noted that when elevation becomes significant, it drives the decision to deforest and convert the area into pasture or agricultural land.

According to LAW No. 12,651, of May 25, 2012, the Brazilian Forest Code, all vegetation located in Permanent Preservation Areas and Legal Reserves must be maintained by the landowner, possessor, or occupant, whether an individual or legal entity, public or private, indicating that the main deforestation agents do not possess legal land use rights, as deforestation rates continue to occur over the years.

Maps of landscape factors, including forest type, soil type, slope, and elevation, were used to define the reference region and ensure its similarity to the project area.

Land tenure was also used to help delineate the RRD. Specifically, municipal, state, and federal forest conservation areas, as well as indigenous reserves, were excluded from the RRD because they differ from the privately owned project area. However, according to the vegetation classification of Brazil by IBGE, both the Project Area and the RRD are entirely covered by native Amazon vegetation, such as Ombrophilous forests.

Module VMD0007 BL-UP states the following regarding infrastructure, such as roads and rivers, which increase the likelihood of deforestation: if these infrastructures historically exist in the RRD, they must be comparable to those expected to exist in the project area. Similarity is assumed between the RRD and respective Agroflorestal areas in terms of transportation networks and human infrastructure. Generally, Brazil's transportation infrastructure is inadequate compared to networks in similarly sized countries. The paved road network represents only a small portion of the total road network and has shown moderate growth in recent years due to limited maintenance and improvement efforts, compromising productivity. It should also be noted that both transportation network density and social infrastructure are concentrated in large urban centers, particularly in the Southeast and South regions, which generally does not apply to northern regions of the country, where the RRD and respective projects are located.

Regarding settlements near forest areas within the project boundaries, these areas are suitable for deforestation agents who have not historically invaded the project area (e.g., cattle ranchers and agricultural producers) but may do so during the project's duration. Therefore, the reference region included areas where such agents have deforested during the historical reference period.

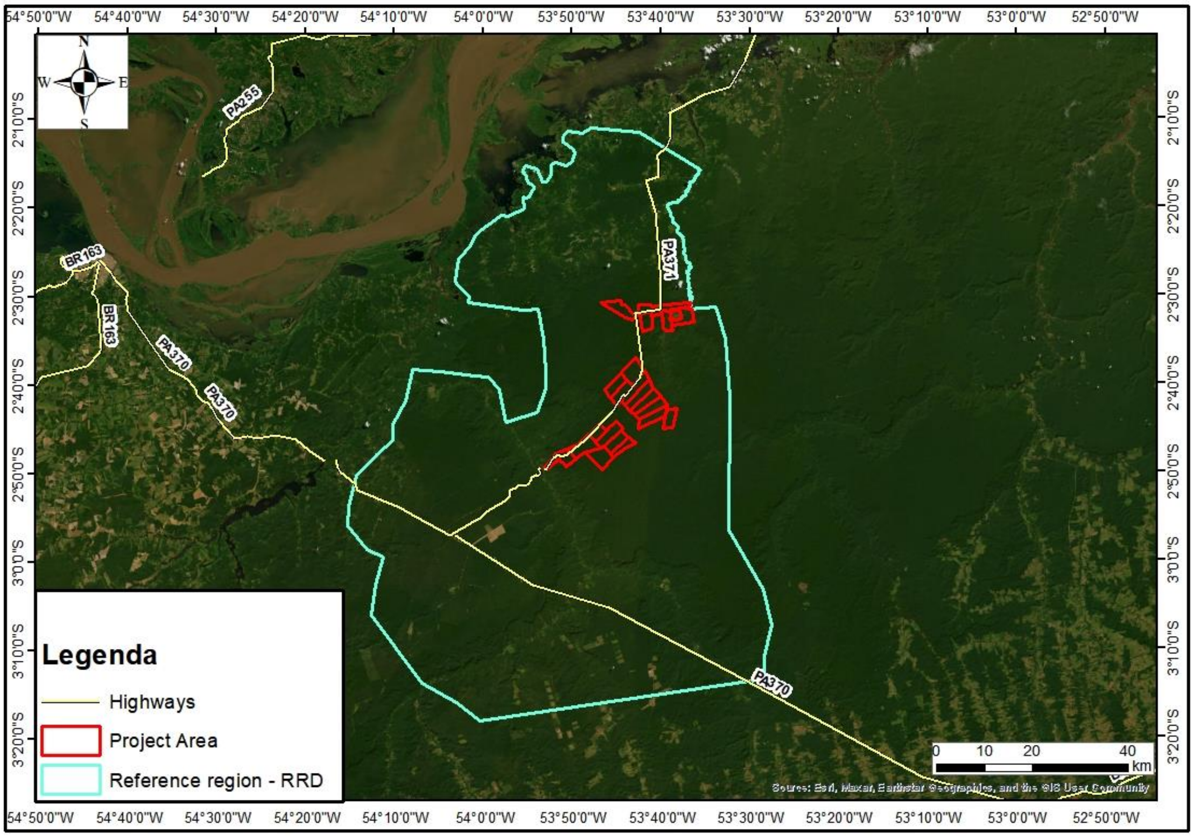


Figure 28. Reference Region Map

Leakage Belt

The leakage belts surround the Agroflorestal project and are located in the most accessible areas with the highest likelihood of deforestation. This is a forested area designed to address the displacement related to "leakage" caused by the activities of the Agroflorestal Project. It meets the following requirements described in module VMD0007 BL-UP: t is the forest area closest to the project area and meets the minimum area requirements (i.e., 90% of the project area).

All parts of the leakage belt are accessible and reachable by deforestation agents. The leakage belt is 100% forested at the start of the project.

Additionally, municipal, state, and federal forest conservation areas, as well as indigenous reserves, were excluded from the leakage belt, as the policies and administrative regulations in these areas differ from private policies.

The Project Area and the Leakage Belt are quantitatively ±20% similar, or qualitatively similar in important aspects, concerning the factors determined by module VMD0007 BL-UP, specifically: forest types; elevation classes; slope classes; soil types; and road density, which were rigorously considered for the allocation of the leakage belt. Evidence of similarity between the Agroflorestal and Leakage Belt areas is shown below in Figure 29.

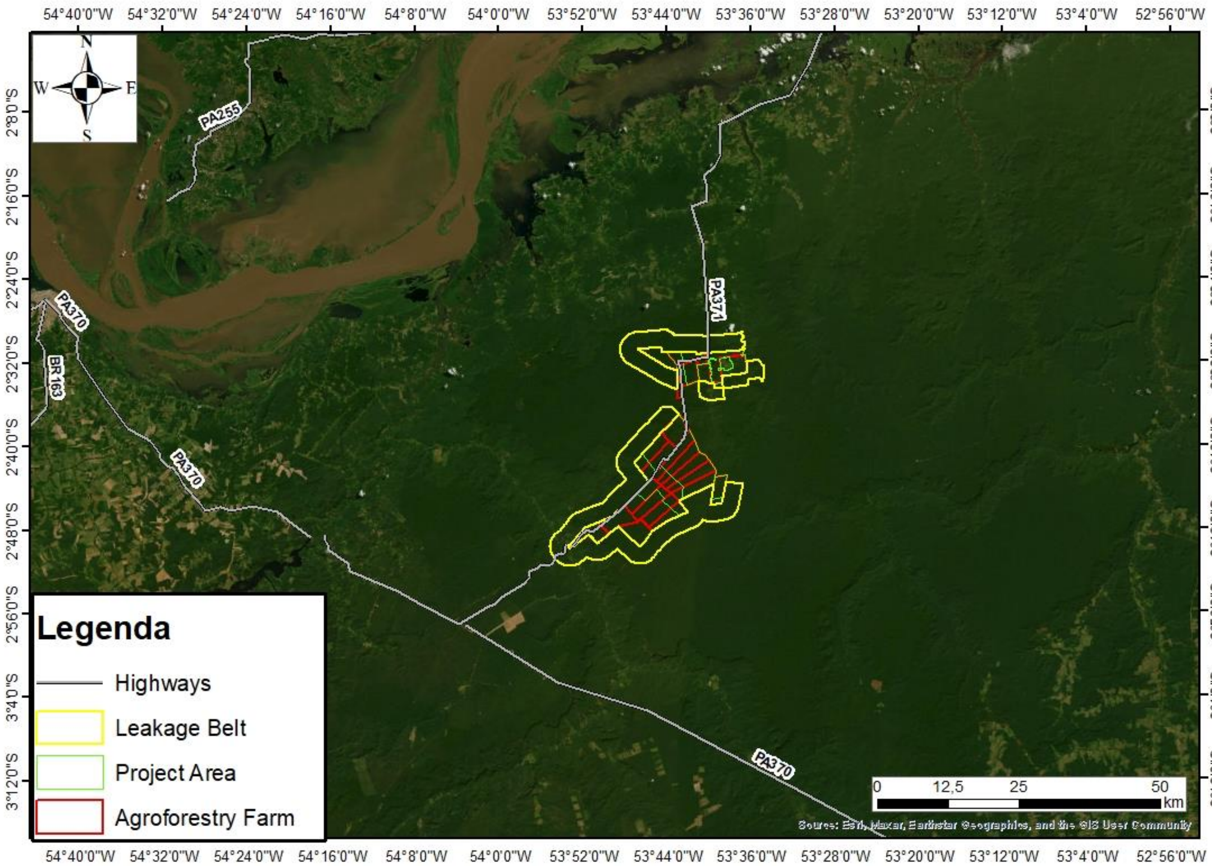


Figure 29. Location of the Leakage Belt in the Agroflorestal Project.

Leakage Management Area

Leakage Management Area (LMA) refers to regions outside the official project boundaries where activities are designed and implemented to mitigate the risk of leakage—essentially, the displacement of deforestation or forest degradation to areas beyond the project's direct control. In the Project Scenario, the LMA plays a crucial role in ensuring the overall effectiveness of forest conservation efforts by addressing deforestation drivers in surrounding areas that could otherwise undermine the project's objectives. The Agroflorestal Novo Horizonte LMA strategy focuses on curbing leakage through a multifaceted approach aimed at both environmental and socio-economic factors. Core activities within the LMA include:

Targeting Key Deforestation Drivers:

The project identifies and addresses the primary causes of deforestation, such as unsustainable agricultural practices, illegal logging, and shifting cultivation. By tackling these root causes, the risk of deforestation "leaking" outside the project zone is reduced.

Promoting Sustainable Livelihoods:

One of the pillars of leakage prevention is the creation of alternative income streams for local communities, reducing their dependence on forest exploitation. This may involve:

Agroforestry Systems: Encouraging the integration of trees into agricultural landscapes, which provides both income and ecosystem services.

Sustainable Agriculture: Training local farmers in climate-smart, low-impact farming methods that enhance productivity while conserving natural resources.

Education and Capacity Building:

Community engagement is central to the success of LMA initiatives. Educational programs are rolled out to raise awareness about the benefits of forest conservation and sustainable land use. Workshops, training sessions, and outreach campaigns help equip local residents with the knowledge and skills needed to adopt practices that align with conservation goals.

Infrastructure and Support Systems:

The project also provides technical and financial assistance to ensure that communities in the LMA can successfully transition to sustainable livelihoods. This includes access to tools, technology, and market linkages that support agroforestry and sustainable agricultural initiatives.

Monitoring and Adaptive Management:

Continuous monitoring of deforestation patterns and socio-economic indicators in the LMA is essential. This allows the project to assess the effectiveness of implemented strategies and adjust actions as necessary. Adaptive management ensures that LMA activities remain responsive to changing circumstances and evolving threats.

The Agroflorestal Novo Horizonte Leakage Management Area was established with consideration of the nearest communities, as depicted in Figure 30.

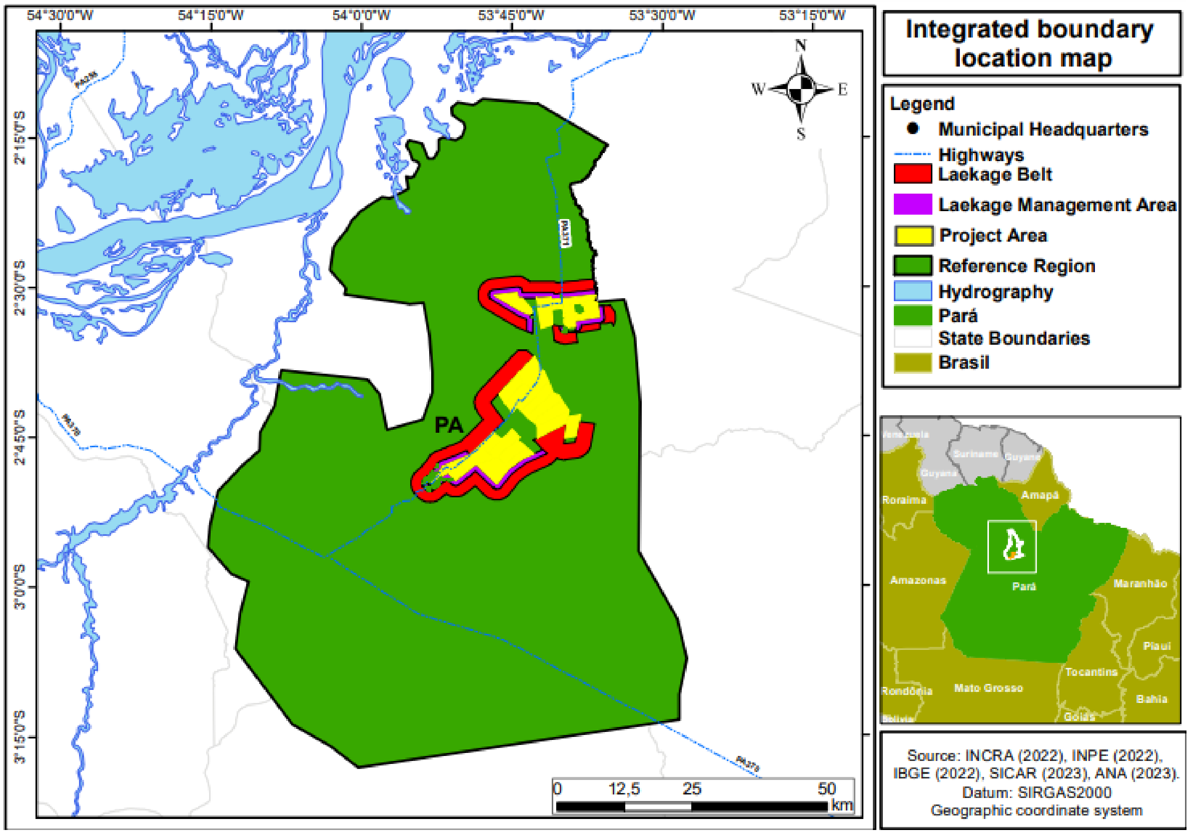


Figure 30. Leakage management areas

Temporal Boundaries

The temporal domain from which information on historical deforestation was extracted, analyzed, and projected into the future was established in accordance with the VMD0007 methodological requirements, using annual data from 2011 to 2020.

The project will begin on 10-October-2021, and end on 09-October-2061, encompassing a total crediting period of 40 years. Therefore, 10-October-2021, and 10-October-2026, represent the start and end dates of the project's first baseline period.

Monitoring Period

The first monitoring period is from 10-October-2021-to-30-September-2024

Carbon Pools

In accordance with the methodology, six carbon pools are considered, as outlined in Table 29. This table provides a comprehensive justification for the inclusion or exclusion of each carbon pool within the project boundary.

Table 29. Carbon pools included or excluded within the boundary of the proposed AUD project activity

Carbon pools	Included / Excluded	Justification / Explanation of choice
Above-ground	Tree: Included	Carbon stock change in this pool is always significant
	Non-Tree: Excluded	Excluded in carbon stocks estimates since adopted carbon stock (Silva, 2014) does not specify non trees in the above ground biomass
Below-ground	Included	Stock change in this pool is significant
Dead wood	Exluded	Excluded for simplification. In the baseline scenario, dead wood is not removed and/or used before the deforestation, as it is often in the process of decomposition in the forest, being left to burn in the baseline case. Therefore, not accounting for this carbon pool is conservative, as it does not consider GHG emissions from deforestation and burning in the baseline.
Harvested wood products	Excluded	Stock change in this pool is not considered in baseline and project scenarios. This is conservative as project scenario includes logging through a sustainable forest management plan.
Litter	Excluded	Excluded as it does not lead to a significant over-estimation of the net anthropogenic GHG emission reductions of the AUD project activity. This exclusion is conservative.

Soil organic carbon	Excluded	Recommended when forests are converted to cropland. Not to be measured in conversions to pasture grasses and perennial crop according to VCS Methodology Requirements, 4.7.

The methodology also evaluates two sources of greenhouse gas (GHG) emissions, as detailed in Table 30. This table includes the rationale for including or excluding these emissions sources within the project boundary.

Table 30. GHG Sources

Source		Gas	Included?	Justification/Explanation
Baseline	Biomass burning	CO_2	Excluded	Excluded as recommended by the applied methodology. Counted as carbon stock change.
		CH_4	Included	Included as non-CO2 emissions from biomass burning in the baseline scenario, according to the methodology.
		N_2O	Included	Included as non-CO2 emissions from biomass burning in the baseline scenario, according to the methodology.
		Other	Excluded	No other GHG gases were considered in this project activity.
	Livestock emissions	CO_2	Excluded	Not a significant source
		CH_4	Excluded	Excluded for simplification. This is conservative.
		N_2O	Excluded	Excluded for simplification. This is conservative.
		Other	Excluded	No other GHG gases were considered in this project activity.

Source		Gas	Included?	Justification/Explanation
Project	Biomass burning	CO_2	Excluded	No biomass burning increase is predicted to occur in the project scenario compared to the baseline case. Therefore, considered insignificant.
		CH_4	Included	Included as non-CO2 emissions from biomass burning in the project scenario, according to the methodology.
		N_2O	Included	Included as non-CO2 emissions from biomass burning in the project scenario, according to the methodology.
		Other	Excluded	No other GHG gases were considered in this project activity.
	Livestock emissions	CO_2	Excluded	Not a significant source
		CH_4	Excluded	No livestock agriculture increase is predicted to occur in the project scenario compared to the baseline case. Therefore, considered insignificant.
		N_2O	Excluded	No livestock agriculture increase is predicted to occur in the project scenario compared to the baseline case. Therefore, considered insignificant.
		Other	Excluded	No other GHG gases were considered in this project activity.

3.4 Baseline Scenario

In the baseline scenario, forested areas are expected to be converted into non-forested areas within the scope of the baseline: Unplanned Avoided Deforestation (AUDD) as a result of dominant land-use changes in the region. These include agriculture and livestock, which result in the removal of primary and secondary forests in the Project Zone.

Estimative of annual areas of unplanned baseline deforestation in the RRD and calcukation of unplanned deforestation rate.

Baseline Modules

VMD0007 "Estimation of baseline carbon stock changes and greenhouse gas emissions from unplanned deforestation and degradation" (BL-UP), v3.3: This module allows for the estimation of changes in carbon stock and GHG emissions related to unplanned deforestation and degradation activities in wetlands within the baseline scenario.

Collection of Appropriate Data Sources

The land use and land cover classes were obtained from the MapBiomas database. Additionally, a mosaic of images from Landsat TM 5 and OLI 8 satellites was used to map the "non-forested vegetation" class. The images cover the historical reference period (2011 to 2020). MapBiomas uses LANDSAT class satellites (30-meter spatial resolution and a 16-day revisit rate) in a mosaic combination designed to minimize cloud coverage issues and ensure interoperability criteria. MapBiomas has a minimum mapped area of 900 m² (30-meter pixel size). A total of 20 mosaic images, composed of two scenes (orbits: 226 and 227 / point: 62), were adopted to map the land-use classes of interest in the reference region within the historical period. The following table describes the data used for the historical analyses of LU/LC changes:

Table 31. Data used for historical LU/LC change analysis

Vector	Sensor	Resolution		Coverage	Acquisition date	Scene	
		Spatial (m)	Spectral (µm)	(Km²)	DD/MM/YY	Path/ Latitude	Row/ Longitude
Satellite	Landsat TM	30	0.45 - 2.35	34.225	2011 - 2011	227	62
Satellite	Landsat OLI	30	0.64 - 1.65	31.110	2012 - 2024	227	62
Satellite	Landsat TM	30	0.45 - 2.35	34.225	2011 - 2011	226	62
Satellite	Landsat OLI	30	0.64 - 1.65	31.110	2012 - 2024	226	62

Table 32. LANDSAT images used for land use mapping.

Number	Date	Projection
LT05_L1TP_226062_20110913_20161006_01_T1	13/09/2011	WSG84
LE07_L1TP_226062_20120806_20200908_02_T1	06/08/2012	WSG84
LC08_L1TP_226062_20131020_20200912_02_T1	20/10/2013	WSG84
LC08_L1TP_226062_20140820_20200911_02_T1	20/08/2014	WSG84
LC08_L1TP_226062_20151026_20200908_02_T1	26/10/2015	WSG84
LC08_L1TP_226062_20160926_20170320_01_T2	26/09/2016	WSG84
LC08_L1TP_226062_20170711_20170726_01_T1	11/07/2017	WSG84
LC08_L1TP_226062_20180831_20180912_01_T1	31/08/2018	WSG84
LC08_L1TP_226062_20191005_20191018_01_T2	05/10/2019	WSG84
LC08_L1TP_226062_20200719_20200911_02_T1	19/07/2020	WSG84
LT05_L1TP_227062_20110616_20200822_02_T1	16/06/2011	WSG84
LE07_L1TP_227062_20120930_20200908_02_T1	30/09/2012	WSG84
LC08_L1TP_227062_20130925_20200912_02_T1	25/09/2013	WSG84
LC08_L1TP_227062_20141030_20200910_02_T1	30/10/2014	WSG84
LC08_L1TP_227062_20150814_20200909_02_T1	14/08/2015	WSG84
LC08_L1TP_227062_20161206_20180528_01_T1	06/10/2016	WSG84
LC08_L1TP_227062_20170718_20170727_01_T1	18/07/2017	WSG84
LC08_L1TP_227062_20180721_20180731_01_T1	21/07/2018	WSG84
LC08_L1TP_227062_20190724_20190801_01_T1	24/07/2019	WSG84
LC08_L1TP_227062_20200624_20200707_01_T1	24/06/2020	WSG84

The land use and land cover data used in the Agroflorestal Project were obtained from the MapBiomas database. The main methodological steps conducted by MapBiomas to map land use and cover in Brazilian territory are described as follows:

Pre-processing: Landsat mosaics were generated based on specific time periods to optimize spectral contrast and help distinguish land use and cover classes. A cloud and shadow removal technique, utilizing the evaluation band and the median reducer, was used to improve data integrity and remove pixels affected by artifacts or cloud contamination.

In Collection 7, new surface reflectance images from the USGS Landsat were used for classification. For each analysis area, a specific temporal mosaic of Landsat images was constructed based on selection criteria that allowed for annual analysis and provided sufficient spectral contrast data to better distinguish land cover and use classes.

The themes of Pasture, Agriculture, Planted Forest, Urban Area, Coastal Area, and Mining processed the Landsat mosaics on an individual scene basis, using a tool developed to evaluate individual images and improve mosaic quality. In the Amazon biome, each Landsat image was classified using the Random Forest algorithm, and the results were reclassified to create the annual land cover and use map.

Classification: The classification scheme was hierarchical, divided into six levels. Each level has subclasses representing different categories of land cover and use. Random Forest and learning algorithms like U-Net were used for classification, with specific parameters and training samples defined for each biome and cross-cutting theme.

Level 1 and Main Classes:

At level 1, there are six main classes: Forest, Non-Forest Formation, Agriculture, Non-Vegetated Area, Water, and Not Observed. At level 2, these classes expand into 16 subclasses, which also combine Land Use and Land Cover (LU/LC) classes.

Level 1 Classes:

Forest is divided into four subclasses: Forest Formation, Savanna Formation, Mangrove, and Restinga Shrub Vegetation.

Non-Forest Formation is subdivided into Wetlands, Pasture Formations, Salt Flats, Rock Outcrops, Other Non-Forest Formations, and the new Herbaceous Vegetation of Sand Dunes class.

Agriculture is subdivided into Pasture, Agriculture, Planted Forests, and Mosaic of Uses.

Water Class includes subdivisions such as River, Lake, Ocean, and Aquaculture.

Post-Processing: Filters are applied to the classified maps to reduce noise and improve accuracy. These filters include:

Gap Filling: Replaces missing or no-data values.

Spatial Filter: Removes isolated pixels that may be misclassified.

Temporal Filters: Corrects land use transitions that are not allowed over time.

Frequency Filter: Eliminates classes with low occurrence.

Incident Filter: Stabilizes noisy pixel trajectories.

Map Integration: Consolidates data from different biomes and cross-cutting themes.

Map Accuracy Assessment: Two validation approaches were used.

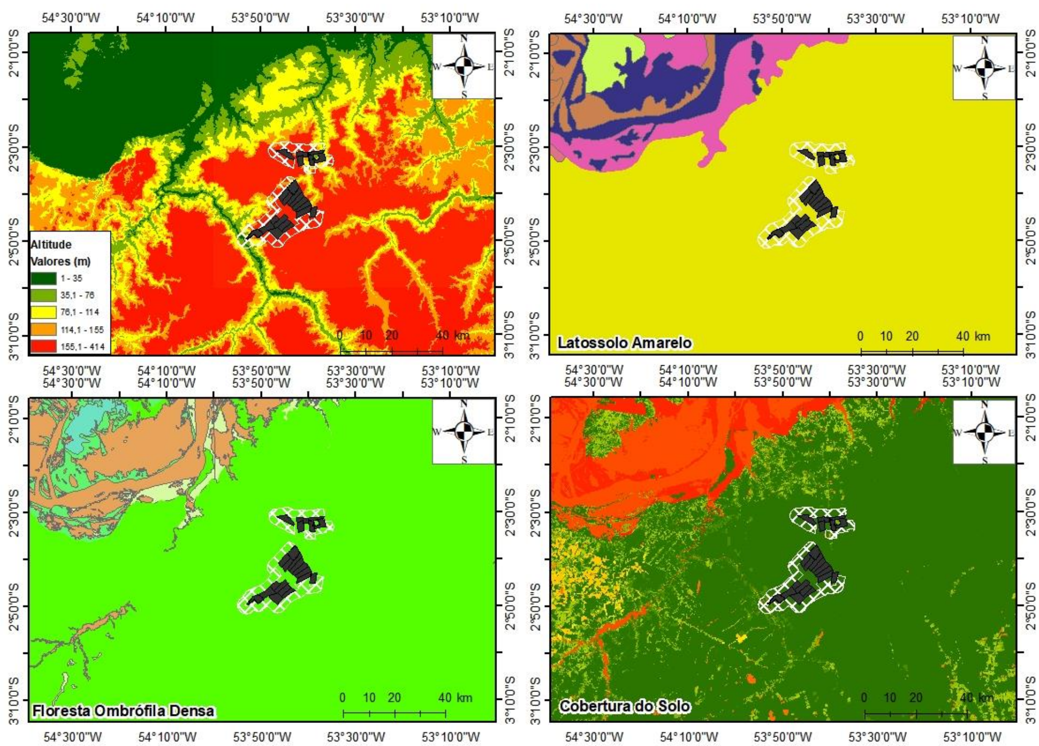
Comparative Analysis: Comparison with reference maps specific to biomes/regions and determined years.

Statistical Accuracy Analysis: Independent sample points were used, with visual interpretation of data across the entire time series.

For the spatial agreement analysis with reference maps, each biome and cross-cutting theme conducted the analyses based on the availability of reference maps. Validation with independent points involved the visual interpretation of Landsat data, MODIS-NDVI time series, and high-resolution Google Earth images, when available.

Reclassification to Meet REDD+ Requirements:

For the main objectives of this project, the original MapBiomas classes were grouped to meet the criteria set by the VMD0007 module (estimation of carbon stock changes and GHG emissions). A reclassification was carried out based on the raster, using deforestation classified year by year. The following table illustrates the reclassification and description of each class, as well as the areas existing before the project's implementation.



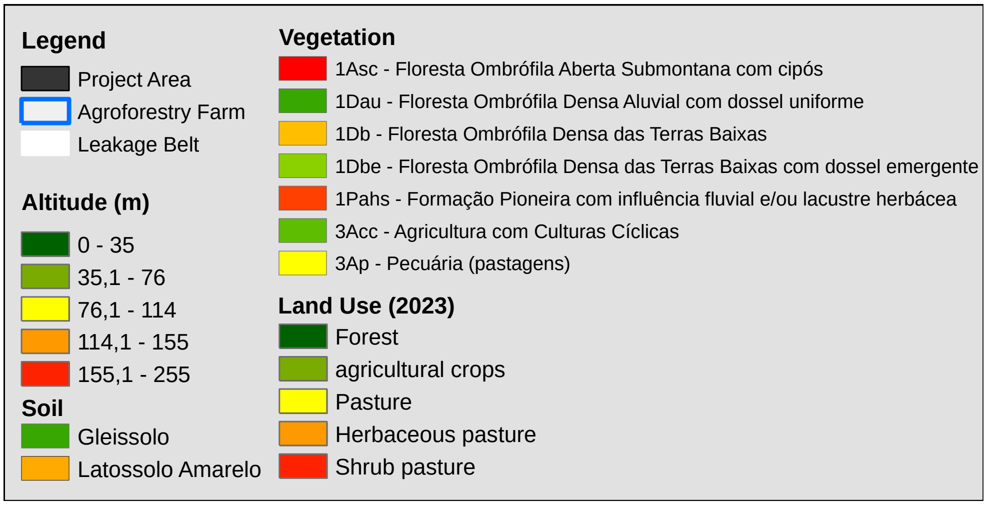


Figure 31. Landscape Analysis

Table 33. Landscape analysis

Landscape configuration and ecological conditions: area in hectares and percentage				
Soil types	RR		PA	
LAd - Latossolo Amarelo Distrófico	583.923,16	100,0%	23.092,85	100,0%
PVAd - Argissolo Vermelho-Amarelo Distrófico		0,0%		0,0%
GXve - Gleissolo Háplico Ta Eutrófico		0,0%		0,0%
Others (Water body, etc.)		0,0%		0,0%
RYve - Neossolo Flúvico Ta Eutrófico		0,0%		0,0%
GXbe - Gleissolo Háplico Tb Eutrófico		0,0%		0,0%
FXd - Plintossolo Háplico Distrófico		0,0%		0,0%
RQg - Neossolo Quartzarénico Hidromórfico		0,0%		0,0%
RYbd - Neossolo Flúvico Tb Distrófico		0,0%		0,0%
Total	583.923,16	100,0%	23.092,85	100,0%
Vegetation	RR		PA	
Dense Ombrophilous Forest (Da, Db)	583.923,16	100,0%	23.092,85	100,0%
Secondary Vegetation		0,0%		0,0%
Open Ombrophylous Forest (Aa, Ab, Ap)		0,0%		0,0%
Savana (Sp)		0,0%		0,0%
Pioneer Formation (Pa)		0,0%		0,0%
Campinarana (Lg)		0,0%		0,0%
Others (Water body, etc.)		0,0%		0,0%

Total	583.923,16	100,0%	23.092,85	100,0%
Rainfall	RR		PA	
Am – Tropical	583.923,16	100,0%	23.092,85	100,0%
Total	583.923,16	100,0%	23.092,85	100,0%
Elevation	RR		PA	
< 50	21.854,00	3,7%	30,51	0,1%
50-100	117.210,76	20,1%	1.537,72	6,7%
100-150	103.835,61	17,8%	4.595,30	19,9%
150-200	251.389,88	43,1%	16.929,33	73,3%
200-250	89.632,91	15,4%	0,00	0,0%
250-300				
Total	583.923,16	100,0%	23.092,85	100,0%
Slope	RR		PA	
Plan	21.854,00	17,2%	30,51	0,1%
Smooth Wavy	117.210,76	38,9%	1.537,72	6,7%
Wavy	103.835,61	35,8%	4.595,30	19,9%
Strong wavy	251.389,88	8,0%	16.929,33	73,3%
Mountanious	89.632,91	0,1%	0,00	0,0%
Steep		0,0%	0,00	0,0%
Total	583.923,16	100,0%	23.092,85	100,0%

Table 34. List of all land use and land cover classes existing at the project start date within the reference region

Class identifier		Trend in carbon stock[1]	Presence in[2]	Baseline activity[3]			Description (including criteria for unambiguous boundary definition)
IDcl	Name			LG	FW	CP	
1	Forest	constant	RR, PA, LK	yes	no	no	According to official classification of the types of vegetation of Brazil (SIVAM) and the high representativeness of the main forest type within the project area, no stratification in different forest classes was conducted. In addition, carbon density is not expected to

							undergo significant changes due to degradation in the baseline case. According to the significance test, carbon stock change due to logging activities in the baseline case is considered insignificant and therefore, trend in carbon stock could be deemed as constant.
2	Non forest	constant	RR, LK	no	no	no	Mosaic of anthropic areas: pasture, annual, perennial crops and roads according to the satellite image classification
3	Hydrography	constant	RR, LK	no	no	no	Presence of rivers and water bodies in the satellite image classification and information from the National Water Agency - ANA

1. Note if "decreasing", "constant", "increasing"
2. RR = Reference region, LK = Leakage belt, LM = Leakage management Areas, PA = Project area
3. LG = Logging, FW = Fuel-wood collection; CP = Charcoal Production (yes/no)
4. Each class shall have a unique identifier (IDcl). The methodology sometimes uses the notation icl (= 1, 2, 3, … Icl) to indicate "initial" (pre-deforestation) classes, which are all forest classes; and fcl (= 1, 2, 3, … Fcl) to indicate final" (post-deforestation) classes. In this table all classes ("initial" and "final") shall be listed.

The classification criteria for the project were the same as those defined by MapBiomas:

Forest Area: Areas of remaining forest. This class includes all areas classified as forest since 2011 that have not changed class during the historical period. In other words, it has been a forest since the beginning of the historical period.

Non-Forest Area: Areas occupied by any anthropogenic use at some point during the historical period. MapBiomas defines deforestation as clear-cutting, which is characterized by the complete removal of forest cover and areas converted to other anthropogenic uses (agriculture, pasture, mining, etc.).

Other Natural Formations: Areas of natural formation that, when vegetated, present a different appearance and physiognomy (Savannah, Scrubland, Natural Grassland, etc.).

Water Bodies: Water bodies, rivers, and streams of any kind.

Mining: Mining areas.

Urbanized Area: Urbanized areas.

As noted in VMD0007, we selected a 10-year period prior to the project start date to define the historical deforestation period, ranging from 2011 to 2020. Various GIS procedures were conducted to select and extract land use values over the years to generate a map containing information on accumulated deforestation and to quantify the forest that was converted into non-forest anthropogenic areas. The map illustrating deforestation during the historical period is shown in the figures below:

At the beginning of the historical period, in 2011, a total of 583,923.16 hectares were covered by forests, and 1,250.42 hectares were occupied by non-forest anthropogenic uses in the RRD. Figure 32 illustrates the reference map of forest cover for the RRD at the beginning of the historical period (2011), and Figure 35 shows it at the end of the historical period (2020).

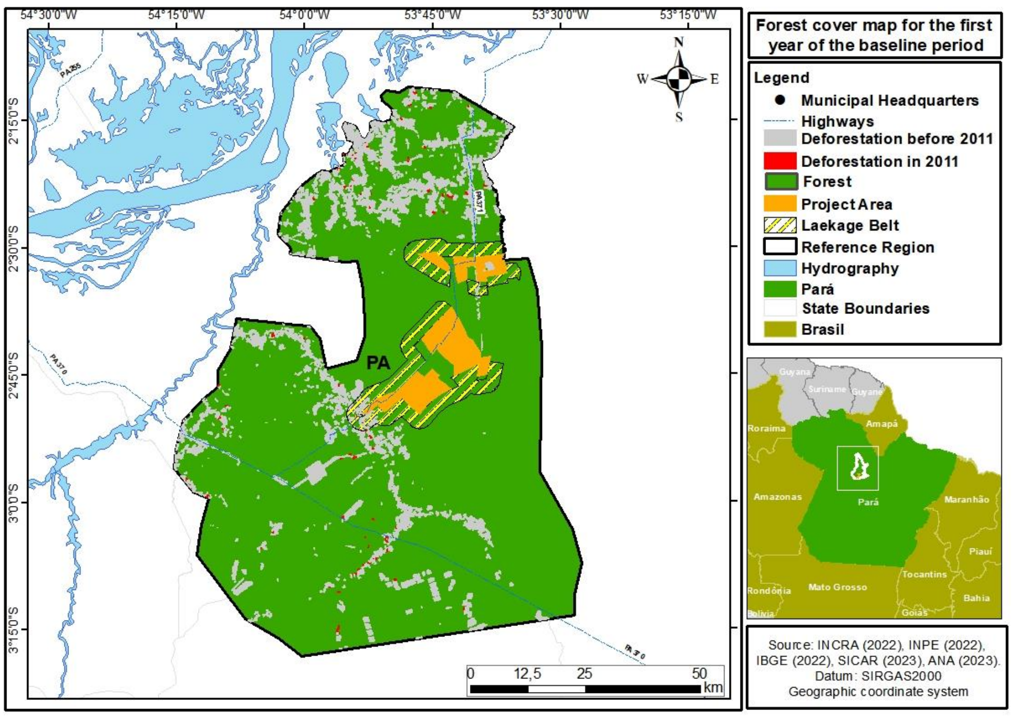


Figure 32. Reference map of forest cover for the first year of the baseline period in PA, LB, and RRD (2011).

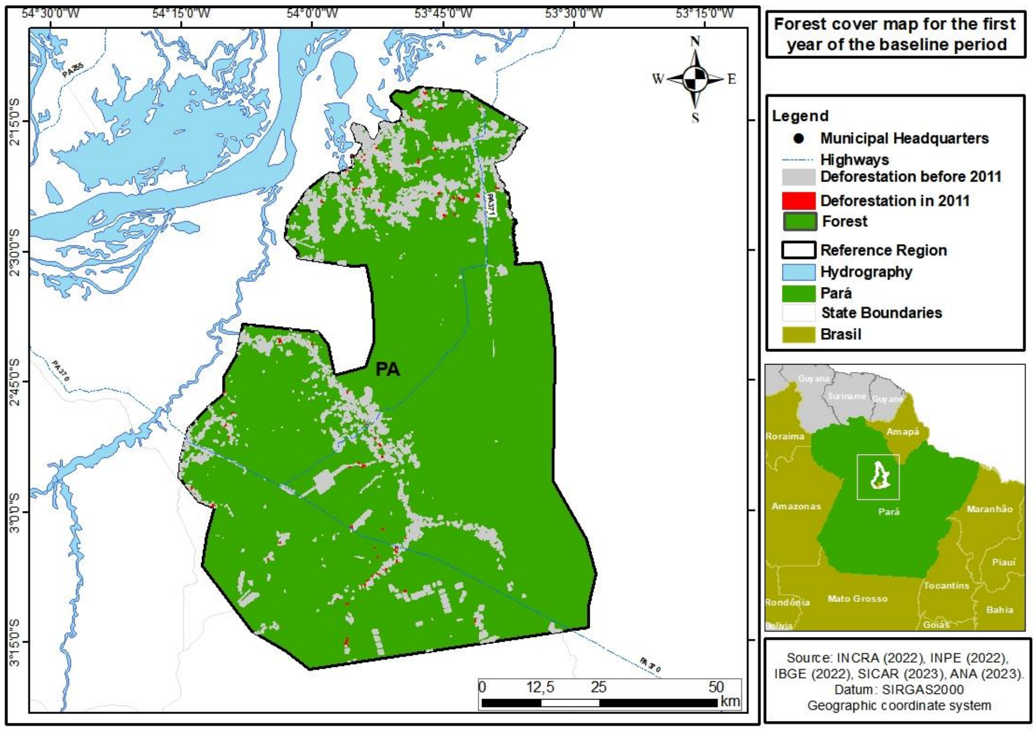


Figure 33. Reference map of forest cover for the first year of the baseline period in REDD (2011).

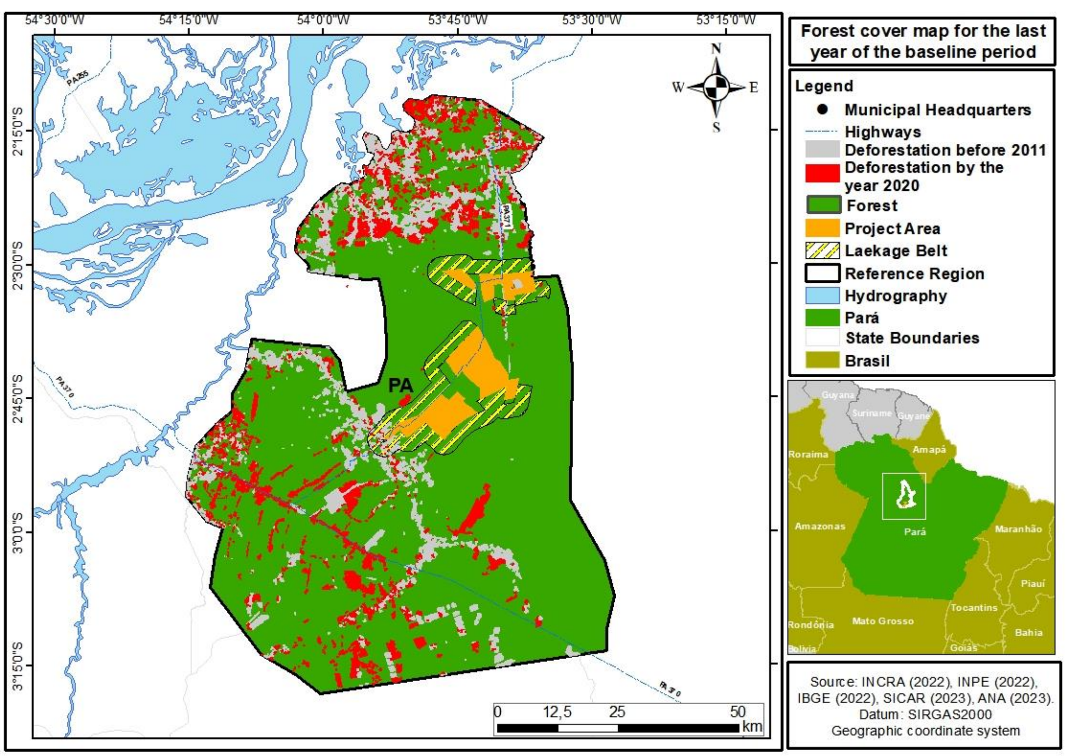


Figure 34. Reference map of forest cover for the last year of the baseline period in PA, LB, and REDD 2020).

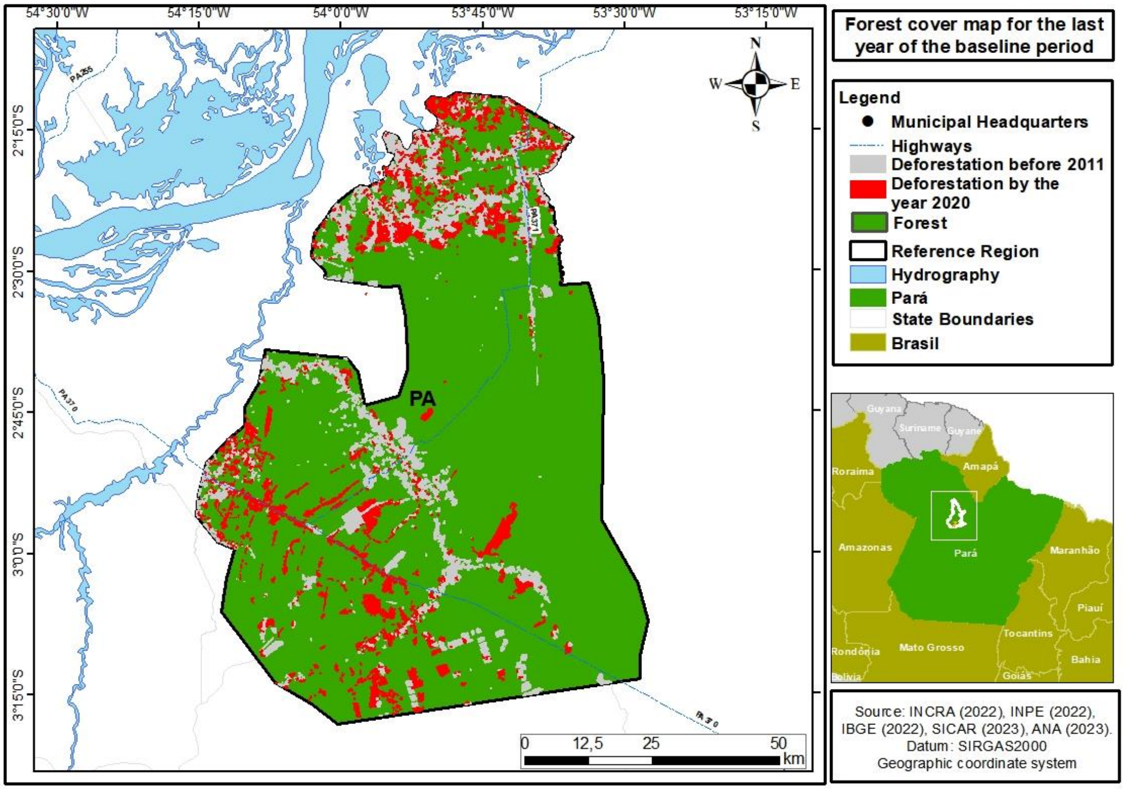


Figure 35. Reference map of forest cover for the last year of the baseline period in REDD (2020).

Estimate of Annual Areas of Unplanned Deforestation in the REDD Area

The methodology provides three approaches for calculating the deforestation area in the REDD area:

Historical average annual deforestation during the historical reference period

A linear regression of deforested area over time

A non-linear regression of deforested area over time

Since the linear relationship did not show a significant trend, the historical average for the historical period was adopted, equal to 0.60% (approximation 1 / Step 2.2 of VMD0007). Table X corresponds to the deforestation areas throughout the historical period, both for the REDD area.

The deforestation rate was derived from an analysis of the deforestation that occurred within the REDD area during the historical reference period, 2011 to 2020. Deforestation was analyzed during the historical reference period within the respective PA, LB, and REDD areas identified based on the following data:

PRODES: The PRODES (Satellite Deforestation Monitoring Project) monitors deforestation via satellite— a forestry practice where all or most trees in an area are uniformly cut down— in the Amazon and Cerrado since 2000.

DETER: This data is generated from a system that alerts to areas of degradation and deforestation starting in 2008 for the Amazon and from 2018 for the Cerrado. The mapping utilizes Landsat satellite images or similar to record and quantify the alert areas produced in the DETER project (Real-Time Deforestation Detection).

SAD: The SAD (Deforestation Alert System) is a monthly, automated monitoring system that issues alerts for the suppression of native vegetation throughout the Cerrado and Amazon biomes, starting in 2022. The system employs advanced image processing techniques from the Sentinel-2 satellite (10 m resolution) to detect, validate, and refine alerts.

MapBiomas Alert: This is a validation and refinement system for deforestation alerts (like SAD and DETER) using high-resolution images, where each alert is validated, refined, and defined within a time occurrence window. For each validated alert, a report is generated identifying before and after images of deforestation.

The annual deforestation estimates for the REDD area were obtained by calculating the amount of area (ha) deforested within the REDD boundaries from the deforestation layer from 2011 to 2020 for the Agroflorestal project. By summing the data described above, we ensured that there were no overlapping deforestation polygons, and the result was the amount of deforestation within the REDD area during the historical reference period year by year.

Table 35. Amount in hectares of year-by-year deforestation during the historical period (2011-2020).

Year	Accumulated Deforestation (ha)	Annual Deforestation (ha)	Average annual rate (%)
2.011	3.898,66	1.250,42	0,21
2.012	2.083,59	833,16	0,14
2.013	2.730,04	1.896,87	0,33
2.014	2.263,37	366,49	0,06
2.015	3.343,23	2.976,74	0,51
2.016	6.570,51	3.593,77	0,62
2.017	7.382,46	3.788,69	0,65
2.018	6.154,48	2.365,80	0,41
2.019	11.791,83	9.426,03	1,62
2.020	17.577,77	8.151,73	1,40

Average	6.379,59	3.464,97	0,60
Total	-	34.649,71	-

Identification of deforestation drivers

Recent analyses suggest that the Amazon region may experience a drastic loss of nearly 50% of its forest cover within the next five years[10]. This alarming forecast highlights the critical need to tackle various degradation factors that will remain significant contributors to carbon emissions until 2050, irrespective of deforestation patterns.

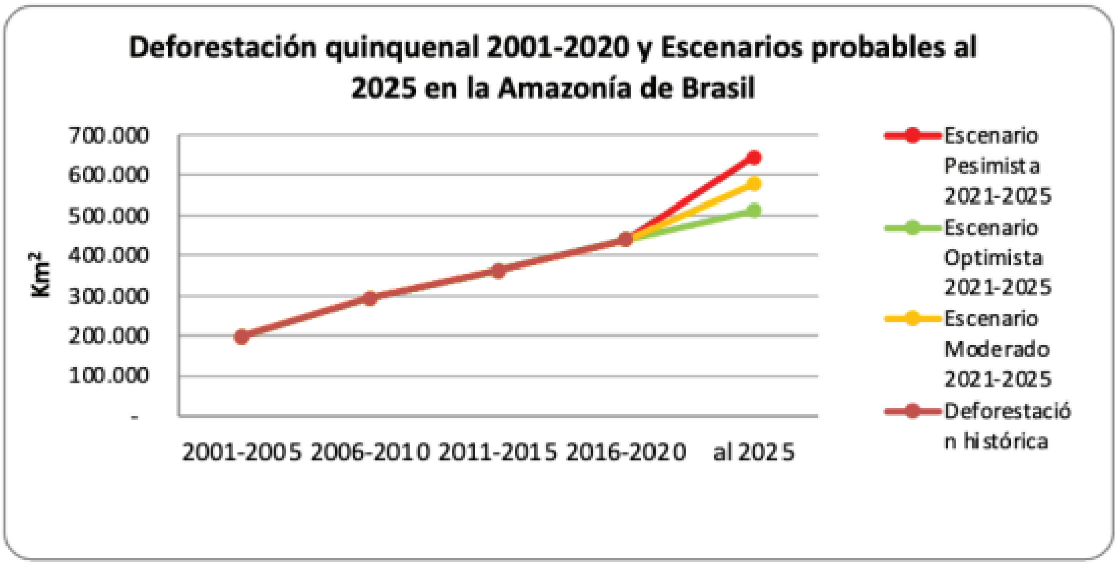


Figure 36. Deforestation Between 2001-2020 and Probable Scenarios for the Brazilian Amazon

Agricultural practices are pivotal in exacerbating deforestation and forest degradation, particularly through their contribution to edge effects in fragmented ecosystems and the use of slash-and-burn techniques. Research has pinpointed four primary drivers of forest degradation: forest fires, edge effects, illegal logging, and severe droughts, which have collectively impacted over 38% of the Amazon's remaining forested areas[11].

Additionally, a 2020 report indicated that 66% of the Amazon is facing significant threats from both deforestation and degradation, with agricultural activities accounting for 84% of the destruction observed in the early 21st century. The expansion of agricultural lands has surged by 50%, now covering 122 million hectares, which constitutes three-quarters of the deforestation in the region.

[10] https://infoamazonia.org/2023/03/21/desmatamento-na-amazonia-passado-presente-e-futuro/

[11] The drivers and impacts of Amazon forest degradation https://www.science.org/doi/10.1126/science.abp8622

The "Floresta em Risco" platform[12] analyzes deforestation in the Brazilian Amazon as a result of the interplay between the vulnerability of native vegetation areas, environmental governance, and macroeconomic factors. Economic variables, particularly agricultural commodity prices and real exchange rates, have a direct impact on the profitability of extractive activities, which in turn influences the demand for marginal lands and leads to deforestation—often facilitated by illegal practices such as land grabbing and mining. Fluctuations in these economic indicators are utilized to estimate expected deforestation levels, identify areas at higher risk of being cleared, and assess the potential for leakage, where deforestation is not entirely eradicated but rather "transferred" to different regions.

Figure 37 illustrates that the municipality of Prainha, which hosts the Agroflorestal Novo Horizonte project, is predominantly located within a region of high deforestation probability.

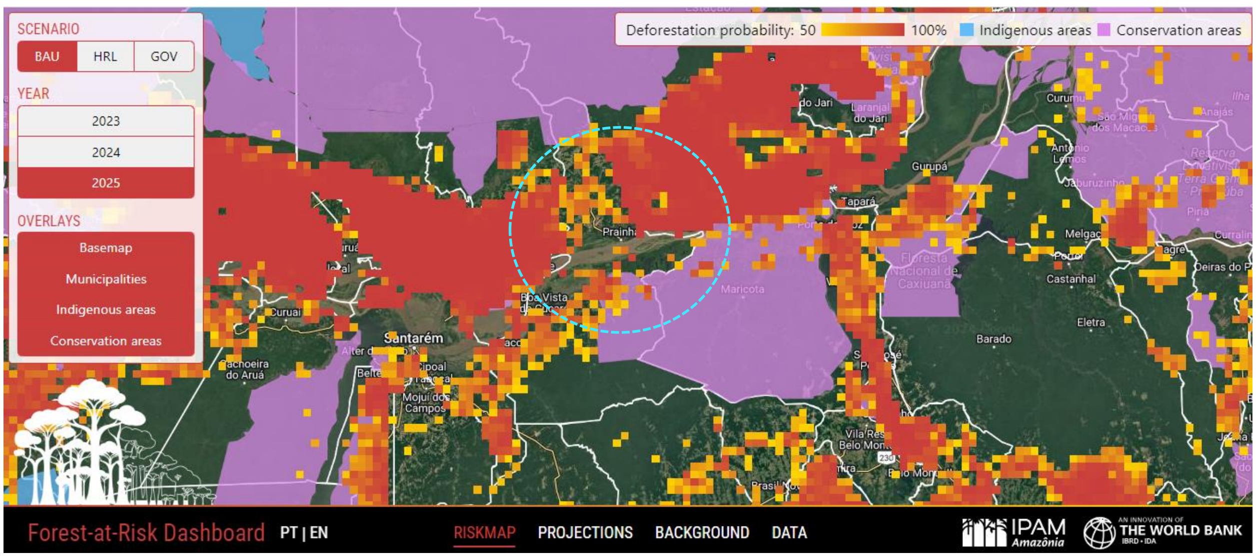


Figure 37. Municipality of Prainha on the 'Forest at Risk' Platform.

The situation in Prainha presented on the platform supports the information outlined in item 1.14, "Condition Prior to Project Initiation."

The socio-economic context of Prainha complicates efforts to mitigate deforestation. The local economy's heavy reliance on land-intensive activities underscores the need for sustainable development strategies that can balance economic growth with environmental conservation. Moreover, Prainha's primary productive activities, such as artisanal fishing, cattle ranching, and cassava cultivation, are supported by local government initiatives aimed at improving the livelihoods of rural workers. However,

[12] https://forestatrisk.ipam.org.br/#map

these activities, particularly cattle ranching, contribute to the ongoing deforestation and environmental degradation in the region.

MapBiomas data[13] indicates that in the municipality of Prainha, since the start of the reference period in 2011, there has been a 62% increase in the area allocated for pasture, rising from approximately 102,000 hectares to 164,000 hectares. Similarly, the area planted with soybeans more than doubled. In 2011, there were only 1.5 hectares planted, which increased to 1,930 hectares by 2023. This expansion occurred primarily over the last six years, coinciding with the financial incentives provided by the previous federal government.

The Agroflorestal Novo Horizonte property is intersected by PA 371, a road that plays a significant role in the process of deforestation. Numerous studies[14,15,16] show that roads are a major driver of deforestation, as they connect previously isolated areas, facilitate the opening of new trails, and promote the expansion of economic activities that can lead to forest degradation. These infrastructures not only increase access to remote regions but also encourage intensive land use and the conversion of forested areas into agricultural and pasture lands.

The available evidence regarding projected future deforestation trends in the Reference Region and Project Area is considered “Sufficient to demonstrate the baseline scenarios.” This indicates that the proposed relationships among agent groups, driving factors, underlying causes, and historical deforestation rates have been substantiated through comprehensive literature reviews and other credible local information sources.

These proposed relationships can be verified through statistical analyses, literature assessments, or other trustworthy sources, including documented insights from local experts, community members, deforestation agents, and other knowledgeable stakeholders familiar with the project area and reference region.

3.5 Additionality

O Projeto REDD+ Agroflorestal Novo Horizonte conduziu uma análise de adicionalidade abrangente, em conformidade com a metodologia aplicada e as ferramentas relevantes.

3.5.1 Regulatory Surplus

Is the project registered or seeking registration in an UNFCCC Annex 1 or Non-Annex 1 country?

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[13] MapBiomas_Link for Prainha

[14] Lima, M., et al. (2021). "Roads, deforestation, and the role of protected areas in the Amazon rainforest." *Environmental Research Letters*, 16(6), 064026. doi:10.1088/1748-9326/ac0741.

[15] Soares-Filho, B.S., et al. (2020). "A global map of road networks and their impact on forest loss." *Nature Sustainability*, 3, 572-581. doi:10.1038/s41893-020-0520-2.

[16] Rufino, M.C., et al. (2022). "The influence of road construction on land-use change and deforestation in tropical forests: A review." *Journal of Environmental Management*, 305, 114375. doi:10.1016/j.jenvman.2021.114375.

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☒ Yes ☐ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes ☒ No

The project includes several non-mandatory activities. However, the main activity of the project is addressing climate change by reducing emissions from deforestation and degradation. This activity encompasses continuous monitoring, and strategies related to economic incentives for local communities and other farmers to combat deforestation and loss of vegetation cover. Considering the deforestation combat activity, it is possible to state that, although Brazil has a robust framework of laws, statutes, and regulations, the implementation of these laws is far from effective.

This situation is historical, but the trend has been exacerbated in the last 10 years by the weakening of public command and control agencies and the encouragement of agricultural practices such as livestock farming and grain production. Additionally, the country has been targeted by orchestrated fires that affect not only native vegetation but also biodiversity and local populations.

The following table presents the legislations related to deforestation and fire use at the federal, state, and municipal levels:

Table 36. legislations related to deforestation and fire use at the federal, state, and municipal levels

Legal Framework	Description	Enforcement Agencies
Forest Code, Law No. 12.651/2012	The Forest Code (Law No. 12.651/2012) regulates deforestation and the use of fire in Brazil. It restricts deforestation in Permanent Preservation Areas (APPs) to protect water resources,	Brazilian Institute of Environment and Renewable Natural Resources (IBAMA) State and Municipal Environmental Secretariats

	biodiversity, and ecological processes (Art. 4), and limits deforestation within Legal Reserves to ensure environmental protection, allowing it only under legal exceptions with specific authorization and compensation (Art. 12). Additionally, property owners must restore illegally deforested areas (Art. 29). The use of fire is generally prohibited to prevent vegetation degradation, except for authorized activities such as agriculture and controlled forest management, which must comply with regulations to avoid uncontrolled spread (Art. 38 and Art. 39).	
Environmental Crimes Law (Law No. 9,605/1998)	The Environmental Crimes Law (Law No. 9,605/1998) addresses illegal deforestation and the improper use of fire in Brazil. It establishes penalties for deforestation carried out without proper authorization and for the use of fire that causes environmental degradation (Art. 38). Additionally, it mandates the enforcement of environmental regulations through monitoring and the application of sanctions (Art. 54).	The Brazilian Institute of Environment and Renewable Natural Resources (IBAMA), the Federal Police, and the Federal Public Prosecutor's Office (MPF) are responsible for implementing and controlling these regulations.

National Integrated Fire Management Policy (Law No. 14,944/2024)	Regulates the use of fire in Brazil, with a particular focus on sensitive areas such as the Amazon. It prohibits unauthorized fire use to prevent environmental damage (Art. 1), requiring specific authorization for any fire-related activities. The law enforcers strict control measures and continuous monitoring to prevent illegal fires (Art. 2). Penalties for non-compliance include significant fines, criminal charges, and mandatory compensations for environmental damage (Art. 3). The law aims to enhance fire management practices and protect Brazil's critical ecosystems from the adverse effects of uncontrolled fires.	Brazilian Institute of Environment and Renewable Natural Resources (IBAMA) State and Municipal Environmental Secretariats Civil Defense Agencies
Para State Law nº 7.813/2018	The Environmental Code of Pará (State law nº 7.813/2018) regulates deforestation and the use of fire in the state of Pará. It defines Permanent Preservation Areas (APPs) and restricts deforestation in these areas to protect water resources, biodiversity, and ecological processes (Art. 50). It also establishes conditions for the suppression of native vegetation, requiring prior authorization and environmental	State Secretariat for Environment and Sustainability of Pará (SEMAS) Brazilian Institute of Environment and Renewable Natural Resources (IBAMA)

	compensation (Art. 51). Additionally, it imposes penalties for illegal deforestation, including fines and other administrative sanctions (Art. 52). The use of fire is regulated to prevent vegetation degradation, requiring prior authorization and compliance with criteria for controlled burning (Art. 60). Penalties for the misuse of fire include fines and other administrative sanctions (Art. 61).	
Para State Decree No. 1.949/2018	The State Decree No. 1.949/2018 establishes guidelines for combating deforestation in the State of Pará. It defines priority areas for the inspection and combat of illegal deforestation (Art. 3), and establishes penalties for illegal deforestation, including fines and other administrative sanctions (Art. 5). Additionally, it mandates the recovery of illegally deforested areas with environmental recovery plans (Art. 7). The use of fire is regulated to prevent vegetation degradation, requiring prior authorization and compliance with criteria for	State Secretariat for Environment and Sustainability of Pará (SEMAS) Brazilian Institute of Environment and Renewable Natural Resources (IBAMA)

	controlled burning (Art. 10). Penalties for the misuse of fire include fines and other administrative sanctions (Art. 12).	

Despite the establishment of a comprehensive legal framework aimed at combating illegal deforestation and the unauthorized use of fire, agencies tasked with enforcing these regulations face a range of significant obstacles that undermine their effectiveness. This can be confirmed by an in-depth analysis of these challenges, drawing on research and case studies to illustrate the multifaceted nature of enforcement difficulties.

The enforcement of environmental laws in the Amazon faces numerous and complex challenges. Insufficient resources, logistical difficulties, political instability, corruption, economic pressures, and lack of coordination all contribute to the ineffectiveness of current enforcement efforts. Addressing these issues requires a multifaceted approach that includes increased funding, improved coordination, enhanced political commitment, and strengthened anti-corruption measures. Only through such comprehensive efforts can the effectiveness of environmental law enforcement in the Amazon be significantly improved.[17][18]

One of the primary impediments to effective environmental law enforcement is the scarcity of resources. As identified by Brito, B (2011)[19], Brazilian environmental agencies such as IBAMA, along with state and municipal secretariats, frequently operate under stringent budgetary constraints. These financial limitations severely curtail their ability to conduct comprehensive monitoring and sustain continuous operational activities. Furthermore, the shortage of qualified and trained personnel exacerbates these challenges. A lack of skilled staff to monitor vast and diverse areas further diminishes the capacity to address illegal activities effectively.

The vast and often remote nature of Brazil's forested regions presents considerable logistical and operational challenges. The extensive coverage of these areas complicates enforcement efforts, making regular monitoring and intervention exceedingly difficult. The inadequacy of infrastructure in these

[17] Crime sem castigo: a efetividade da fiscalização ambiental para o controle do desmatamento ilegal na Amazônia./ Jair Schmitt Brasília, 2015. 188 p. : il.
[18] https://igarape.org.br/wp-content/uploads/2022/04/The-ecosystem-of-environmental-crime-in-the-Amazon.pdf
[19] Brito, Brenda & Barreto, Paulo. (2011). Enforcement Against Illegal Logging in the Brazilian Amazon. 10.4337/9781781000946.00032.

isolated regions further impedes field operations and the collection of crucial evidence needed for law enforcement[20][21].

Political factors also play a significant role in shaping the effectiveness of environmental law enforcement. Changes in government administration and shifts in policy priorities can lead to inconsistent application of environmental laws and fluctuating resource allocation. Political instability impacts the continuity of enforcement efforts[22][23]. Additionally, when political priorities shift towards economic development at the expense of environmental protection, the enforcement of environmental regulations can be compromised.

Corruption at various levels of government represents a major obstacle to effective enforcement. Corruption can undermine enforcement efforts by allowing illegal activities to persist unchecked[24]. Moreover, a lack of genuine political will to enforce environmental regulations can result in inconsistent application of laws, further weakening enforcement[25,26].

Economic pressures and conflicting interests also pose significant challenges to environmental law enforcement. Activities such as agriculture and livestock farming frequently conflict with environmental regulations. These economic interests can drive efforts to circumvent or disregard environmental laws. Additionally, powerful economic interest groups may exert influence to weaken enforcement or avoid rigorous application of regulations.

Finally, the lack of effective coordination and integration between various levels of government and different agencies contributes to enforcement challenges[27]. Poor coordination results in enforcement gaps and ineffective implementation of laws. The absence of integrated monitoring and information-sharing systems further complicates the detection and response to illegal activities.

3.5.2 Additionality Methods

[20] GANDOUR, Clarissa; ASSUNÇÃO, Juliano. Policy Brief. Brazil Knows What To Do To Fight Deforestation in the Amazon: Monitoring and Law Enforcement Work and Must be Strengthened. Rio de Janeiro: Climate Policy Initiative, 2019

[21] de Freitas, V. (2023) The Challenges of Environmental Protection in the Brazilian Amazon. *Beijing Law Review*, **14**, 324-340.

[22] https://www.oc.eco.br/en/brazilian-agency-only-spent-41-of-environmental-inspection-budget/

[23] Nunes, F.S.M., Soares-Filho, B.S., Oliveira, A.R. *et al.* Lessons from the historical dynamics of environmental law enforcement in the Brazilian Amazon. *Sci Rep* **14**, 1828 (2024). https://doi.org/10.1038/s41598-024-52180-7

[24] Governança Fundiária Frágil, Fraude e Corrupção: Um terreno fértil para a grilagem de terras. Transparência Internacional, 2021. Link to access

[25] Barbosa, L. G., Alves, M. A. S., & Grelle, C. E. V. (2021). Actions against sustainability: Dismantling of the environmental policies in Brazil. Land Use Policy, 104, 105384.

[26] https://news.mongabay.com/2019/06/brazil-guts-environmental-agencies-clears-way-for-unchecked-deforestation/

[27] Benatti, J.H., da Cunha Fischer, L.R. New trends in land tenure and environmental regularisation laws in the Brazilian Amazon. Reg Environ Change 18, 11–19 (2018). https://doi.org/10.1007/s10113-017-1162-0

Additionality was analyzed following the methodological tool VT0001, version 3.0, for the demonstration and assessment of additionality in VCS Agriculture, Forestry and Other Land use (AFOLU) project activities. The following steps were developed to demonstrate additionality:

Step 1. Identification of alternative land-use scenarios to the proposed VCS AFOLU project Activity

Step 2. Investment analysis

Step 3. Barrier analysis; and

Step 4. Common practice analysis

a) Step 1. Identification of alternative land use scenarios to the proposed VCS AFOLU project activity

The implementation of a REDD+ project (Reducing Emissions from Deforestation and Forest Degradation) requires demonstrating additionality, meaning that the proposed activities would result in additional conservation benefits that would not occur in the absence of the project. To achieve this, it is essential to identify and analyze realistic alternative land use scenarios that could occur within the project boundaries without the planned intervention. Below are three plausible alternative scenarios grounded in the socioeconomic, environmental, and legal context of Prainha, Pará.

Sub-Step 1a: Identify credible alternative land-use scenarios to the proposed VCS

<u>Scenario 1: Continuation of the Sustainable Forest Management (Business as Usual) until 2025.</u>

In this scenario, Agroflorestal Novo Horizonte would continue with its FSC-certified Sustainable Forest Management (SFM) without implementing the REDD AUD project activities. The observed deforestation patterns, driven by the interaction of local agents, drivers, and underlying causes over time, are expected to remain key contributors to future deforestation in the region. Moreover, the financial viability of SFM is further undermined by the presence of an illegal timber market, which operates at significantly lower costs due to non-compliance with regulations. This market makes illegal logging more financially attractive to some landowners, reducing the incentive to follow sustainable practices.

Even with the implementation of sustainable management practices, the project area remains vulnerable to external pressures such as invasions, illegal resource extraction, and unauthorized agricultural expansion. This vulnerability is particularly pronounced in regions where government enforcement is weak or inconsistent. The lack of formal registration as a VCS AFOLU project exacerbates this issue, as it limits the additional support and resources typically provided to enhance monitoring and enforcement efforts. Furthermore, Agroflorestal Novo Horizonte would face limitations in investing in local community development programs. These programs, which are a critical component of REDD+ projects, provide sustainable economic alternatives that help reduce pressures on forests. Without these resources, the climate benefits are not fully recognized or accounted for in global climate change mitigation efforts.

It is important to note that Agroflorestal Novo Horizonte's sustainable management areas are set to conclude in 2026, after which they will enter a period of fallow until 2050, making it even more essential for the carbon project to be established. Without sustained land use and economic activity, it will become increasingly difficult to maintain the social, environmental, and monitoring work carried out by Agroflorestal. The project area will become more vulnerable to illegal deforestation and forest conversion, primarily driven by the expansion of agriculture, particularly cattle ranching and soybean cultivation.

This trend is already evident in the region. According to MapBiomas data, between 2011 and 2023, the area allocated to pastures in Prainha increased by 62%, from approximately 102,000 hectares to 164,000 hectares. This substantial growth reflects the regional trend of expanding cattle ranching as the dominant economic activity, often linked to illegal deforestation and environmental degradation. During the same period, soybean cultivation in Prainha saw a dramatic rise, with the planted area expanding from a mere 1.5 hectares in 2011 to 1,930 hectares in 2023. This rapid increase, especially pronounced over the past six years, correlates with previous government financial incentives aimed at promoting extensive agriculture in the region. The expansion of agriculture is further facilitated by the PA-371 state highway, which crosses the property in question. The highway plays a critical role in providing access and enabling the transportation of agricultural products, thereby contributing to the expansion of deforestation. Numerous studies have demonstrated that roads are one of the main drivers of deforestation in the Amazon, as they open up previously isolated areas to occupation and illegal resource extraction. The accessibility provided by infrastructure such as PA-371 accelerates the pressure on forested lands, leading to more extensive and disorganized land use changes.

Despite the existence of robust environmental legislation, such as the Brazilian Forest Code (Law No. 12,651/2012), which sets parameters for forest conservation and sustainable land use, enforcement remains a significant challenge. The effectiveness of these laws is undermined by insufficient enforcement and widespread impunity in cases of illegal deforestation. As a result, these destructive practices persist and even intensify. Data from the Brazilian Institute of Environment and Renewable Natural Resources (IBAMA) underscores this enforcement gap, revealing that between 2019 and 2022, only 10.2% of illegally deforested areas were subject to legal action or effective embargoes. This highlights the considerable difficulty in curbing deforestation, even with existing legislation.

The economic pressures driving deforestation are compounded by the pursuit of quick financial returns through agricultural and cattle ranching expansion. The lack of viable sustainable economic alternatives in the region makes extensive agricultural practices particularly attractive, further entrenching the likelihood of continued deforestation. In the absence of effective conservation interventions, such as the REDD+ project, this scenario suggests an ongoing and potentially escalating trend of environmental degradation in Prainha. Without the REDD+ project, the continuation and likely intensification of deforestation in the project area would result in substantial forest cover loss, a decline in biodiversity, and significant greenhouse gas emissions. This underscores the urgent need for additional interventions to alter the current trajectory of environmental degradation in the region.

Scenario 2: Discontinuation of Sustainable Management and Risk of Land Conversion or Sale

In this scenario, the landowner imediateally ceases forest management activities and either opens the remaining 20% of the area for conversion or decides to sell the property. Under the Brazilian Forest Code (Law No. 12,651/2012), properties in the Amazon must keep 80% of their area as a Legal Reserve, preserving native vegetation and natural resources. However, if the landowner imediatlily abandons forest management, this Legal Reserve remains unchanged, but the remaining 20% could be subject to conversion or sale. The conversion of this land is driven by economic incentives, such as the expansion of agriculture, which has been a major driver of deforestation in the Amazon[28].

Alternatively, selling the property can also lead to increased deforestation. New owners might convert the land to more profitable uses, such as cattle ranching or soybean cultivation, which are known to cause extensive deforestation and environmental degradation. The presence of roads and infrastructure can further facilitate the expansion of such activities, exacerbating the impact on the forest.

The abandonment of sustainable forest management without pursuing other conservation measures will lead to a loss of ecological integrity in the area. The lack of active monitoring and protection increases the risk of illegal logging and land invasions, further accelerating forest degradation. Moreover, without a formal conservation strategy or sustainable development programs, the local communities will lose potential benefits from sustainable land use practices and will continue to face economic pressures that may lead to environmentally harmful practices.

Thus, the cessation of forest management activities, combined with the potential opening of the remaining 20% of the area or the sale of the property, poses a significant threat to forest conservation efforts. The resulting deforestation and land conversion will likely have severe consequences for biodiversity, carbon emissions, and the overall environmental health of the region.

Scenario 3: Maintenance of Project Activities Without Supplementary Income by REDD registration

[28] Soares-Filho, B., et al. (2014). Cracking Brazil's Forest Code. *Science*, 344(6182), 363-364.

In this scenario, the landowners are committed to maintaining the preserved forest area and implementing sustainable activities within the project boundary. However, these efforts are undertaken without any supplementary income from external sources, such as carbon credits or financial incentives. The landowners focus their monitoring and surveillance measures primarily on the Legal Reserve, which constitutes 80% of the property, and also on preventing potential land invasions by external agents.

Dedicated employees would be be assigned to patrol and monitor the area, equipped with essential resources such as food, water, motorcycles, and mobile phones to support their activities. This ensures that the monitoring teams are well-prepared to conduct regular patrols and respond promptly to any incidents that may arise. Their presence is crucial for the ongoing protection of the forest and the enforcement of conservation measures.

In addition to physical patrols, satellite-based monitoring will be employed to detect fire alarms and other potential threats, as indicated by MapBiomas Alert data. This geospatial analysis enables early detection and tracking of fires, allowing for timely responses and effective mitigation. By identifying patterns of deforestation or regions with higher risk near the properties, the analysis provides valuable information for developing targeted strategies to prevent and address potential threats to the forest. To further mitigate risks, the project proponent will collaborate with the other landowners and local stakeholders in a cooperative effort to address potential carbon leakage. This approach aims to promote innovative forest and land use practices in the region, reducing the risk of carbon loss due to activities outside the project area. The involvement of local stakeholders is key to fostering sustainable practices and enhancing the overall effectiveness of conservation efforts. The actions are presented on item 1.12.

A notable characteristic of this scenario is that all conservation activities are fully financed by the landowners. Without additional revenue sources, such as carbon credits, the financial burden of maintaining and monitoring the preserved area falls solely on them. This financial constraint could potentially limit the scope and effectiveness of conservation efforts, as the landowners are responsible for covering all costs related to monitoring, patrolling, and equipping their teams.

Despite the absence of supplementary income, the landowners' commitment to maintaining the ecological integrity of the preserved area is evident through their implementation of robust monitoring and surveillance measures. The combination of physical patrols, satellite monitoring, and leakage control reflects a comprehensive approach to forest conservation, demonstrating their dedication to safeguarding the environment. However, the lack of additional revenue may pose challenges to the project's long-term sustainability. While the measures are effective in the short term, ongoing financial support is essential for sustaining conservation activities and addressing emerging challenges. Without supplementary income, the landowners may struggle to maintain the necessary level of effort and resources required for continued success.

In conclusion, while the maintenance of the preserved forest area and the implementation of sustainable activities underscore a strong commitment to conservation, the reliance on self-financing presents significant challenges. The absence of external financial support could limit the project's long-term effectiveness and sustainability. The financial burden on the landowners highlights the need for securing additional revenue streams to support and enhance conservation efforts over time.

Sub-step 1b. Consistency of credible land use scenarios with enforced mandatory applicable laws and regulations

Scenario 1: In this scenario, Agroflorestal Novo Horizonte would continue with its FSC-certified Sustainable Forest Management (SFM) without implementing the REDD AUD project activities. Law enforcement is a critical issue that can undermine the effectiveness of the scenario.

Scenario 2: The discontinuation of sustainable management and the possibility of 20% land conversion is allowed from the perspective of the Brazilian Forest Code, which requires maintaining 80% of the area as Legal Reserve. However, the conversion of the 20% area or the sale of the property may lead to deforestation and habitat loss, negatively impacting biodiversity and carbon stocks. The decision to stop sustainable management could result in significant deforestation, especially if the land is sold to new owners who may adopt harmful practices. The lack of monitoring and active protection accelerates forest degradation. Discontinuing sustainable management and potential sale of the land represent a high-risk scenario for conservation and sustainability, increasing the likelihood of deforestation and environmental degradation. Land conversion and sale may be inconsistent with environmental legislation, but non-compliance is common due to ineffective enforcement. Evidence shows that land conversion and sales frequently occur in areas with economic pressures.

Scenario 3: Maintaining sustainable activities without additional financial support is legally acceptable but may be financially unsustainable. Maintaining and monitoring the Legal Reserve, as well as active surveillance, are important, but the lack of external funding may limit long-term effectiveness and sustainability. Despite a commitment to conservation, the lack of additional revenue may restrict the landowners' ability to effectively maintain monitoring and protection efforts. Dependence on self-funding may be insufficient to address ongoing and emerging conservation challenges. Scenario 4 is viable in the short term, but the absence of external financial support may limit the long-term sustainability and effectiveness of conservation activities.

<u>Outcome of Sub-step 1b:</u>

Scenario 1 and Scenario 3 are plausible and in compliance with mandatory regulations, though both face significant challenges related to enforcement and funding.

Scenario 2 are plausible but faces substantial issues with legal non-compliance and insufficient enforcement, making them less likely under the current regional context.

Conclusion: The credible scenarios for the REDD+ project that comply with mandatory laws and regulations are Scenario 1 and Scenario 3, although both present challenges related to enforcement and funding. Scenarios 2, while realistic, faces substantial issues with legal non-compliance and insufficient enforcement. Therefore, the proposed REDD+ project is additional because the alternative scenarios do not provide the same conservation and emission reduction benefits as the project. The analysis of gaps and difficulties in the enforcement of environmental laws in Brazil has been presented previously.

Sub-step 1c. Selection of the baseline scenario:

<u>Scenario 1: Continuation of the Sustainable Forest Management without project activities (Business as usual)</u>

FSC-certified sustainable forest management is a legally acceptable and realistic land use scenario. FSC certification ensures that forest management complies with rigorous environmental standards and is a viable alternative to land conversion for other uses. This approach is consistent with established practices in the region and current environmental policies. Despite its legal compliance, Scenario 1 faces significant challenges related to law enforcement and a lack of financial incentives.

Competition with the illegal market and difficulties in enforcement can undermine the effectiveness of anti-deforestation efforts. The lack of financial resources is a critical obstacle that limits the ability to maintain effective forest management practices. The challenge of ensuring ongoing monitoring and control of illegal deforestation and fires due to insufficient resources makes Scenario 1 less robust in the long term.

The expansion of cattle ranching and soybean activities near the project area increases pressure on forested areas, making the maintenance of monitoring and socio-environmental activities even more challenging. Agroflorestal Novo Horizonte will need to secure additional funding to ensure the sustainability of conservation efforts. It is important to note that the areas dedicated to sustainable forest management will conclude in 2026, after which they will enter a period of fallow until 2050. Therefore, additional financial resources will be crucial to ensure the continuity of conservation and the effectiveness of long-term monitoring and control activities.

Step 2. Investment analysis

According to the VT0001 tool, the objective of this step is to "assess whether the proposed project activity, in the absence of revenue from the sale of carbon credits, is economically or financially less attractive compared to at least one alternative land use scenario." To achieve this, an investment analysis was conducted.

Sub-step 2a. Determine appropriate analysis method

The Agroflorestal REDD+ AUD Project benefits financially from both the sale of VCUs and profits from timber harvesting through its sustainable forest management plan. To verify the project's additionality, an investment comparison (Option II) will be performed to determine if, without the revenue from GHG credits, the project activities would be less financially or economically favorable than other potential land use options.

Sub-step 2b. – Option II. Apply investment comparison analysis

Investment analysis can be conducted using indicators such as Net Present Value (NPV) and Internal Rate of Return (IRR). NPV measures the difference between the present value of future cash flows and the initial investment cost, making it ideal for decisions aimed at maximizing absolute value, especially in long-term scenarios or when considering the cost of capital. In contrast, IRR calculates the annual expected rate of return, facilitating direct comparisons between projects and being useful in preliminary evaluations where simplicity is needed. NPV is preferred for maximizing value creation, while IRR is advantageous for comparing the profitability of different projects. Thus, for comparing the scenarios, IRR will be used.

Sub-step 2c. – Calculation and comparison of financial indicators

An investment analysis was conducted to compare scenarios 1 e 3, as these represent the most plausible legal activities for the project proponent.

In Scenario 1, the focus is on Sustainable Forest Management (SFM) practices, considering the permissible annual exploitation yield until the project's end in 2025. It also assumes that SFM will resume in 2050 after a 25-year pause. During this paused period (2025-2050), the scenario accounts for maintenance costs without any revenue generation. Importantly, no project activities, such as carbon offset initiatives, are factored into this scenario.

In contrast, Scenario 3 follows the same assumptions regarding SFM but includes project activities that have been implemented without registration with VERRA.

Additionally, alongside these two scenarios, we present a comparison of investment parameters considering only the implementation of the carbon project, excluding any SFM activities.

Table 37 provides a comparative financial analysis, showing the associated maintenance costs, internal rate of return (IRR), net present value (NPV), and cost-benefit ratio over the 40-year project lifespan. All data used in this analysis was supplied by the farm owners and will be available for auditor review.

Table 37. Comparative financial analysis

Scenarios / Parameters	Scenario 1 - Forest Management - No Project Activities	Scenario 3 - Forest Management with project activities but without REDD AUD register on VERRA	Only 'Registered REDD AUD Project Scenario
Implementation Costs	$623.885,92	$695.186,12	$935.650,10
Total Costs	$27.550.800,67	$30.376.724,75	$2.896.572,08
Total Revenue from timber sale	$136.256.684,49	$136.256.684,49	$-
Total Revenue from Carbon Credits	$ -	$ -	$ 6.648.314,18
Forest conservation and social environmental activities	$ -	$ 2.896.572,08	$ 2.896.572,08
NPV	$43.784.808,96	$43.269.937,76	$2.123.159,27
IRR	24,44	21,85	0,14
Cost-Benefit Relation	6,7	6,1	2,8

The analysis shows that Scenario 3 has lower economic indicators compared to Scenario 1, which does not include project activities. The scenario with only registered project activities and no SFM presents significantly weaker economic performance than those that include SFM. This demonstrates that the costs associated with implementing and maintaining project activities are higher than in Scenario 1 (without the project), leading to reduced revenues and making Scenario 3 less financially viable. Moreover, implementing the proposed AFOLU project actions for registration proves to be even less attractive financially.

In conclusion, it is financially unfeasible to implement project activities aimed at protecting and monitoring forested areas, maintaining security, and ensuring the integrity of these areas over the 40-year project period, particularly during the 25-year pause in SFM. The profitability of cattle ranching, linked to the sale of illegally harvested timber, incentivizes land invasions by external actors due to its high financial returns, which in turn leads to illegal deforestation. However, in a scenario where carbon credits are sold, the financial risks for landowners decrease, thus encouraging the continuation of project actions, such as area monitoring, internal training, and socio-environmental and biodiversity initiatives.

Sub-step 2d: Sensitivity analysis

The sensitivity analysis is a crucial step in verifying the additionality of VCS AFOLU (Agriculture, Forestry, and Other Land Use) projects, especially when applying Option II. It evaluates the robustness of the project's financial attractiveness in response to reasonable variations in critical assumptions used in the investment analysis. For a project to be considered financially additional, the conclusion must consistently show that, across a realistic range of financial assumptions, the project would not be financially attractive without the benefits provided by the VCS.

In the context of Option II, sensitivity analysis is essential to determine whether the proposed project would remain financially unattractive in the absence of the VCS financial incentives. If, after this analysis, it is proven that the project would not be the most financially appealing option without these incentives, the project can move forward to the next evaluation step (Common Practice Analysis). Conversely, if the project is shown to be financially viable without the VCS benefits, it cannot be considered additional solely based on financial analysis.

The importance of this step lies in the fact that financial assumptions (such as costs, revenues, or return rates) can influence the project's economic viability. Sensitivity analysis ensures that the decision on additionality is not based on static or overly optimistic assumptions. In summary, this analysis enhances the integrity of the process by ensuring the project is deemed additional only if it truly relies on VCS financial incentives to be feasible.

As critical variables, a 10%, 20%, and 50% reduction in timber prices was analyzed to assess the impact on key financial metrics like Net Present Value (NPV), Internal Rate of Return (IRR), and the cost-benefit ratio.

Table 38. Sensitivity analysis

Scenário 1			
	NPL	IRR	Cost x Benefits Ratio
	43.784.808,96	24,44	6,73
-0,1	31.134.282,61	21,58	6,67
-0,2	30.082.895,28	18,73	5,93
-0,5	16.266.358,70	10,15	3,70
Scenário 3			
	NPL	IRR	Cost x Benefits Ratio
	$43.269.937,76	21,85	$6,15
-0,1	$38.105.193,32	19,28	$6,06

-0,2	R$ 33.039.129,90	16,72	$5,39
-0,5	$17.840.939,67	9,02	3,37

The results indicate that even with a 50% reduction in timber prices in both scenarios, key financial indicators such as Net Present Value (NPV) and Internal Rate of Return (IRR) remain more favorable compared to an area solely dedicated to a registered REDD AUD project on VERRA. However, it is important to emphasize that **Scenario 3**, which involves project activities without formal registration, demonstrates a weaker financial performance than **Scenario 1** – without any project activity. This comparison underscores the potential benefits of carbon credit revenue, which could support the improvement of forest monitoring, social and biodiversity programs, and more robust risk management measures. By relying on carbon credits, the financial viability of project activities may be significantly enhanced, allowing for better long-term sustainability and an expanded scope of environmental and community benefits.

Additionally, the NPV and IRR analyses do not fully account for the 25-year fallow period associated with sustainable forest management, during which no economic activities generate revenue. Despite the absence of income during this time, ongoing maintenance, protection, and monitoring efforts must continue, adding costs that are not reflected in the financial models.

According to the Brazilian Institute of Forests[29], the price of tropical timber, such as Ipê, experienced minimal negative variation between 2011 and 2021. The most notable decline was -16% in 2020, when the price fell to $172.00, likely due to the COVID-19 pandemic. The price subsequently rebounded in 2021 and surged significantly in 2022 (+66%) and 2023 (+38%). This trend indicates a relatively stable market for Ipê timber.

Table 39. Price of tropical timber

Year	Price ($)	Variation (%)
2013	159	-
2014	177	11,3

[29] https://www.ibflorestas.org.br/conteudo/novidade-boletim-preco-da-madeira-monitoramento-e-mercado

2015	153	-13,6
2016	172	12,4
2017	212	23,3
2018	212	0,0
2019	206	-2,8
2020	172	-16,5
2021	199	15,7
2022	330	65,8
2023	457	38,5

This historical price data suggests that, despite fluctuations, the profitability of timber exploitation remains robust. The significant price increases observed in 2022 and 2023 reflect strong demand, making timber exploitation potentially very lucrative.

However, this profitability also poses a risk of incentivizing illegal activities, particularly in regions with weak enforcement mechanisms and high demand for timber. The elevated prices create additional motivation for unauthorized exploitation, as the potential financial rewards outweigh the risks involved.

Therefore, it is essential to address both the economic incentives and the enforcement challenges when evaluating the viability and risks associated with timber exploitation. Implementing comprehensive monitoring and control measures is critical to curbing illegal activities and ensuring that timber extraction is conducted sustainably and in compliance with legal standards.

Step 3. Barrier analysis

Sub-step 3a. Identify barriers that would prevent the implementation of the type of proposed project activity

The barrier analysis aims to demonstrate that the proposed project faces significant obstacles that would prevent its implementation without the revenue from carbon credits. To this end, two main barriers have been identified and documented:

Investment Barriers

The sustainable forest management practiced in the Agroflorestal Novo Horizonte area will enter a 25-year fallow period starting in 2026. During this period, the area will experience a halt in economic activities and revenue, which increases vulnerability to deforestation agents. This interruption in economic activities creates a significant investment barrier for the implementation and maintenance of the project.

Furthermore, the project will require considerable investments in the application of technologies, hiring technical staff, acquiring equipment for patrolling and monitoring, as well as carrying out socio-environmental and biodiversity actions. These investments are significantly higher in the project scenario compared to the current situation, where revenue from economic activities helps cover operational and monitoring costs.

These high investment requirements represent a crucial barrier for implementing similar activities by private entities, as adequate financing for such projects is often scarce, especially during extended periods of economic inactivity. Therefore, the need for substantial investments in technology and infrastructure, coupled with the lack of revenue during the fallow period, highlights the significant financial barrier that must be overcome to ensure the project's viability and sustainability.

Institutional Barriers

Application of Legislation - Lack of enforcement of laws related to forests or land use.

As described in item 3.5.1 of this document, in Brazil, there is a significant barrier related to the enforcement of environmental legislation against illegal deforestation activities. In the case of the project, the risk is exacerbated by the highway that crosses the territory, which further facilitates the movement and access of illegal loggers.

Sub-step 3b. Show that the identified barriers would not prevent the implementation of at least one of the alternative land use scenarios (except the proposed project activity):

Considering the arguments related to high monitoring costs associated with deforestation and the low enforcement of legislation, which have been extensively discussed in this document, it is evident that these barriers do not impede the implementation of at least one alternative land use scenario. On the contrary, they contribute to illegal activities rather than restrict them.

The inadequate enforcement of Brazilian legislation does not prevent the project from occurring. The investment barrier, however, presents a significant challenge for the implementation of project activities, especially in the absence of REDD+ carbon project resources during the 25-year fallow period. Nevertheless, other land use scenarios, such as those involving continued illegal deforestation or alternative forms of agriculture, may not be as strongly affected by this investment barrier, as they do not require the same level of financial commitment and infrastructure.

Therefore, while the investment barrier poses a substantial obstacle to the proposed project activity, it does not entirely prevent the implementation of alternative land use scenarios. These alternative scenarios, although potentially less desirable from an environmental perspective, remain viable options under the current conditions.

Step 4. Common practice analysis

There are no carbon projects or initiatives similar to those proposed by Agroflorestal Novo Horizonte in the region. There are private companies conducting forest management activities near the project area, but due to high costs, socio-environmental and forest conservation actions with a high level of monitoring, as conducted by Agroflorestal, are rare in the region. Sustainable forest management activities in the region are still at risk of being replaced by more profitable but less sustainable activities, such as soybean cultivation, cattle ranching, or illegal timber extraction. A study that cross-referenced timber exploitation mapped by satellite images with activity authorizations in Pará showed a 22% increase in illegality. Rural properties registered in the Land Management System (Sigef), the Rural Environmental Registry (CAR), the National System of Property Certification (SNCI), or the Terra Legal Program were responsible for 72% of illegal timber extraction in Pará between August 2022 and July 2023. Isso sugere que a ilegalidade ocorre em grande parte em áreas privadas.[30]

Regarding other crops, several farms in the region are converting their forests for grain cultivation, particularly soybeans. According to a researcher, in 2022, for a gross total revenue of approximately $1,903.38 per hectare/year (R$ 10,338.90), generated by grain production sales in Brazil, the financial return rate (net profit), after deducting production costs (with soybean production as a benchmark), could reach around 70% per year, which is roughly $1,288.70 per hectare/year (R$ 7,000.00). There is no alternative investment, especially in the financial market, that can compete with such a high potential return.

3.6 Methodology Deviations

Not applicable. No methodological deviation was proposed.

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

The Agroflorestal Novo Horizonte project is currently undergoing validation alongside the verification of its first monitoring period. Ecolance Privated Limited serves as the validation and verification body.

[30] https://idesam.org/wp-content/uploads/2022/07/Evolucao-do-Setor-Madeireiro-na-Amazonia-de-1980-a-2020.pdf

The implemented actions are detailed throughout this document. Monitoring indicators have been established to assess the social, environmental, and climate benefits, aimed at reducing GHG emissions. Below are the activities that have already been conducted:

Satellite Image Monitoring Implementation Status (2021–2024)

Implemented. Satellite images are analyzed biannually, resulting in monitoring reports.

Surveillance and Monitoring Implementation Status (2021–2024)

Since 2021, Agroflorestal Novo Horizonte has conducted monthly patrols at key points on the farm. The installation of the communication tower marked a significant advancement, facilitating communication between areas and speeding up information exchange. In 2021, a new pickup truck was purchased to support surveillance activities.

Fire Prevention Implementation Status (2021–2024)

Fire brigades have been formed, and annual training sessions are provided for staff and the local community.

Road and Firebreak Maintenance Implementation Status (2021–2024)

Roads and firebreaks are being consistently maintained.

Reinforcement of Property Boundaries Implementation Status (2021–2024)

Maintenance and repair of fences and signs are regularly carried out to reinforce property boundaries.

Socio-environmental Activities Implementation Status (2021–2024)

Since 2021, Agroflorestal Novo Horizonte has undertaken community actions, including socioeconomic diagnostics, training, technical assistance for agriculture, and the organization of the local school. Additional planned activities will be implemented following project validation, once the necessary funding is secured.

Biodiversity implementation status (2021-2024)

A fauna survey was conducted in 2024 and will be monitored biannually using camera traps.

Partnerships with Government Entities, Educational Institutions, and Civil Society Groups Implementation Status (2021–2024)

Agroflorestal Novo Horizonte has established partnerships with the municipality of Prainha and ATER agencies to provide support to local communities. Additional partnerships are expected with the fire department, education department, health department, Sebrae, and ICMBio to expand benefits to local communities.

During the monitoring period from October 10th, 2021, to September, 30th, 2024, the actual deforestation rate within the Reference Region was 37.123,64 hectares, representing approximately 6,83% of the total area. In contrast, the baseline scenario predicted a deforestation of 20.485,57 hectares, or 3,77% of the Reference Region.

Throughout the crediting period, the project has continuously operated, focusing on illegal logging prevention, monitoring, and surveillance. There was no deforestation within the Project Area during the same monitoring period, underscoring the effectiveness of the implemented measures.

As this is the inaugural monitoring period, the project's activities align with the descriptions provided in Sections 1.12 (Description of the Project Activity), 1.18 (Sustainable Development Contributions).

5 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

Based on the selected baseline approach, the historical average method was used, and the results of the projected annual baseline deforestation areas in the reference region, leakage belt, and project area are described and presented in the tables below.

The projection of future deforestation within the reference region follows four key steps designed to determine where deforestation is most likely to occur under the baseline scenario, matching the projected deforestation locations with carbon stocks. This analysis is crucial to estimating the annual areas of baseline deforestation within the project area and leakage belt.

Definition of Model Assumptions: This step defines the modeled deforestation dynamics, including assumptions about spatial drivers and deforestation agents' behaviors. The analysis assumes that deforestation is not random but influenced by bio-geophysical and economic attributes that make certain areas more attractive for deforestation.

Organization of the Spatial and Non-Spatial Database: This involves the collection, selection, and standardization of spatial driver variables (SDVi) or "predisposing factors" that influence deforestation, such as soil fertility, slope, proximity to roads, and access to markets. These factors are identified through empirical analysis of spatial correlations between historical deforestation patterns and geo-referenced variables.

Calibration and Validation of the Model: This step involves combining spatial variables and validating the model against historical data to ensure its accuracy. Calibration is performed by evaluating the spatial correlation between historical deforestation events and the selected variables, allowing the model to predict future deforestation risk with greater reliability.

Development of Scenarios: Future scenarios are developed using historical trends under a Business-as-Usual scenario. Factor maps are created using GIS, overlaying spatial driver variables with historical deforestation data to generate a risk map showing areas most susceptible to future deforestation. High-risk areas, typically those with fertile soils, flat terrain, and close proximity to roads and markets, are assumed to be deforested first.

The variables and deforestation patterns shown in the table below were analyzed collectively to produce the risk map. Factor maps were generated using an empirical approach, where deforestation likelihood was estimated based on the percentage of pixels deforested during the analysis period.

Table 40. Table 12. List of variables, maps and factor maps

Factor Map		Source	Variable represented		Meaning of categories or pixel value			Other maps or variables used to create the Factor Map		Algorithm or equation used	Comments
ID	File Name		Unit	Description	Range		Meaning	ID	File Name		
1	d_estradas2.tif	IBGE/Imazon	Meters	Distances to paved and unpaved roads	1	9115	Lower values indicate greater proximity		MERGE_estradas_rst_Focal_Desmat20.tif	Euclidean Distance (ArcGIS 10.8)	Quantitative variable
2	d_assentamentos.tif	INCRA		Distances to rural settlements	0	17838	Lower values indicate greater proximity		Assentamento_INCRA	Euclidean Distance (ArcGIS 10.8)	Quantitative variable
3	d_rios_mbiomas.tif	MapBiomas	Meters	Distance to rivers mapped by MapBiomas	0	29502	Lower values indicate greater proximity		Rios_MapBiomas	Euclidean Distance (ArcGIS 10.6)	Quantitative variable

4	d_rios.tif	ANA	Meters	Distance to rivers	0	20774	Lower values indicate greater proximity		Rios_ANA	Euclidean Distance (ArcGIS 10.6)	Quantitative variable
5	d_urbana.tif	IBGE	Meters	Distance to urban centers	1435,6	59039	Lower values indicate greater proximity		Cidades_IBGE	Euclidean Distance (ArcGIS 10.6)	Quantitative variable
6	dem.tif	SRTM	Meters	Average change in altitude	197	608	Lower values indicate lower altitude				Quantitative variable
7	slope_perc.tif	SRTM	Percentage	Average variation of land slope	0	228,04	Lower values indicate less slope			Slope (ArcGIS 10.8)	Quantitative variable
8	Tis.tif	FUNAI		Demarcated indigenous lands	1	2	1 are areas outside indigenous lands and 2 are areas within indigenous lands		Tis.tif		Qualitative variable
9	Ucs	MMA (Ministery of the Environment)		National and State Protected Areas	1	2 and 3	1 are areas outside Ucs, 2 are full use UCs and 3 are strictly protected UCs		UCs.tif		Qualitative variable

The location analysis of future deforestation in the Reference Region was conducted using the most suitable model to determine the annual deforestation areas within the Project Area and Leakage Belt (as VM0015). Once the location analysis was completed, the proportions of annual baseline deforestation areas within the project area and leakage belt were determined using Geographic Information Systems (GIS) analysis. As previously mentioned in item 3.1.3, the historical reference period used for the baseline scenario analysis and baseline emissions calculations covers the years 2011 to 2021. The accumulated projected baseline deforestation in the Reference Region was 20.485,57 hectares during the first baseline period (Table 41).

Table 41. Annual areas of baseline deforestation in the Reference Region

Project year t	Stratum i in the reference region	Total	
	1	annual	cumulative
	ABSLRRi,t	ABSLRRi,t	ABSLRRi,t

	ha/yr	ha/yr	ha/yr
2021	3.464,97	3.464,97	3.464,97
2022	3.444,53	3.444,53	6.909,50
2023	3.424,20	3.424,20	10.333,70
2024	3.404,00	3.404,00	13.737,70
2025	3.383,92	3.383,92	17.121,62
2026	3.363,95	3.363,95	20.485,57

The projected accumulated baseline deforestation within the Project Area during the first baseline period is estimated to be 870,61 hectares (Table 42).

Table 42. Table 10: Annual Areas of Baseline Deforestation in the Project Area

Project year t	Stratum i in the reference region	Total	
	1	annual	cumulative
	ABSLPAi,t	ABSLPAi,t	ABSLPAi,t
	ha/yr	ha/yr	ha/yr
2021	147,26	147,26	147,26
2022	146,39	146,39	293,65
2023	145,52	145,52	439,17
2024	144,67	144,67	583,84
2025	143,81	143,81	727,65
2026	142,96	142,96	870,61

The projected accumulated baseline deforestation within the Leakage belt during the first baseline period is estimated to be 1329,59 hectares (Table 43).

Table 43. Table 11: Annual Areas of Baseline Deforestation in the Leakage Belt

Project year t	Stratum i in the reference region	Total

	1	annual	cumulative
	ABSLLKi,t	ABSLLKi,t	ABSLLKi,t
	ha/yr	ha/yr	ha/yr
2021	233,20	233,20	233,20
2022	221,88	221,88	455,08
2023	220,57	220,57	675,65
2024	219,27	219,27	894,92
2025	217,98	217,98	1112,90
2026	216,69	216,69	1329,59

With the area and specific locations of future deforestation now identified, we can assess pre-deforestation carbon stocks by aligning the predicted deforestation sites with forest classes that have established carbon stock values. It is important that the pre-deforestation carbon stocks reflect the conditions that are expected to exist in the year when deforestation is projected to occur. This approach ensures that the carbon stocks considered accurately represent the anticipated status at the time of deforestation.

The objective of this step is to determine the activity data for the initial forest classes (icl) designated for deforestation, as well as the activity data for the subsequent land classes (fcl) that will emerge in the baseline scenario following deforestation. For post-deforestation carbon stocks, these may be calculated either as a historical area-weighted average of carbon stock values or through spatial modeling techniques.

Based on the analyses conducted through the previously outlined methodologies, we have projected the extent of baseline land-use and land-cover change throughout the first baseline period across the reference region, project area, and leakage belt, categorized within each relevant stratum.

According to Step 5.1, titled “Calculation of Baseline Activity Data by Forest Class,” the annual baseline deforestation maps generated in the previous phase can be integrated with the current Land Use and Land Cover (LU/LC) map. This combination results in a detailed representation of deforestation by forest class within the baseline scenario.

For the Agroflorestal project, the changes in baseline LU/LC during the project crediting period primarily involve the conversion of the dense submontane forest typology to a final land use classification of ‘non-forest.’ The outcome of this analysis will include maps illustrating the polygons of the dense submontane forest that are expected to be deforested each year, assuming no intervention from the AUD project activity.

The extracted data will detail the number of hectares of dense submontane forest expected to be lost, with the results presented in the relevant tables: Table 44 for the reference region, Table 45 for the project area, and Table 46 for the leakage belt. This analysis should cover at least the baseline validity period and may extend to the project crediting period, as necessary.

Table 44. Annual Areas Deforested Per Forest Class icl Within the Reference Region in the Baseline Case (Baseline Activity Data Per Forest Class)

Area deforested per forest class icl within the reference region		Total baseline deforestation in the reference region	
IDicl>	1		
Name >	Managed Ombrophilous Dense Forest	ABSLRRt	ABSLRR
		annual	cumulative
Project year t	ha	ha/yr	ha
2021	3.464,97	3.464,97	3.464,97
2022	3.444,53	3.444,53	6.909,50
2023	3.424,20	3.424,20	10.333,70
2024	3.404,00	3.404,00	13.737,70
2025	3.383,92	3.383,92	17.121,62
2026	3.363,95	3.363,95	20.485,57

Table 45. Annual Areas Deforested Per Forest Class icl Within the Project Area in the Baseline Case (Baseline Activity Data Per Forest Class)

Area deforested per forest class icl within the project area		Total baseline deforestation in the project area	
IDicl>	1		
Name >	Managed Ombrophilous Dense Forest	ABSLPAt	ABSLPA
		annual	cumulative
Project year t	ha	ha/yr	ha
2021	147,26	147,26	147,26
2022	146,39	146,39	293,65
2023	145,52	145,52	439,17
2024	144,67	144,67	583,84
2025	143,81	143,81	727,65

2026	142,96	142,96	870,61

Table 46. Annual Areas Deforested Per Forest Class icl Within the Leakage Belt in the Baseline Case (Baseline Activity Data Per Forest Class)

Area deforested per forest class icl within the leakage belt		Total baseline deforestation in the leakage belt	
IDicl>	1		
Name >	Managed Ombrophilous Dense Forest	ABSLLKt	ABSLLK
		annual	cumulative
Project year t	ha	ha/yr	ha
2021	233,20	233,20	233,20
2022	221,88	221,88	455,08
2023	220,57	220,57	675,65
2024	219,27	219,27	894,92
2025	217,98	217,98	1112,90
2026	216,69	216,69	1329,59

According to the methodology VM00145, the Historical LU/LC Change (Method 1) was applied to determine the LU/LC class that would replace the forest cover in the baseline scenario. The table below presents the area of Zone 1, which encompasses areas of possible post-deforestation LU/LC classes within the reference region.

Table 47. Zones of the Reference Region Encompassing Different Combinations of Potential Post-Deforestation LU/LC Classes

Zone		Name:		Total of all other LU/LC classes present in zone		Total area of zone	
		F1					
		IDfcl	1				
		Area	% of Zone	Area	% of Zone	Area	% of Zone
iDz	Name	ha	%	ha	%	ha	%
1	Zone 1	583923,16	0,00	583923,16	0,00	583923,16	0,00
Total area of each class fcl		583923,16	0,00	583923,16	0,00	583923,16	0,00

Tables 48, 49, and 50 below depict the annual areas deforested in each zone in the baseline case within the reference region, project area, and leakage belt, respectively:

Table 48. Annual Areas Deforested in Each Zone Within the Reference Region in the Baseline Case (Baseline Activity Data Per Zone)

		Total baseline deforestation in the reference region	
iDz >	1		
Name >	Zone 1	ABSLRRt	ABSLRR
Project year t	ha	ha	ha
2021	3464,97	3464,97	3464,97
2022	3444,53	3444,53	6909,50
2023	3424,20	3424,20	10333,70
2024	3404,00	3404,00	13737,70
2025	3383,92	3383,92	17121,62
2026	3363,95	3363,95	20485,57

Table 49. Annual Areas Deforested in Each Zone Within the Project Area in the Baseline Case (Baseline Activity Data Zone)

Area established after deforestation per zone within the project area		Total baseline deforestation in the project area	
iDz >	1		
Name >	Zone 1	ABSLPAt	ABSLPA
Project year t	ha	ha	ha
2021	147,26	147,26	147,26
2022	146,39	146,39	293,65
2023	145,52	145,52	439,17
2024	144,67	144,67	583,84
2025	143,81	143,81	727,65
2026	142,96	142,96	870,61

Table 50. Annual Areas Deforested in Each Zone Within the Leakage Belt in the Baseline Case (Baseline Activity Data Per Zone)

		Total baseline deforestation in the leakage belt	
iDz >	1		
Name >	Zone 1	ABSLLKt	ABSLLK
Project year t	ha	ha	ha
2021	233,20	233,20	233,20
2022	221,88	221,88	455,08
2023	220,57	220,57	675,65
2024	219,27	219,27	894,92
2025	217,98	217,98	1112,90
2026	216,69	216,69	1329,59

Average carbon stocks were calculated based on the forest stratum present within the project area. Aboveground carbon stocks were measured through an in-situ forest inventory, while belowground carbon stocks were estimated using relevant scientific literature. Changes in carbon stocks due to wood products were deemed negligible and therefore excluded from the project boundary.

To determine the carbon stocks (ton CO_2/ha) of the managed area, 40 permanent forest inventory plots (125 m x 20 m = 250 m^2) were randomly allocated within the project area. All tree individuals with a Diameter at Breast Height (DBH) greater than 10 cm were measured, and data were collected on their variables, including Circumference at Breast Height (CBH in centimeters), estimated Height (H in meters), stem quality, and additional observations.

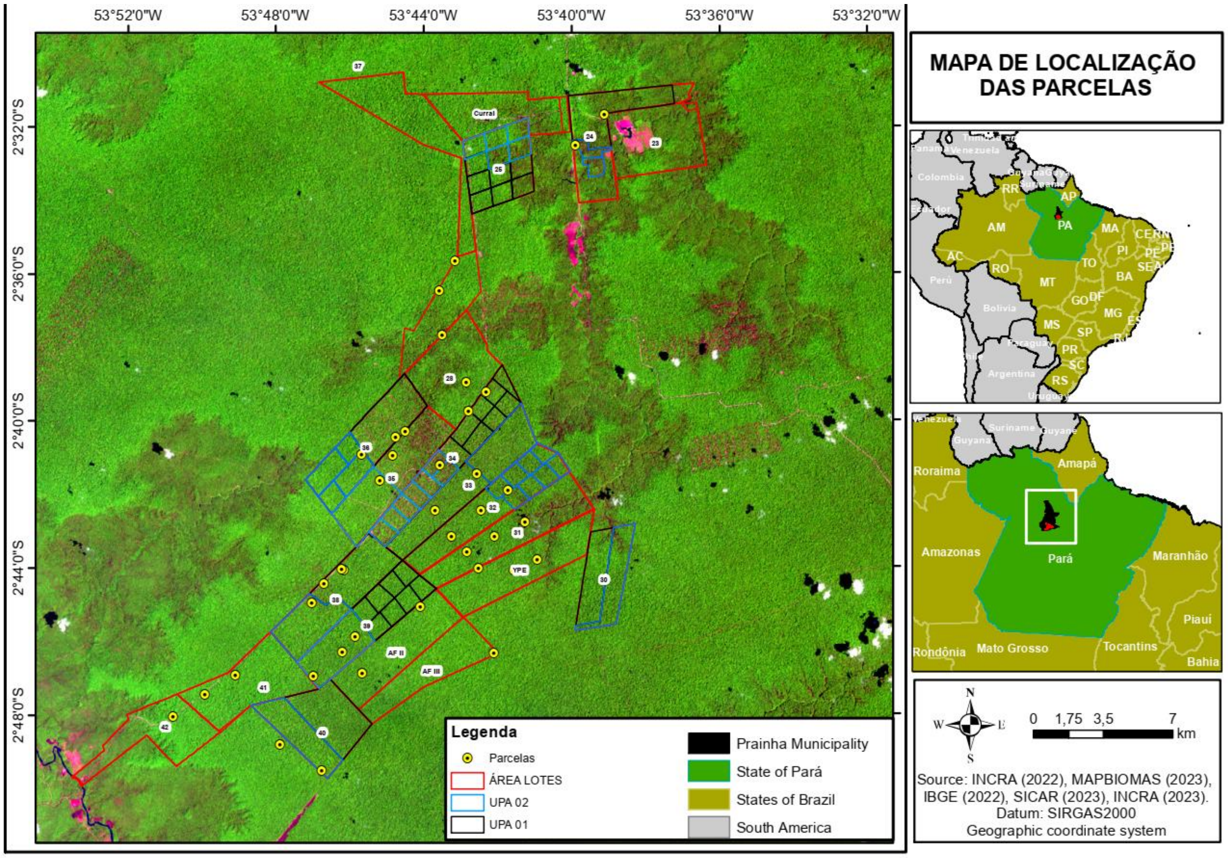

Figure 38. Location of Biomass Inventory Plots

Following the inventory, the collected data were tabulated in an Excel spreadsheet (.xls), where the carbon determination equation from the National Institute for Amazon Research (INPA) was applied for each inventoried tree:

$$CO_2eq = 2.247\ x\ (DBH^{1.92})\ x\ 0.584\ x\ 0.485\ x\ \frac{3.667\ x\ IDH}{1000}$$

Where:

DBH = Diameter at Breast Height;

IDH = index of dominant height correction based on data from nearby forests, as per the scientific publication on the Tapajós flora[31]

Using the INPA carbon equation and the adjustment for the forest's dominant height, the average value of tons of CO2 equivalent for each plot (250 m²) was determined and extrapolated to a hectare.

In the 40 plots, an average of 586.05 tons CO_2 per hectare was found, with a confidence interval of ±41.26 tons CO2 per hectare and an uncertainty of 7.04% in the data sampling.

[31] (http://bg.copernicus.org/articles/10/8385/2013/bg-10-8385-2013.pdf).

Table 51. Table 23: Carbon Stocks Per Hectare of Initial Forest Classes icl Existing in the Project Area and Leakage Belt – Estimated Values

Project year t	Initial forest class icl											
	Name	Managed Ombrophilous Dense Forest										
	IDicl											
	Average carbon stock per hectare +-90% CI											
	Cabicl		Cbbicl		Cwpicl Total carbon stocks in wood products in post-deforestation land uses are considered insignificant						Ctoticl	
					Short lived (0–3 years)		Medium lived (3–100 years)		Long lived (>100 years)			
	C stock (t CO2e/ ha)	+-90% CI (t CO2e/ ha)	C stock (t CO2e/ ha)	+-90% CI (t CO2e/ ha)	C stock (t CO2e/ ha)	+-90% CI (t CO2e/ ha)	C stock (t CO2e/ ha)	+-90% CI (t CO2e/ ha)	C stock (t CO2e/ ha)	+-90% CI (t CO2e/ ha)	C stock (t CO2e/ ha)	+-90% CI (t CO2e/ ha)
0												
1		627,3		194,5		0,0		4,0		3,5		822,3
2	586,05		181,7		0,0		4,0		3,5		768,2	
3												
...		544,8		168,9		0,0		4,0		3,5		714,1
T												

Table 52. Carbon Stocks Per Hectare of Initial Forest Classes icl Existing in the Project Area and Leakage Belt – Values After Discounting for Uncertainty

Project year t	Initial forest class icl					
	Name	Managed Ombrophilous Dense Forest				
	IDicl					
	Average carbon stock per hectare +-90% CI					
	Cabicl	Cbbicl	Cwpicl Total carbon stocks in wood products in post-deforestation land uses are considered insignificant			Ctoticl
			Short lived (0–3 years)	Medium lived (3–100 years)	Long lived (>100 years)	

	C stock (t CO2e/ ha)	+-90% CI (t CO2e/ ha)	C stock (t CO2e/ ha)	+-90% CI (t CO2e/ ha)	C stock (t CO2e/ ha)	+-90% CI (t CO2e/ ha)	C stock (t CO2e/ ha)	+-90% CI (t CO2e/ ha)	C stock (t CO2e/ ha)	+-90% CI (t CO2e/ ha)	C stock (t CO2e/ ha)	+-90% CI (t CO2e/ ha)
0	586,05		181,68		0,00		3,99		3,53		771,7	
1		627,31		194,47		0,00		0,00		0,00		826,0
2												
3												
...		544,79		168,89		0,00		0,00		0,00		717,4
T												

The VM0015 methodology employs standard linear functions to account for the changes in carbon stock within initial forest classes (icl) and the subsequent increase in carbon stock in post-deforestation classes (fcl).

1) Above-Ground Biomass:

Initial Forest Classes (icl): It is assumed that there is an immediate release of 100% of the carbon stock during the year t = t* (the year deforestation occurs).

Post-Deforestation Classes (fcl): A linear increase from 0 t CO2e/ha in year t = t* to 100% of the long-term average carbon stock (over 20 years) is expected by year t = t* + 9. This suggests that one-tenth of the final carbon stock accumulates each year over the 10 years following deforestation.

2) Below-Ground Biomass:

Initial Forest Classes (icl): An annual release of one-tenth of the initial carbon stock is anticipated each year from t = t* to t = t* + 9.

Post-Deforestation Classes (fcl): A similar linear increase from 0 t CO2e/ha in year t = t* to 100% of the long-term average carbon stock is assumed for the 10-year period post-deforestation, with 1/10th of the final carbon stock accumulating annually.

Table 53 and 54 outlines the carbon stock change factors associated with land-use change categories. These factors were subsequently utilized to compute the changes in baseline carbon stock. The calculations were based on the assumptions laid out in the VM0015 methodology, utilizing Method 1.

Table 53. Emission Factors for Initial Forest Classes icl (Method 1)

Managed Ombrophilous Dense Forest						
Year after deforestation		∆Cabctz,t	∆Cbbctz,t	Cwpicl - short lived	Cwpicl - medium lived	Cwpicl - long lived
1	t*	-586,0	-18,2	0,0	4,0	3,5
2	t*+1	0,0	-18,2	0,0	0,0	0,0
3	t*+2	0,0	-18,2	0,0	0,0	0,0
4	t*+3	0,0	-18,2	0,0	0,0	0,0
5	t*+4	0,0	-18,2	0,0	0,0	0,0
6	t*+5	0,0	-18,2	0,0	0,0	0,0
7	t*+6	0,0	-18,2	0,0	0,0	0,0
8	t*+7	0,0	-18,2	0,0	0,0	0,0
9	t*+8	0,0	-18,2	0,0	0,0	0,0
10	t*+9	0,0	-18,2	0,0	0,0	0,0
11	t*+10	0,0	0,0	0,0	0,0	0,0
12	t*+11	0,0	0,0	0,0	0,0	0,0
13	t*+12	0,0	0,0	0,0	0,0	0,0
14	t*+13	0,0	0,0	0,0	0,0	0,0
15	t*+14	0,0	0,0	0,0	0,0	0,0
16	t*+15	0,0	0,0	0,0	0,0	0,0
17	t*+16	0,0	0,0	0,0	0,0	0,0
18	t*+17	0,0	0,0	0,0	0,0	0,0
19	t*+18	0,0	0,0	0,0	0,0	0,0
20	t*+19	0,0	0,0	0,0	0,0	0,0
21	t*+20	0,0	0,0	0,0	0,0	0,0

Table 54. Long-term (20-years) average carbon stocks per hectare of post-deforestation LU/LC classes present in the reference region

Project year t	Post deforestation class *fcl*			
	Name:	Pasture		
	ID $_{fcl}$	1		
	Average carbon stock per hectare ± 90% CI			
	Cab $_{fcl}$		Cbb $_{fcl}$	
	average stock	± 90% CI	*average stock*	± 90% CI
	t CO_2e ha^{-1}	t CO_2e ha^{-1}	t CO_2e ha^{-1}	t CO_2e ha^{-1}
t*	0,00	0,00	0,00	0,00
t*+1	0,00	0,00	0,00	0,00
t*+2	0,00	0,00	0,00	0,00
t*+3	0,00	0,00	0,00	0,00
t*+4	0,00	0,00	0,00	0,00
t*+5	0,00	0,00	0,00	0,00
t*+6	0,00	0,00	0,00	0,00
t*+7	0,00	0,00	0,00	0,00
t*+8	0,00	0,00	0,00	0,00
t*+9	0,00	0,00	0,00	0,00
t*+10	0,00	0,00	0,00	0,00
t*+11	0,00	0,00	0,00	0,00
t*+12	0,00	0,00	0,00	0,00
t*+13	0,00	0,00	0,00	0,00
t*+14	0,00	0,00	0,00	0,00
t*+15	0,00	0,00	0,00	0,00
t*+16	0,00	0,00	0,00	0,00
t*+17	0,00	0,00	0,00	0,00
t*+18	0,00	0,00	0,00	0,00
t*+19	0,00	0,00	0,00	0,00
t*+20	0,00	0,00	0,00	0,00
...	0,00	0,00	0,00	0,00
t*+30	0,00	0,00	0,00	0,00
Average	0,00	0,00	0,00	0,00

The following tables detail the baseline changes in carbon stock across the components—above-ground biomass and below-ground biomass within the Project Area. Calculations utilize carbon stock change factors from earlier sections. The resulting carbon stock changes for initial forest classes in the reference region, project area, and leakage belt are also summarized in the subsequent tables, highlighting the impact of land-use changes on carbon dynamics.

Table 55. Baseline Carbon Stock Change for Selected Carbon Pools in the Reference Region (Method 1) - Aboveground

Carbon Stock Changes in Aboveground Biomass per Initial Forest Class icl		Total Carbon Stock Change in Aboveground Biomass of Initial Forest Classes in the Reference Region		Carbon Stock Changes in Aboveground Biomass Per Post-deforestation Zone z		Total Carbon Stock Change in Aboveground Biomass of Post-deforestation Zones in the Refence Region		Total Net Carbon Stock Change in Aboveground Biomass of Post-deforestation Zones in the Reference Region	
ID icl	1	ΔCab BSLRRicl,t Annual	ΔCab BSLRR icl Cumulative	ID z	1	ΔCab BSLRRz,t Annual	ΔCabBSLRRz Cumulative	ΔCab BSLRRt Annual	ΔCab BSLRR Cumulative
Name	Managed Ombrophilous Dense Forest			Name					
Project year	t CO2e	t CO2e	t CO2e	Project year	t CO2e	t CO2e	t CO2e	t CO2e	t CO2e
2021	-2030644,63	-2030644,63	-2030644,63	2021	0,0	0	0	-2030644,63	-2030644,63
2022	-2018665,77	-2018665,77	-4049310,40	2022	0,0	0	0	-2018665,77	-4049310,40
2023	-2006751,38	-2006751,38	-6056061,78	2023	0,0	0	0	-2006751,38	-6056061,78
2024	-1994913,18	-1994913,18	-8050974,96	2024	0,0	0	0	-1994913,18	-8050974,96
2025	-1983145,30	-1983145,30	-10034120,26	2025	0,0	0	0	-1983145,30	-10034120,26
2026	-1971441,89	-1971441,89	-12005562,15	2026	0,0	0	0	-1971441,89	-12005562,15

Table 56. Baseline Carbon Stock Change for Selected Carbon Pools in the Reference Region (Method 1) - Belowground

Carbon Stock Changes in Below ground Biomass per Initial Forest Class icl	Total Carbon Stock Change in Aboveground Biomass of Initial Forest Classes in the Reference Region	Carbon Stock Changes in Below ground Biomass Per Post-	Total Carbon Stock Change in belowround Biomass of Post-deforestation	Total Net Carbon Stock Change in Below ground Biomass of Post-deforestation Zones in the Reference Region

ID icl	1	ΔCbb BSLRRicl,t Annual	ΔCbb BSLRR icl Cumulative
Name	Managed Ombrophilous Dense Forest		
Project year	t CO2e	t CO2e	t CO2e
2021	-62949,98	-62949,98	-62949,98
2022	-125528,62	-125528,62	-188478,61
2023	-187737,92	-187737,92	-376216,52
2024	-249580,22	-249580,22	-625796,75
2025	-311057,73	-311057,73	-936854,47
2026	-372172,43	-372172,43	-1309026,90

deforestation Zone z		Zones in the Refence Region	
ID z	1	ΔCbb BSLRRz,t Annual	ΔCbbBSLRRz Cumulative
Name			
Project year	t CO2e	t CO2e	t CO2e
2021	0,0	0	0
2022	0,0	0	0
2023	0,0	0	0
2024	0,0	0	0
2025	0,0	0	0
2026	0,0	0	0

ΔCbb BSLRRt Annual	ΔCbb BSLRR Cumulative
t CO2e	t CO2e
-62949,98	-62949,98
-125528,62	-188478,61
-187737,92	-376216,52
-249580,22	-625796,75
-311057,73	-936854,47
-372172,43	-1309026,90

Table 57. Baseline Carbon Stock Change for Selected Carbon Pools in the Project Area (Method 1) - Aboveground

Carbon Stock Changes in Aboveground Biomass per Initial Forest Class icl		Total Carbon Stock Change in Aboveground Biomass of Initial Forest Classes in the Project Area	
ID icl	1	ΔCab BSLPAicl,t Annual	ΔCab BSLPA icl Cumulative
Name	Managed Ombrophilous Dense Forest		
Project year	t CO2e	t CO2e	t CO2e
2021	-86301,68	-86301,68	-86301,68
2022	-85791,82	-85791,82	-172093,49
2023	-85281,95	-85281,95	-257375,45
2024	-84783,81	-84783,81	-342159,26
2025	-84279,81	-84279,81	-426439,06

Carbon Stock Changes in Aboveground Biomass Per Post-deforestation Zone z		Total Carbon Stock Change in Aboveground Biomass of Post-deforestation Zones in the Project Area	
ID z	1	ΔCab BSLPAz,t Annual	ΔCabBSLPAz Cumulative
Name			
Project year	t CO2e	t CO2e	t CO2e
2019	0,0	0	0
2020	0,0	0	0
2021	0,0	0	0
2022	0,0	0	0
2023	0,0	0	0

Total Net Carbon Stock Change in Aboveground Biomass of Post-deforestation Zones in the Project Area	
ΔCab BSLPAt Annual	ΔCab BSLPA Cumulative
t CO2e	t CO2e
-86301,68	-86301,68
-85791,82	-172093,49
-85281,95	-257375,45
-84783,81	-342159,26
-84279,81	-426439,06

2026	-83781,67	-83781,67	-510220,73	2024	0,0	0	0	-83781,67	-510220,73

Table 58. Baseline Carbon Stock Change for Selected Carbon Pools in the Project Area (Method 1) - Belowground

Carbon Stock Changes in Below ground Biomass per Initial Forest Class icl		Total Carbon Stock Change in Below ground Biomass of Initial Forest Classes in the Project Area		Carbon Stock Changes in Below ground Biomass Per Post-deforestation Zone z		Total Carbon Stock Change in Below ground Biomass of Post-deforestation Zones in the Project Area		Total Net Carbon Stock Change in Below ground Biomass of Post-deforestation Zones in the Project Area	
ID icl	1	ΔCbb BSLPAicl,t Annual	ΔCbb BSLPA icl Cumulative	ID z	1	ΔCbb BSLPAz,t Annual	ΔCbbBSLPAz Cumulative	ΔCbb BSLPAt Annual	ΔCbb BSLPA Cumulative
Name	Managed Ombrophilous Dense Forest			Name					
Project year	t CO2e	t CO2e	t CO2e	Project year	t CO2e	t CO2e	t CO2e	t CO2e	t CO2e
2021	-2675,35	-2675,35	-2675,35	2021	0,0	0,0	0,0	-2675,35	-2675,35
2022	-5334,90	-5334,90	-8010,25	2022	0,0	0,0	0,0	-5334,90	-8010,25
2023	-7978,64	-7978,64	-15988,89	2023	0,0	0,0	0,0	-7978,64	-15988,89
2024	-10606,94	-10606,94	-26595,83	2024	0,0	0,0	0,0	-10606,94	-26595,83
2025	-13219,61	-13219,61	-39815,44	2025	0,0	0,0	0,0	-13219,61	-39815,44
2026	-15816,84	-15816,84	-55632,28	2026	0,0	0,0	0,0	-15816,84	-55632,28

Table 59. Table 57. Baseline Carbon Stock Change for Selected Carbon Pools in the Leakage Belt (Method 1) - Aboveground

Carbon Stock Changes in Aboveground Biomass per Initial Forest Class icl	Total Carbon Stock Change in Aboveground Biomass of Initial Forest Classes in the Leakage Belt	Carbon Stock Changes in Aboveground Biomass Per Post-deforestation Zone z	Total Carbon Stock Change in Aboveground Biomass of Post-deforestation Zones in the Leakage Belt	Total Net Carbon Stock Change in Aboveground Biomass of Post-deforestation Zones in the Leakage Belt

ID icl	1	ΔCab BSLLKicl,t Annual	ΔCab BSLLK icl Cumulative
Name	Managed Ombrophilous Dense Forest		
Project year	t CO2e	t CO2e	t CO2e
2021	-136666,79	-136666,79	-136666,79
2022	-130032,71	-130032,71	-266699,50
2023	-129264,98	-129264,98	-395964,48
2024	-128503,12	-128503,12	-524467,60
2025	-127747,11	-127747,11	-652214,71
2026	-126991,11	-126991,11	-779205,82

ID z	1	ΔCab BSLLKz,t Annual	ΔCabBSLLKz Cumulative
Name			
Project year	t CO2e	t CO2e	t CO2e
2021	0,0	0,0	0,0
2022	0,0	0,0	0,0
2023	0,0	0,0	0,0
2024	0,0	0,0	0,0
2025	0,0	0,0	0,0
2026	0,0	0,0	0,0

ΔCab BSLLKt Annual	ΔCab BSLLLK Cumulative
t CO2e	t CO2e
-136666,79	-136666,79
-130032,71	-266699,50
-129264,98	-395964,48
-128503,12	-524467,60
-127747,11	-652214,71
-126991,11	-779205,82

Table 60. Baseline Carbon Stock Change for Selected Carbon Pools in the Leakage Belt (Method 1) - Belowground

Carbon Stock Changes in Below ground Biomass per Initial Forest Class icl		Total Carbon Stock Change in Aboveground Biomass of Initial Forest Classes in the Leakage Belt	
ID icl	1	ΔCbb BSLLKicl,t Annual	ΔCbb BSLLK icl Cumulative
Name	Managed Ombrophilous Dense Forest		
Project year	t CO2e	t CO2e	t CO2e
2021	-4236,67	-4236,67	-4236,67
2022	-8267,68	-8267,68	-12504,35
2023	-12274,90	-12274,90	-24779,25
2024	-16258,50	-16258,50	-41037,75
2025	-20218,66	-20218,66	-61256,41
2026	-24155,38	-24155,38	-85411,79

Carbon Stock Changes in Below ground Biomass Per Post-deforestation Zone z		Total Carbon Stock Change in Aboveground Biomass of Post-deforestation Zones in the Leakage Belt	
ID z	1	ΔCbb BSLLKz,t Annual	ΔCbbBSLLKz Cumulative
Name			
Project year	t CO2e	t CO2e	t CO2e
2021	0,0	0,0	0,0
2022	0,0	0,0	0,0
2023	0,0	0,0	0,0
2024	0,0	0,0	0,0
2025	0,0	0,0	0,0
2026	0,0	0,0	0,0

Total Net Carbon Stock Change in Below ground Biomass of Post-deforestation Zones in the Leakage Belt	
ΔCbb BSLLKt Annual	ΔCbb BSLLK Cumulative
t CO2e	t CO2e
-4236,67	-4236,67
-8267,68	-12504,35
-12274,90	-24779,25
-16258,50	-41037,75
-20218,66	-61256,41
-24155,38	-85411,79

In analyzing the REDD AUD carbon project based on the VM0015 methodology, it is essential to understand the dynamics of emissions resulting from the use of fire in converting forests to non-forest areas. This process represents a significant source of non-CO2 gas emissions, such as methane (CH4) and nitrous oxide (N2O). Historical data reveal a consistent increase in fire alerts, with a greater concentration in areas that have been recently deforested, primarily due to the practice of deforestation through slash-and-burn.

The calculation of greenhouse gas (GHG) emissions from biomass burning was conducted in accordance with the requirements of the VM0015 methodology, specifically item 8.1.5.2.

The parameters used are presented in the table below:

Table 61. Parameters used to calculate non-CO2 emissions from forest fires

Initial Forest Class										
IDcl	Name	% Fburnt icl	tCO2e ha-1 Cab	% Pburnt ab, icl	% CE ab, icl	tCO2e ha-1 ECO2-ab	tCO2e ha-1 EBBCO2-tot	tCO2e ha-1 EBBN2O icl	tCO2e ha-1 EBBCH4 icl	tCO2e ha-1 EBBtot icl
1	Managed Ombrophilous Dense Forest	0,45	586,05	0,72	0,50	94,94	102,85	0,82	12,57	13,39

Table 62. Baseline Non-CO2 Emissions from Forest Fires in the Project Area

	Emissions of non-CO2 gasses from baseline forest fires		Total baseline non-CO2 emissions from forest fire in the project area	
	ID icl = 1			
Project year t	ABSLPA icl,t	EBBBSLtot icl	Annual EBBBSLPAt	Cumulatiuve EBBBSLPA

	ha	CO2e ha-1	tCO2e	tCO2e
2021	147,26	1971,81	1971,81	1971,81
2022	146,39	1960,16	1960,16	3931,97
2023	145,52	1948,51	1948,51	5880,49
2024	144,67	1937,13	1937,13	7817,62
2025	143,81	1925,62	1925,62	9743,23
2026	142,96	1914,23	1914,23	11657,47

5.2 Project Emissions

Ex ante estimated net carbon stock change in the project area under the project scenario is showed in Table 63 below.

Table 63. Ex Ante Estimated Net Carbon Stock Change in the Project Area Under the Project Scenario

	Total carbon stock decrease due to planned logging activities		Total carbon stock increase due to growth without harvest		Total carbon stock decrease due to unavoided unplanned deforestation		Total carbon stock change in the project case	
Project year	annual $\Delta CPLdPA_t$ tCO_2-e	cumulative $\Delta CPLdPA$ tCO_2-e	annual $\Delta CPLiPA_t$ tCO_2-e	cumulative $\Delta CPLiPA$ tCO_2-e	annual $\Delta CUDdPA_t$ tCO_2-e	cumulative $\Delta CUDdPA$ tCO_2-e	annual $\Delta CPSPA_t$ tCO_2-e	cumulative $\Delta CPSPA$ tCO_2-e
2021	0,00	0,00	0,00	0,00	13346,55	13346,55	13346,55	13346,55
2022	0,00	0,00	0,00	0,00	13669,01	27015,56	13669,01	27015,56
2023	0,00	0,00	0,00	0,00	13989,09	41004,65	13989,09	41004,65
2024	0,00	0,00	0,00	0,00	14308,61	55313,26	14308,61	55313,26
2025	0,00	0,00	0,00	0,00	14624,91	69938,18	14624,91	69938,18
2026	0,00	0,00	0,00	0,00	14939,78	84877,95	14939,78	84877,95

As forest fires were included in the baseline scenario, non-CO2emissions from biomass burning should also be included in the project scenario

Table 64. Total Ex Ante Estimated Actual Emissions of Non- CO2 Gases Due to Forest Fires in the Project Area

Project year	Total baseline non-CO2 emissions from forest fire in the project area	
	A n n u al E B B B SL tCO2e	C u m ul at iv tCO2e
2021	295,77	295,77
2022	294,02	589,80
2023	292,28	882,07
2024	290,57	1172,64
2025	288,84	1461,49
2026	287,14	1748,62

The expected *ex ante* net carbon stock changes and non-CO2 emissions in the Project area under the project scenario is summarized in Table 65 below.

Table 65. Total Ex Ante Estimated Actual Net Carbon Stock Changes and Emissions of Non-CO2 Gases in the Project Area

Project year	Total Ex Ante Carbon Stock Decrease due to Planned Activities		Total Ex Ante Carbon Stock Decrease due to Unavoided Unplanned Deforestation		Total Ex Ante Net Carbon Stock Change		Total Ex Ante Estimated Actual Non-CO2 Emissions from Forest Fires in the Project Area	
	annual ΔCPAdPAt tCO_2-e	cumulative ΔCPAdPA tCO_2-e	annual ΔCUDdPAt tCO_2-e	cumulative ΔCUDdPA tCO_2-e	annual $ΔCPSPA_t$ tCO_2-e	cumulative ΔCPSPA tCO_2-e	annual EBBPSPAt tCO_2-e	cumulative EBBPSPA tCO_2-e
2021	-	-	13346,55	13346,55	13346,55	13346,55	295,77	295,77
2022	-	-	13669,01	27015,56	13669,01	27015,56	294,02	589,80
2023	-	-	13989,09	41004,65	13989,09	41004,65	292,28	882,07
2024	-	-	14308,61	55313,26	14308,61	55313,26	290,57	1172,64
2025	-	-	14624,91	69938,18	14624,91	69938,18	288,84	1461,49
2026	-	-	14939,78	84877,95	14939,78	84877,95	287,14	1748,62

5.3 Leakage Emissions

This step provides an ex ante estimate of the possible decrease in carbon stock and increase in greenhouse gas (GHG) emissions due to leakage. The applied methodology considers two sources of leakage: a) decrease in carbon stocks and increase in GHG emissions associated with leakage prevention measures; and b) decrease in carbon stocks and increase in GHG emissions resulting from activity displacement. To mitigate the risk of displacement, baseline deforestation agents may participate in activities within the project area and leakage management area, aiming to replace their income and livelihoods, thereby reducing deforestation. If a significant decrease in carbon stock or increase in GHG emissions occurs, this change must be accounted for and monitored ex post. If the net sum of changes in carbon stock is positive, leakage prevention measures are not causing a decrease in carbon stock, and this increase will not be considered in the net GHG reductions of the project. Conversely, if the net sum is negative, it must be accounted for if significant.

The planned interventions in this project did not result in decreases in carbon stocks or increases in GHG emissions due to activities in the leakage management area, as they do not involve practices such as agricultural intensification or fertilization.

Table 66. 'Ex Ante Estimated Leakage due to Activity Displacement

Year	Total ex ante estimated decrease in carbon stocks due to displaced deforestation		Total ex ante estimated increase in GHG emissions due to displaced forest fires	
	annual ∆CADLKt tCO2-e	cumulative ∆CADLK tCO2-e	annual EADLKt tCO2-e	cumulative EADLK tCO2-e
2021	2001,98	2001,98	44,37	44,37
2022	2050,35	4052,33	44,10	88,47
2023	2098,36	6150,70	43,84	132,31
2024	2146,29	8296,99	43,59	175,90
2025	2193,74	10490,73	43,33	219,22
2026	2240,97	12731,69	43,07	262,29

Table 67.'Ex Ante Estimated Total Leakage

Project year	Total Ex Ante GHG Emissions from Increased Grazing Activities annual EgLKt tCO$_2$-e	cumulative EgLK tCO$_2$-e	Total Ex Ante Increase in GHG Emissions due to Displaced Forest Fires annual EADLKt tCO$_2$-e	cumulative EADLK tCO$_2$-e	Total Ex Ante Decrease in Carbon Stocks due to Displaced Deforestation annual ∆CADLKt tCO$_2$-e	cumulative ∆CADLK tCO$_2$-e	Carbon Stock Decrease due to Leakage Prevention Measures annual ∆CLPMLKt tCO$_2$-e	cumulative ∆CLPMLK tCO$_2$-e	Total Net Carbon Stock Change due to Leakage annual ΔCLK_t tCO$_2$-e	cumulative ∆CLK tCO$_2$-e	Total Net Increase in Emissions due to Leakage annual ELKt tCO$_2$-e	cumulative ELK tCO$_2$-e
2021	0,00	0,00	44,37	44,37	2001,98	2001,98	0,00	0,00	2001,98	2001,98	44,37	44,37
2022	0,00	0,00	44,10	88,47	2050,35	4052,33	0,00	0,00	2050,35	4052,33	44,10	88,47
2023	0,00	0,00	43,84	132,31	2098,36	6150,70	0,00	0,00	2098,36	6150,70	43,84	132,31
2024	0,00	0,00	43,59	175,90	2146,29	8296,99	0,00	0,00	2146,29	8296,99	43,59	175,90
2025	0,00	0,00	43,33	219,22	2193,74	10490,73	0,00	0,00	2193,74	10490,73	43,33	219,22
2026	0,00	0,00	43,07	262,29	2240,97	12731,69	0,00	0,00	2240,97	12731,69	43,07	262,29

5.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

The present REDD project is expected to avoid a predicted 1060,92 ha of deforestation, equating to 586,692.4 tCO_2e in emissions reductions over the first baseline period.

Table 68. Summary of emissions reductions of Agroflorestal Novo Horizonte

Vintage period	Estimated baseline emissions (tCO$_2$e)	Estimated project emissions (tCO$_2$e)	Estimated leakage emissions (tCO$_2$e)	Estimated buffer pool allocation (tCO$_2$e)	Estimated reduction VCUs (tCO$_2$e)	Estimated removal VCUs (tCO$_2$e)	Estimated total VCU issuance (tCO$_2$e)
2021	1971,81	295,77	44,37	9031,22	75260,17	0	66228,95
2022	1960,16	294,02	44,10	9243,53	77029,39	0	67785,86
2023	1948,51	292,28	43,84	9454,26	78785,53	0	69331,27
2024	1937,13	290,57	43,59	9664,66	80538,82	0	70874,16
2025	1925,62	288,84	43,33	9872,91	82274,22	0	72401,31
2026	1914,23	287,14	43,07	10080,22	84001,79	0	73921,58

Sum	11657,47	1748,62	262,29	57346,79	477889,92	0	420543,13

The summary of GHG reductions and removals in the Agroflorestal Novo Horizonte project is included in the table below, which includes estimates of GHG emissions reduction (REDDt), calculations of buffer and leakage, and the calculation of tradable Verified Carbon Units (VCUt).

Ex Ante Estimated Net GHG Emission Reductions: (ΔREDDt) and Verified Carbon Units (VCUt)

Project year	Baseline carbon stock changes		Baseline GHG emissions		Ex ante project carbon stock changes		Ex ante project GHG emissions		Ex ante leakage carbon stock changes		Ex ante leakage GHG emissions		Ex ante net anthropogenic GHG emission reductions		Ex ante VCUs tradable		Ex ante buffer credits	
	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative
	$\Delta CBSLPA_t$	$\Delta CBSLPA$	$EBBBSLPA_t$	EBBBSLPA	$\Delta CPSPA_t$	$\Delta CPSPA$	EBBPSPAt	EBBPSPA	ΔCLK_t	ΔCLK	ELKt	ELK	$\Delta REDD_t$	$\Delta REDD$	VCU_t	VCU	VBC_t	VBC
	tCO_2-e	tCO_2-e	tCO_2-e	tCO_2-e	tCO_2-e	tCO_2-e	tCO_2-e	tCO_2-e	tCO_2-e	tCO_2-e	tCO_2-e	tCO_2-e	tCO_2-e	tCO_2-e	tCO_2-e	tCO_2-e	tCO_2-e	tCO_2-e
2021	88977,03	88977,03	1971,81	1971,81	13346,55	13346,55	295,77	295,77	2001,98	2001,98	44,37	44,37	140807,3	140807,3	123.910	123.910	16896,9	16896,9
2022	91126,71	180103,74	1960,16	3931,97	13669,01	27015,56	294,02	589,80	2050,35	4052,33	44,10	88,47	85536,8	226344,1	75.272	199.183	10264,4	27161,3
2023	93260,59	273364,34	1948,51	5880,49	13989,09	41004,65	292,28	882,07	2098,36	6150,70	43,84	132,31	87369,5	313713,6	76.885	276.068	10484,3	37645,6
2024	95390,75	368755,08	1937,13	7817,62	14308,61	55313,26	290,57	1172,64	2146,29	8296,99	43,59	175,90	89189,6	402903,2	78.487	354.555	10702,7	48348,4
2025	97499,42	466254,50	1925,62	9743,23	14624,91	69938,18	288,84	1461,49	2193,74	10490,73	43,33	219,22	90997,1	493900,3	80.077	434.632	10919,6	59268,0
2026	99598,51	565853,01	1914,23	11657,47	14939,78	84877,95	287,14	1748,62	2240,97	12731,69	43,07	262,29	92792,1	586692,4	81.657	516.289	11135,0	70403,1

6 MONITORING

6.1 Data and Parameters Available at Validation

Data / Parameter	CF
Data unit	tC/tdm
Description	Default value of carbon fraction in biomass
Source of data	Nogueira et al 2008 - "Estimates of forest biomass in the Brazilian Amazon: New allometric equation and adjustments to biomass from wood-volume inventories"
Value applied:	0.48
Justification of choice of data or description of measurement methods and procedures applied	The consistency of the adopted data can be assured, as the conversion factor used is the same one applied by Silva (2014), which is the official reference for the Agroflorestal Novo Horizonte Project's carbon stock. Additionally, Silva (2014) is more recent than the IPCC (2003) and IPCC (2006) studies. Therefore, the National Forestry Service (SNIF) recommends using 0.48 as a conservative conversion factor, based on the assertion that carbon constitutes 50% of biomass.
Purpose of data	This parameter is used to calculate the baseline, project and leakage emissions from deforestation occurred in the baseline and project scenarios. Provides an estimate of the carbon content of the vegetation biomass within the project reference region
Comments	If new and more accurate carbon fraction data become available, these can be used to estimate the net anthropogenic GHG emission reduction of the subsequent fixed baseline period.

Data / Parameter	Ctotfcl
Data unit	tCO2e/ha
Description	carbon stock per hectare in anthropic areas in equilibrium of post-deforestation class fcl in tCO2e/ha
Source of data	Local plots

Value applied:	586,05
Justification of choice of data or description of measurement methods and procedures applied	To determine the carbon stocks (ton CO2/ha) of the managed area, 40 permanent forest inventory plots (125 m x 20 m = 250 m²) were randomly allocated within the project area. All tree individuals with a Diameter at Breast Height (DBH) greater than 10 cm were measured, and data were collected on their variables, including Circumference at Breast Height (CBH in centimeters), estimated Height (H in meters), stem quality, and additional observations. Following the inventory, the collected data were tabulated in an Excel spreadsheet (.xls), where the carbon determination equation from the National Institute for Amazon Research (INPA) was applied for each inventoried tree: $$CO_2eq = 2.247\ x\ (DBH^{1.92})\ x\ 0.584\ x\ 0.485\ x\ \frac{3.667\ x\ IDH}{1000}$$ Using the INPA carbon equation and the adjustment for the forest's dominant height, the average value of tons of CO2 equivalent for each plot (250 m²) was determined and extrapolated to a hectare. In the 40 plots, an average of 586.05 tons CO_2 per hectare was found, with a confidence interval of ±41.26 tons CO2 per hectare and an uncertainty of 7.04% in the data sampling.
Purpose of data	This parameter is used to calculate the baseline emissions from deforestation occurred in the baseline scenario. Provides an average of the post-deforestation carbon stock per hectare within the reference region.
Comments	N/A

Data / Parameter	DLF
Data unit	%
Description	Displacement Leakage Factor
Source of data	VCS Standard – Version 4, Section 3.15.14 – Item 1
Value applied:	15

Justification of choice of data or description of measurement methods and procedures applied	According to VCS requirements, where the applied methodology requires the quantification of activity shifting leakage, projects may apply the optional default activity-shifting leakage deduction of 15 percent to the gross GHG emission reductions and/or removals.
Purpose of data	This parameter is used to calculate leakage emissions in the baseline scenario due to activity displacement leakage, providing an ex-ante estimation of the decrease in carbon stocks and increase in GHG emissions. This value was calculated based on the percent of deforestation expected to be displaced outside the project boundary due to the implementation of the AUD project activity.
Comm ents	This parameter will be updated at each renewal of fixed baseline period

Data / Parameter	ΔCBSLLKt
Data unit	tCO2e
Description	Annual carbon stock changes in leakage management areas in the baseline case at year t.
Source of data	Calculated according to methodology VM0015 v1.1
Value applied:	0.000
Justification of choice of data or description of measurement methods and procedures applied	Leakage prevention activities generating a decrease in carbon stocks should be estimated ex ante and accounted. The leakage prevention measures proposed by the present project do not include decrease in carbon stocks due to activities implemented in the leakage management area.
Purpose of data	This parameter was used to calculate leakage emissions in the baseline scenario due to leakage prevention measures implemented in the leakage management area. It provides an ex-ante estimation of the decrease in carbon stocks due to the activities implemented.
Comments	This parameter will be updated at each renewal of fixed baseline period

Data / Parameter	EBBBSLPAt
Data unit	tCO_2e
Description	Sum of (or total) baseline non-CO_2 emissions from forest fire at year t in the project area.
Source of data	Calculated according to methodology VM0015 v1.1
Value applied:	355,1 (Annual average actual non-CO_2 emissions due to biomass burning within the project area during the crediting period)
Justification of choice of data or description of measurement methods and procedures applied	Non-CO_2 emissions from biomass burning are calculated according to requirements of methodology VM0015 v1.1. In order to estimate non-CO_2 emissions from forest fires, the average percentage of the area which contemplates the municipalities within the RR that was cleared by burning for other land uses involving deforestation, such as cattle raising and farming (Fburnt), and the average of biomass that has commercial value, and could be removed prior to clear cutting and burning (Pburnt,p) were estimated, either for the baseline and project case.
Purpose of data	This parameter is used to calculate non-CO_2 emissions due to forest fires within the project area in the baseline scenario, providing an ex-ante estimation.
Comments	EBBBSLPAt

Data / Parameter	Fburnticl
Data unit	%
Description	Proportion of forest area burned during the historical reference period in the forest class.
Source of data	Measured or estimated from literature. Fburnt data source:

Value applied:	45
Justification of choice of data or description of measurement methods and procedures applied	Value of "Biomass combustion factors by forest class group in the Amazon biome" for Open Tropical Rainforest available at the 4th National Communication from Brazil to the United Nations Framework Convention on Climate Change.
Purpose of data	This parameter is the average percentage of the area within the RR that was cleared by burning for other land uses involving deforestation, such as cattle raising and farming, and is used to calculate baseline and project non-CO2 emissions from forest fire at year t in the project area (parameter EBBBSLPAt).
Comments	Monitoring is done only once at project start.

Data / Parameter	Pburntp,icl
Data unit	%
Description	Average proportion of mass burnt in the carbon pool in the forest class
Source of data	Measured or estimated from literature. Pburnt data source:
Value applied:	72
Justification of choice of data or description of measurement methods and procedures applied	Pburnt was estimated using the average biomass per hectare that has commercial value and could be removed prior to clear cutting and burning. However, due to the lack of literature estimates, a study from the Brazilian Amazon in the Cerrado vegetation was used for the comparison. This study reported that the total biomass consumed by fires varies from 72% to 84% (average 78%) in denser Cerrado types, which is a forest vegetation.
Purpose of data	This parameter is the average of biomass that has commercial value, and could be removed prior to clear cutting and burning, and is used to

	calculate baseline and project non-CO2 emissions from forest fire at year t in the project area (parameter EBBBSLPAt).
Comments	Monitoring is done only once at project start.

Data / Parameter	EI
Data unit	%
Description	Ex ante estimated effectiveness index.
Source of data	Estimate based on verified reports of similar VM0015 REDD projects in Brazil up to date available in the VCS database.
Value applied:	85
Justification of choice of data or description of measurement methods and procedures applied	Based on the comparison between ex post and ex ante deforestation of similar REDD projects developed in Brazil, available in verified reports in VERRA database up to date.
Purpose of data	This parameter is used to calculate project emissions in the baseline scenario.
Comments	This parameter will be updated at each renewal of fixed baseline period.

6.2 Data and Parameters Monitored

Data / Parameter	abicl
Data unit	Mg/ha
Description	Average biomass stock per hectare in the above-ground biomass pool of initial forest class icl in Mg/ha.
Source of data	Average values for the above-ground biomass were taken from direct measures on permanent plots
Description of measurement methods and procedures applied	The following sources will be monitored: Biomass stock surveys Periodic reports from area supervisor Local forest inventories (if applicable/available)

Frequency of monitoring/recording	At each monitoring report or whenever replacement is necessary to
Value applied:	586,05
Monitoring equipment	No monitoring equipment is used to determine this parameter.
QA/QC procedures applied	Data shall be in accordance with VM0015 v1.1 requirements.
Purpose of data	This parameter is used to calculate baseline emissions, project emissions and leakage emissions in both baseline and project scenarios.
Calculation method	Permanent plots inventory
Comments	The values will be reassessed every 6 years or when data is more than 6 years old, whichever occurs first.

Data / Parameter	bbicl
Data unit	Mg/ha
Description	Average biomass stock per hectare in the below-ground biomass pool of initial forest class icl in Mg/ha.
Source of data	Average values for the below-ground biomass were taken from the applied methodology VM0015 v1.1, which estimates a root-to-shoot ratio of 0.31 according to Nogueira et al 2008 - "Estimates of forest biomass in the Brazilian Amazon: New allometric equation and adjustments to biomass from wood-volume inventories"
Description of measurement methods and procedures to be applied	The following sources will be monitored: Biomass stock surveys Periodic reports from area supervisor Local Forest Inventories
Frequency of monitoring/recording	At each monitoring report or whenever replacement is necessary to

Value applied:	181,68
Monitoring equipment	No monitoring equipment is used to determine this parameter.
QA/QC procedures to be applied	Data shall be in accordance with VM0015 v1.1 requirements
Purpose of data	This parameter is used to calculate baseline, project and leakage emissions in the baseline and project scenarios.
Calculation method	Average values for the below-ground biomass were taken from the applied methodology VM0015 v1.1, which estimates a root-to-shoot ratio of 0.31 according to Nogueira et al 2008 - "Estimates of forest biomass in the Brazilian Amazon: New allometric equation and adjustments to biomass from wood-volume inventories"
Comments	The values will be reassessed every 6 years or when data is more than 6 years old, whichever occurs first.

Data / Parameter	ACPAt
Data unit	ha
Description	Annual area within the Project Area affected by catastrophic events at year t.
Source of data	Remote sensing data and GIS.
Description of measurement methods and procedures to be applied	In addition to field data from the management team, when applicable, the following sources will also be monitored: INMET INPE
Frequency of monitoring/recording	At the occurrence of catastrophic events, when significant.
Value applied	The value will be calculated and applied at the occurrence of catastrophic events, when significant.

Monitoring equipment	Remote sensing and GIS.
QA/QC procedures to be applied	Best practices in remote sensing and GIS. Furthermore, the following sources will be also monitored to confirm the data obtained from remote sensing and GIS: INMET INPE Field data from the management team, when applicable.
Purpose of data	This parameter is used to calculate project emissions in the project scenario. Provides an ex-post estimation of the area affected by catastrophic events within the project area.
Calculation method	Remote sensing and GIS.
Comments	Decrease in carbon stocks and increases in GHG emissions (e.g. in case of forest fires) due to natural disturbances (such as hurricanes, earthquakes, volcanic eruptions, tsunamis, flooding, drought, fires, tornados or winter storms) or man-made events, including those over which the project proponent has no control (such as acts of terrorism or war), are subject to monitoring and must be accounted under the project scenario, when significant.

Data / Parameter	ABSLLKt
Data unit	ha
Description	Annual area of deforestation within the leakage belt at year t.
Source of data	Remote sensing and GIS.
Description of measurement methods and procedures to be applied	Deforestation in the leakage belt area may be considered activity displacement leakage. Activity data for the leakage belt area will be determined using the same methods applied to monitoring deforestation activity data in the project area.
Frequency of monitoring/recording	Annually

Value applied	249,5
Monitoring equipment	Remote sensing and GIS.
QA/QC procedures to be applied	Best practices in remote sensing.
Purpose of data	This parameter is used to calculate leakage emissions in the project scenario. Provides the ex-post value of the deforested area within the leakage belt.
Calculation method	Analysis of satellite images and maps.

Data / Parameter	ABSLPAt
Data unit	ha
Description	Annual area of deforestation in the project area at year t.
Source of data	Remote sensing and GIS.
Description of measurement methods and procedures to be applied	Forest cover change due to deforestation will be monitored through periodic assessment of classified satellite imagery covering the project area.
Frequency of monitoring/recording	Annually
Value applied	176,81
Monitoring equipment	No equipment were used
QA/QC procedures to be applied	Best practices in remote sensing.
Purpose of data	This parameter is used to calculate PA emissions in the project scenario. Provides the ex-post value of the deforested area within the PA
Calculation method	Analysis of satellite images and maps.
Comments	

Data / Parameter	ABSLRRt
Data unit	ha
Description	Annual area of deforestation in the reference region at year t.
Source of data	Remote sensing and GIS.
Description of measurement methods and procedures to be applied	Forest cover change due to deforestation will be monitored through periodic assessment of classified satellite imagery covering the reference region.
Frequency of monitoring/recording	Annually
Value applied	4483,8
Monitoring equipment	No equipment were used
QA/QC procedures to be applied	Best practices in remote sensing.
Purpose of data	This parameter will be used to calculate baseline emissions and project emissions in both baseline and project scenarios. Provides the ex-ante and ex post values of the deforested area per forest class within the reference region.
Calculation method	Analysis of satellite images and maps.
Comments	N/A

Data / Parameter	∆CADLKt
Data unit	tCO2e
Description	Total decrease in carbon stocks due to displaced deforestation at year t
Source of data	Calculated according to methodology VM0015 v1.1.
Description of measurement methods and procedures to be applied	Deforestation in the leakage belt area may be considered activity displacement leakage.

Frequency of monitoring/recording	Annually
Value applied	2605,4 (Annual average projected)
Monitoring equipment	Remote sensing and GIS to assess deforestation in the leakage belt.
QA/QC procedures to be applied	Best practices in remote sensing.
Purpose of data	This parameter will be used to calculate leakage emissions in the project scenario.
Calculation method	Emissions from deforestation at each forest class are estimated by multiplying the detected area of forest loss by the average forest carbon stock per unit area.
Comments	

Data / Parameter	ΔCPSLKt
Data unit	tCO_2e
Description	Total annual carbon stock change in leakage management areas in the project case at year t
Source of data	Calculated according to methodology VM0015 v1.1.
Description of measurement methods and procedures to be applied	The planned activities in leakage management areas that result in carbon stock decrease will be subject to monitoring, when significant.
Frequency of monitoring/recording	Annually
Value applied	0.00
Monitoring equipment	Remote sensing and GIS to assess deforestation in the leakage management areas.
QA/QC procedures to be applied	Best practices in remote sensing.

Purpose of data	This parameter will be used to calculate leakage emissions in the project scenario
Calculation method	Emissions from deforestation are estimated by multiplying the detected area of forest loss by the average forest carbon stock per unit area.
Comments	The leakage prevention measures proposed by the present project do not include decrease in carbon stocks due to activities implemented in the leakage management area.

Data / Parameter	ΔCUDdPAt
Data unit	tCO2e
Description	Total actual carbon stock change due to unavoided unplanned deforestation at year t in the project area.
Source of data	Calculated according to methodology VM0015 v1.1
Description of measurement methods and procedures to be applied	Forest cover change due to unplanned deforestation will be monitored through periodic assessment of classified satellite imagery covering the project area.
Frequency of monitoring/recording	Annually
Value applied	17369,7 (Annual average).
Monitoring equipment	Remote sensing and GIS to assess deforestation in the project area.
QA/QC procedures to be applied	Best practices in remote sensing.
Purpose of data	This parameter will be used to calculate project emissions in the project scenario. Provides the ex-post value of the change in carbon stocks due to unavoided unplanned deforestation within the project area.
Calculation method	Emissions from deforestation at each forest class are estimated by multiplying the detected area of forest loss by the average forest carbon stock per unit area.

Comments	N/A

Data / Parameter	EADLKt
Data unit	tCO2e
Description	Total ex post increase in GHG emissions due to displaced forest fires at year t.
Source of data	Calculated according to methodology VM0015 v1.1.
Description of measurement methods and procedures to be applied	Forest fires in the leakage belt area may be considered activity displacement leakage. GHG emissions due displaced forest fires will be subjected to monitoring, when significant.
Frequency of monitoring/recording	Annually
Value applied	53,3
Monitoring equipment	Remote sensing and GIS to assess deforestation in the leakage belt.
QA/QC procedures to be applied	Best practices in remote sensing and GIS.
Purpose of data Calculation method Comments	This parameter will be used to calculate leakage emissions in the baseline and project scenario. Provides the ex post value of the increase in GHG emissions due to displaced forest fires in the leakage belt. GHG emissions from deforestation are estimated by multiplying the detected area of forest loss in the leakage belt times the average forest carbon stock per unit area.

Data / Parameter	EBBPSPAt
Data unit	tCO_2e
Description	Sum of (or total) of actual non-CO_2 emissions from forest fire at year t in the project area
Source of data	Calculated according to methodology VM0015 v1.1
Description of measurement methods and procedures to be applied	If forest fires occur, these non-CO_2 emissions will be subject to monitoring and accounting, when significant. In addition to remote sensing data and GIS, which can identify the area affected by forest fire, the local team could also confirm the obtained data. No forest fire will be used by the project owner. However, it is expected that some unplanned deforestation within the project area will occur during the crediting period, which conversion of forest to non-forest may involve fire. The effect of fire on carbon emissions is counted in the estimation of carbon stock changes in the parameter ΔCUDdPAt; therefore CO_2 emissions from forest fires were ignored to avoid double counting. However, non-CO_2 emissions (CH_4 and N_2O) from forest fires must be counted in the project scenario, when they are significant. To be conservative, it will be assumed that all unplanned deforestation within the project area will involve fire. Therefore, non-CO_2 emissions from forest fires will be quantified and deducted from emission reductions
Frequency of monitoring/recording	Annually
Value applied	2,130.86 (annual average actual non-CO_2 emissions due to biomass burning within the project area during the crediting period)
Monitoring equipment	Remote sensing and GIS to assess deforestation in the project area.

QA/QC procedures to be applied	Best practices in remote sensing and GIS.
Purpose of data	This parameter will be used to calculate non-CO2 emissions due to forest fires within the project area in the project scenario, providing an estimate of the ex-post value for each vegetation type.
Calculation method	If forest fires occur, non-CO2 emissions from biomass burning will calculated according to requirements of methodology VM0015 v1.1. Therefore, this parameter will be calculated as the multiplication of the annual area of deforestation of initial forest classes in the project area in the project scenario times the total GHG emission from biomass burning in initial forest classes (EBBtoticl,t), when significant.
Comments	N/A

Data / Parameter	EBBtoticl,t
Data unit	tCO2e/ha
Description	Total GHG emission from biomass burning in forest class icl at year t.
Source of data	Calculated according to methodology VM0015 v1.1
Description of measurement methods and procedures to be applied	This parameter was calculated according to requirements and default values established by the VM0015 v1.1 methodology. In order to estimate non-CO2 emissions from forest fires, the average percentage of the area which contemplates the municipalities within the RR that was cleared by burning for other land uses involving deforestation, such as cattle raising and farming (Fburnt), and the average of biomass that has commercial value, and could be removed prior to clear cutting and burning (Pburnt,p) were estimated, either for the baseline and project case. These average percentage values are assumed to remain the same in the future, according to the applied methodology

Frequency of monitoring/recording	Annually
Value applied	13,39
Monitoring equipment	Remote sensing and GIS.
QA/QC procedures to be applied	Best practices in remote sensing and GIS.
Purpose of data	This parameter is used to calculate the baseline, project and leakage non-CO2 emissions from biomass burning occurred in the baseline and project scenarios
Calculation method	This parameter was calculated according to requirements and default values established by the VM0015 v1.1 methodology.
Comments	GWP for CH4 and N2O were obtained according to the most recent version of the VCS Standard.

Data / Parameter	RFt
Data unit	%
Description	Risk factor used to calculate VCS buffer credits
Source of data	VCS Non-Permanence Risk Report; Remote sensing data and GIS; Literature data.
Description of measurement methods and procedures to be applied	All sources of data from the VCS Non-Permanence Risk Report will be used to measure the various risk factors.
Frequency of monitoring/recording	Annually
Value applied	12

Monitoring equipment	Remote sensing and GIS.
QA/QC procedures to be applied	Best practices in remote sensing and GIS. The VCS Non-Permanence Risk Report will be verified together with the monitoring report at each verification event.
Purpose of data	This parameter represents the non-permanence risk rating of the project, which was used to determine the number of buffer credits that shall be deposited into the AFOLU pooled buffer account.
Calculation method	This parameter was calculated using the last available version of the AFOLU Non-Permanence Risk Tool. All the risk factors described in the VCS Non-Permanence Risk Report will be assessed.
Comments	N/A

6.3 Monitoring Plan

This monitoring plan has been developed according to the VCS Methodology VM0015 version 1.1.

Organizational structure

Agroflorestal Novo Horizonte is responsible for the costing and implementation and/or maintenance of the project's forest management and activities to reduce deforestation and degradation, surveillance, fire prevention, illegal extraction of wood, prevention of invasions, among others, implementation and maintenance of social and environmental activities to reduce leakage, decrease the risks of non-permanence of carbon and any additional Standards for the assessment of social and environmental co-benefits.

In addition, it is responsible for keeping all documentation required by the project in order, as well as project maintenance expenses; Execute, monitor and maintain in full operation the structure that authorizes and serves as the basis for the development of the Project, ensuring the reduction of deforestation and degradation, the implementation and maintenance of social and environmental activities (or designating and hiring third parties responsible for the activities).

Baseline Revision

The baseline will be reassessed every 6 years or when new regulatory requirements are published, and it will be validated alongside the subsequent verification. During the review, information on agents, drivers, and underlying causes of deforestation in the reference region will be collected to enhance future projections and project activity design. The projected annual deforestation areas of the baseline will be adjusted as necessary, and the locations of projected deforestation will be reassessed using updated spatial data. Areas already credited for avoided deforestation in previous periods will be excluded from future projections to ensure they are not credited again. Baseline monitoring will strictly follow the VM0015 methodology, version 1.1, or the most recent applicable version, ensuring accuracy and compliance with required standards.Data to be collected

Monitoring of Deforestation and Project Emissions

Changes in forest cover due to unplanned deforestation are monitored through periodic assessments of classified satellite images covering the project area. Deforestation emissions are estimated by multiplying the detected area of forest loss by the average forest carbon stock per unit area.

The project boundary, as defined in the PD, will serve as the initial "forest cover reference map" against which changes in forest cover will be assessed during each monitoring period. The total project area was demonstrated to meet the definition of forest at the start of the crediting period. For subsequent monitoring periods, changes in forest cover will be assessed against the previously classified forest cover map, which marks the beginning of the monitoring interval. The resulting classified image is compared to the previous one to detect changes in forest cover during the monitoring period and becomes the updated reference map for the next monitoring interval. Thus, the forest reference map is updated at each monitoring event.

Monitoring of Leakage

The calculation procedure for estimating leakage emissions in the project scenario will involve monitoring the following leakage sources:

Carbon Stock Changes and GHG Emissions Associated with Leakage Prevention Activities:

Monitoring of the leakage belt and leakage management area will be conducted in the same manner as within the project area and reference region. The most recent VCS guidelines on this subject will be applied. Its boundary will need to be reassessed at the end of each fixed baseline period using the same methodological approaches as in the previous period.

Decrease in Carbon Stock or Increase in GHG Emissions due to Leakage Prevention Measures:

These changes, which are likely to occur within the leakage management area, will be monitored through documentation and field assessments. In areas where carbon stock improvement is observed, the project conservatively assumes stable stocks, and biomass monitoring is not conducted.

Decrease in Carbon Stock and Increase in GHG Emissions Due to Activity Displacement Leakage:

Deforestation within the leakage belt can be considered activity displacement leakage. Activity data for the leakage belt will be determined using the same methods applied to monitor deforestation activity data within the project area. Leakage will be calculated by comparing ex-ante and ex-post assessments. However, if strong evidence can be gathered that deforestation within the leakage belt is attributable to deforestation agents not linked to the project area, the detected deforestation will not be attributed to the project activity and, therefore, will not be considered leakage

Monitoring of Natural Disturbances and Catastrophic Events

Carbon stock losses within the project area due to natural disturbances such as wildfires and other catastrophic events will be promptly estimated following the event. Decreases in carbon stocks and increases in GHG emissions resulting from natural or man-made disturbances, including those beyond the project proponent's control (e.g., acts of terrorism or war), are subject to monitoring when significant. If the affected area generated Verified Carbon Units (VCUs) in past verifications, the total net change in carbon stocks and GHG emissions will be assessed, and an equivalent amount of VCUs will be canceled from the VCS buffer. No new VCUs will be issued until all carbon stock losses and GHG emissions increases have been fully offset. Monitoring and reassessment of related parameters will occur at each verification and revalidation point, with the instance owner responsible for training and management of personnel involved in monitoring and safety. In the event of forest fires, non-CO2 emissions will be accounted for when significant, and the project proponent must notify Verra within 30 days of identifying any likely loss event, ensuring updates to carbon project reports and non-permanence risk assessments.

Update of Forest Carbon Stock Estimates

Forest inventories are conducted annually in accordance with the forest management plan, and this information can be utilized to update estimates of forest carbon stocks. If new and more accurate carbon stock data become available, they may be employed to estimate the net anthropogenic GHG emission reductions for the subsequent fixed baseline period. For the current fixed period, new data will only be incorporated if validated by an accredited VCS verifier. Should new data be utilized during the current fixed period, the baseline will be recalculated using this updated information, ensuring compliance with the requirements established by the applied methodology, VM0015. Only validated data will be considered for estimating carbon stock changes and GHG emissions.

Methods for Generating, Recording, Aggregating, Collecting, and Reporting Data on Monitored Parameters

All data sources and procedures for processing, classification, and change detection will be meticulously documented and stored in a dedicated long-term electronic archive maintained by

Agroflorestal Novo Horizonte. Due to the long-term nature of the project and the rapid advancement of software updates and data storage technology, electronic files will be periodically updated or converted to formats compatible with future software applications as needed. All maps and records generated during project implementation will be securely stored and made available for inspection by VCS verifiers during verification. Furthermore, data collected from ground-truth points—including GPS coordinates, identified land-use classes, and supporting photographic evidence—will be systematically recorded and archived.

Monitored data will be retained for two years following the conclusion of the crediting period or the last issuance of carbon credits for this project activity, whichever occurs later. Agroflorestal Novo Horizonte will hold the authority for registration, monitoring, measurement, and reporting. The monitored parameters will be tracked with the frequency described in subsequent sections.

Quality Assurance/Quality Control

To ensure the consistency and quality of results, spatial analysts engaged in image processing, interpretation, and change detection will strictly adhere to the protocols outlined in the VCS Methodology. The reliable data collected and documented will serve as a technical support tool for informed decision-making, aimed at enhancing project outcomes and adapting to current needs and realities. All project activities must align with the management plans detailed in the Project Design Document (PD).

Project implementation will be continuously monitored using remote sensing techniques, supplemented by field studies. Land-use monitoring will utilize remote sensing methods, incorporating images from sources such as INPE (PRODES), Landsat, ResourceSat, and others. These images will undergo digital processing for the interpretation and classification of land cover classes.

The management structure includes local staff from the Agroflorestal Novo Horizonte management team, who are present in the project area during most of the harvesting season and reside in nearby cities. They will provide periodic reports to the project proponent, responsible for overseeing monitoring, quality control, and quality assessment procedures. All monitored parameters will be meticulously checked.

Additionally, Agroflorestal Novo Horizonte will implement sustainability reporting in accordance with the CCB Standards, focusing on environmental and social activities within both the Project Area and the Leakage Management Area.

Procedures for Handling Internal Auditing and Non-Conformities

Internal auditing and non-conformities at Agroflorestal Novo Horizonte will follow established procedures guided by a Sustainable Forest Management Plan developed by third-party experts. This plan includes processes for identifying non-conformities and conducting staff training.

Agroflorestal Novo Horizonte forest management is certified under FSC standards. In the event of measurement errors, Agroflorestal will investigate the root causes and implement corrective actions to ensure conservative data reporting.

A project information quality management system will minimize errors and ensure reliable monitoring results, including staff training and in-field verification of adherence to methodological guidelines. Any identified non-conformities during audits will be addressed promptly.

Forest cover changes due to unplanned deforestation will be monitored using satellite imagery, with emissions estimated based on the area of forest loss and average carbon stocks. The project boundary will serve as the initial benchmark for assessing changes in forest cover, which has been verified to meet forest definitions at the start of the crediting period. Subsequent monitoring will compare changes against prior classified forest cover maps.

Monitoring will be done by the project proponent and outsourced to a third party having sufficient capacities to perform the monitoring tasks. To ensure the operation of the monitoring activities, the operational and managerial structure will be established according to the table below.

Table 69. Monitoring responsible and frequency

Variable to be monitored	*Responsible*	*Frequency*
Revision of the baseline	Agroflorestal Novo Horizonte and another external institution qualified for the monitoring	Every 6 years or if new requirements is lauched
Monitoring Deforestation and Project Emissions	Agroflorestal Novo Horizonte and another external institution qualified for the monitoring	Monthly
Monitoring of non-CO2 emissions from forest fires	Agroflorestal Novo Horizonte and another external institution qualified for the monitoring	Monthly
Monitoring Leakage	Agroflorestal Novo Horizonte and another external institution qualified for the monitoring	Monthly

Monitoring of Natural Disturbance and catastrophic events	Agroflorestal Novo Horizonte and another external institution qualified for the monitoring	When a natural event occurs
Updating Forest Carbon Stocks Estimates	Agroflorestal Novo Horizonte and another external institution qualified for the monitoring	At least, every 6 years, only if necessary.
Quality Assurance and Quality control	Agroflorestal Novo Horizonte and another external institution qualified for the monitoring	Prior to each verification.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

7.1 Data and Parameters Monitored

Data / Parameter	$ACPA_{icl,t}$
Data unit	ha
Description	Annual area within the project area affected by catastrophic events in class *icl* at year *t*
Value applied:	0
Comments	Each time a catastrophic event occurs

Data / Parameter	Cabicl
Data unit	t CO2e/ha
Description	Average carbon stock per hectare in aboveground biomass carbon pool per zone z
Observation	Measured
Value applied:	586,05
Comments	Only once at project start and where mandatory

Data / Parameter	Cbb_{icl}
Data unit	t CO2e/ha
Description	Average carbon stock per hectare in belowground biomass carbon pool of initial forest class *icl*
Observation	Nogueira et al 2008 - "Estimates of forest biomass in the Brazilian Amazon: New allometric equation and adjustments to biomass from wood-volume inventories"
Value applied:	181,7
Comments	

Data / Parameter	*Ctotcl*
Data unit	t CO2e/ha
Description	Average carbon stock per hectare in all accounted carbon pools of LU/LC class *cl*
Observation	768,2
Value applied:	Calculated
Comments	

Data / Parameter	ΔCADLK
Data unit	tCO_2e
Description	Cumulative total decrease in carbon stocks due to displaced deforestation
Observation	ΔCADLK tCO_2-e 0,00 24923,96 31673,25 51551,32
Value applied:	Calculated

Comments	

Data / Parameter	ΔCBSLPAft
Data unit	tCO_2e
Description	Total baseline carbon stock change in final classes within project area at year *t*
Observation	
Value applied:	ΔCBSLPA$_t$ tCO_2-e 88.977,03 91.126,71 93.260,59 95.390,75
Comments	

Data / Parameter	ΔCBSLPA
Data unit	t CO2e
Description	Total baseline carbon stock changes in project area
Observation	
Value applied:	368.755,08
Comments	

Data / Parameter	*$ABSLPA_{ct,t}$*
Data unit	ha
Description	Area of category *ct* deforested at time *t* within project area in baseline case
Observation	
Value applied:	<table><tr><th rowspan="2">Project year t</th><th>ABSLPA icl,t</th></tr><tr><th>ha</th></tr><tr><td>2021</td><td>0,00</td></tr><tr><td>2022</td><td>41,25</td></tr><tr><td>2023</td><td>9,93</td></tr><tr><td>2024</td><td>31,36</td></tr></table>
Comments	

Data / Parameter	ΔCBSLLK
Data unit	t CO2e
Description	Cumulative carbon stock changes in leakage management zones in baseline case
Observation	
Value applied:	ΔC

	BSLLK Cumulative t CO2e 0,00 -24923,96 -31673,25 -51551,32
Comments	

Data / Parameter	ΔCFCdPA
Data unit	t CO2e
Description	Cumulative decrease in carbon stock due to forest fires and catastrophic events at year *t* in project area
Observation	Ex post
Value applied:	ΔCFCdPA tCO_2-e 0 0 0 0
Comments	

Data / Parameter	ΔCFCdPAt
Data unit	t CO2e
Description	Total decrease in carbon stock due to forest fires and catastrophic events at year *t* in project area
Observation	Ex post
Value applied:	*0*
Comments	

Data / Parameter	ΔCFCiPA
Data unit	t CO2e
Description	Cumulative increase in carbon stock due to forest fires and catastrophic events at year *t* in project area
Observation	Ex post
Value applied:	*0*
Comments	

Data / Parameter	ΔCFCiPAt
Data unit	t CO2e
Description	Total increase in carbon stock due to forest fires and catastrophic events at year *t* in project area

Observation	Ex post
Value applied:	*0*
Comments	

Data / Parameter	ΔCLK
Data unit	t CO2e
Description	Total cumulative decrease in carbon stocks within leakage belt at year *t*
Observation	
Value applied:	ΔCLK tCO_2-e 0,0 0,0 0,0 0,0
Comments	

Data / Parameter	ΔCLKt
Data unit	t CO2e
Description	Total decrease in carbon stocks within leakage belt at year *t*

Observation	
Value applied:	ΔCLK_t tCO_2-e 0,0 0,0 0,0 0,0
Comments	

Data / Parameter	ΔCLPMLK
Data unit	t CO2e
Description	Cumulative carbon stock decrease due to leakage prevention measures
Observation	Ex ante and ex post
Value applied:	ΔCLPMLK tCO_2-e 0,0 0,0 0,0 0,0
Comments	

Data / Parameter	ΔCLPMLKt

Data unit	t CO2e
Description	Carbon stock decrease due to leakage prevention measures at year *t*
Observation	Ex ante and ex post
Value applied:	ΔCLPMLKt tCO_2-e 0,0 0,0 0,0 0,0
Comments	

Data / Parameter	ΔCPAdPA
Data unit	t CO2e
Description	Cumulative decrease in carbon stock due to all planned activities at year *t* in project area
Observation	Ex ante and ex post
Value applied:	*0*
Comments	

Data / Parameter	ΔCPSPA
Data unit	t CO2e

Description	Cumulative project carbon stock change within project area at year *t*
Observation	Ex ante and ex post
Value applied:	tCO_2-e 0,0 0,0 0,0 0,0
Comments	

Data / Parameter	ΔCUCdPAt
Data unit	t CO2e
Description	Total decrease in carbon stock due to catastrophic events at year *t* in project area
Observation	Ex post
Value applied:	*0*
Comments	

Data / Parameter	ΔCUCiPA
Data unit	t CO2e
Description	Cumulative increase in carbon stock in areas affected by catastrophic events (after such events) at year *t* in project area

Observation	Ex post
Value applied:	*0*
Comments	

Data / Parameter	ΔCUCiPAt
Data unit	t CO2e
Description	Total increase in carbon stock in areas affected by catastrophic events (after such events) at year *t* in project area
Observation	Ex post
Value applied:	*0*
Comments	

Data / Parameter	ΔCUCdPA
Data unit	t CO2e
Description	Cumulative decrease in carbon stock due to catastrophic events at year *t* in project area
Observation	Ex post
Value applied:	*0*

Comments	

Data / Parameter	ΔCUDdPAt
Data unit	t CO2e
Description	Total actual carbon stock change due to unavoided unplanned deforestation at year *t* in project area
Observation	Ex ante and ex post
Value applied:	*0*
Comments	

Data / Parameter	ΔCUFdPA
Data unit	t CO2e
Description	Cumulative total decrease in carbon stock due to unplanned (and planned where applicable) forest fires in project area
Observation	
Value applied:	*0*
Comments	

Data / Parameter	ΔCUFdPAt

Data unit	t CO2e
Description	Total decrease in carbon stock due to unplanned (and planned where applicable) forest fires at year *t* in project area
Observation	Ex post
Value applied:	*0*
Comments	

Data / Parameter	ΔCUFiPA
Data unit	t CO2e
Description	Cumulative increase in carbon stock in areas affected by forest fires (after such events) in project area
Observation	Ex post
Value applied:	*0*
Comments	

Data / Parameter	ΔCUFiPAt
Data unit	t CO2e
Description	Total increase in carbon stock in areas affected by forest fires (after such events) at year *t* in project area

Observation	Ex post
Value applied:	*0*
Comments	

Data / Parameter	ΔREDD
Data unit	t CO2e
Description	Cumulative net anthropogenic GHG emission reduction attributable to AUD project activity
Observation	Ex ante and ex post
Value applied:	ΔREDD tCO_2-e 90948,84 184035,72 279244,82 376572,70
Comments	

Data / Parameter	ΔREDDt
Data unit	t CO2e
Description	Net anthropogenic GHG emission reduction attributable to AUD project activity at year *t*

Observation	Ex ante and ex post
Value applied:	annual $\Delta REDD_t$ tCO$_2$-e / $\Delta REDD_t$ tCO$_2$-e tCO$_2$-e / 90948,84 75260,17 / 93086,88 77029,39 / 95209,10 78785,53 / 97327,88
Comments	

Data / Parameter	*EADLK*
Data unit	t CO2e
Description	Cumulative total increase in GHG emissions due to displaced forest fires
Observation	Ex ante and ex post
Value applied:	EADLK tCO$_2$-e / EADLK tCO$_2$-e 3122,5 / 0 6093,5 / 0 9047,0 / 0 11983,0 / 0
Comments	

Data / Parameter	*EBBBSPA*
Data unit	t CO2e
Description	Cumulative baseline non-CO_2 emissions from forest fires at year *t* in project area
Observation	ex post
Value applied:	*0*
Comments	

Data / Parameter	*EBBBSLtot$_{icl}$*
Data unit	t CO2e
Description	Sum of (or total) actual non-CO_2 emissions from forest fires at year *t* in stratum *i* in forest class *icl*
Observation	Ex ante and ex post
Value applied:	*Leakage belt*

Project year t	ABSLPA icl,t	EBBBSLtot icl	Annual EBBBSLPAt	Cumulatiuve EBBBSLPA
	ha	CO2e ha-1	tCO2e	tCO2e

	2021	0,00	0,00	0,00	0,00
	2022	41,25	552,34	552,34	552,34
	2023	9,93	132,96	132,96	685,30
	2024	31,36	419,91	419,91	1105,21
Comments					

Data / Parameter	*EBBPSPA*
Data unit	t CO2e
Description	Cumulative (or total) actual non-CO_2 emissions from forest fires at year *t* in project area
Observation	ex post
Value applied:	EBBBSLPA tCO_2-e 1.971,81 3.931,97 5.880,49 7.817,62
Comments	

Data / Parameter	*EBBPSPAt*
Data unit	t CO2e
Description	Sum of (or total) actual non-CO_2 emissions from forest fires at year *t* in project area
Observation	Ex ante and ex post

Value applied:	$EBBBSLPA_t$ tCO_2-e 1.971,81 1.960,16 1.948,51 1.937,13
Comments	

Data / Parameter	*EgLK*
Data unit	t CO2e
Description	Cumulative emissions from grazing animals in leakage management zones at year *t*
Observation	
Value applied:	EgLK tCO_2-e 0,0 0,0 0,0 0,0
Comments	

Data / Parameter	*EgLKt*
Data unit	t CO2e

Description	Emissions from grazing animals in leakage management zones at year *t*
Observation	
Value applied:	EgLKt tCO_2-e 0,0 0,0 0,0 0,0
Comments	

Data / Parameter	*ELK*
Data unit	t CO2e
Description	Cumulative sum of ex ante estimated leakage emissions at year *t*
Observation	
Value applied:	ELK tCO_2-e 0 0 0 0
Comments	

Data / Parameter	*ELKt*
Data unit	t CO2e
Description	Sum of ex ante estimated leakage emissions at year *t*
Observation	
Value applied:	ELKt tCO_2-e 0,0 0,0 0,0 0,0
Comments	

Data / Parameter	*VBCt*
Data unit	t CO2e
Description	Number of buffer credits deposited in the VCS buffer pool at time *t*
Observation	
Value applied:	VBC_t tCO_2-e 10913,86 11170,43 11425,09 11679,35

Comments	

Data / Parameter	*VCUt*
Data unit	t CO2e
Description	Number of Verified Carbon Units (VCUs) to be made available for trade at time *t*
Observation	
Value applied:	VCU_t tCO_2-e **80034,98** **81916,45** **83784,01** **85648,53**
Comments	

7.2 Baseline Emissions

Temporal Boundaries

The temporal domain from which information about historical deforestation for the verification period was extracted and analyzed according to the VMD0007 methodological requirements includes annual data from 2021 to 2024. Therefore, October 10, 2021, and 30 September, 2024, mark the start and end dates of the first verification period for the project’s carbon credits.

In this scenario, it is expected that forested areas will be converted into non-forested areas, with Unplanned Deforestation Avoided (AUDD) occurring continuously as a result of dominant land use changes in the region. These changes include agriculture and livestock farming, which lead to the removal of primary and secondary forests in the Project Zone.

Estimation of annual areas of unplanned deforestation in RRD and calculation of the deforestation rate of the first verification report

Collection of Appropriate Data Sources

Land use and land cover classes were obtained from the MapBiomas database. Additionally, a mosaic of Landsat OLI 8 satellite images was created to map the "non-forest vegetation" class. The images cover the verification period (2021 to 2024). MapBiomas utilizes LANDSAT-class satellites (30-meter spatial resolution and 16-day revisit rate) in a mosaic combination aimed at minimizing cloud coverage issues and ensuring interoperability criteria. MapBiomas has a minimum mapped area of 900 m^2 (30-meter pixel size). A total of 11 mosaic images, composed of two scenes (orbits: 226 and 227 / point: 62), were adopted to map the land use classes of interest in the reference region within the years 2021 to 2024. The following table describes the data used for historical LU/LC change analyses:

Number	Date	Projection
LC08_L1TP_226062_20211010_20211019_02_T1	10/10/2021	WGS84
LC08_L1TP_226062_20210807_20210818_02_T1	07/08/2021	WGS84
LC09_L1TP_226062_20220717_20230407_02_T1	17/07/2022	WGS84
LC09_L1TP_226062_20220818_20230402_02_T1	18/08/2022	WGS84
LC08_L1TP_226062_20230930_20231010_02_T1	30/09/2023	WGS84
LC08_L1TP_226062_20240221_20240229_02_T1	21/02/2024	WGS84
LC09_L1TP_226062_20240229_20240229_02_T1	29/02/2024	WGS84
LC09_L1TP_226062_20240924_20240924_02_T1	24/09/2024	WGS85
LC08_L1TP_227062_20211001_20211013_02_T1	01/10/2021	WGS84
LC08_L1TP_227062_20220817_20220824_02_T1	17/08/2022	WGS84
LC08_L1TP_227062_20231210_20231214_02_T1	10/12/2023	WGS84

The land use and cover data used in the Agroflorestal Project were obtained from the MapBiomas database. The main methodological steps conducted by MapBiomas to map land use and cover in Brazil are described as follows:

Pre-processing: Landsat mosaics were generated based on specific time periods to optimize spectral contrast and help discriminate land use and cover classes. Cloud and shadow removal techniques utilizing assessment bandwidth and median reducers were employed to enhance data integrity and eliminate pixels affected by artifacts or cloud contamination. In Collection 7, new Landsat surface reflectance images from USGS were used for classification. A specific temporal mosaic of Landsat images was constructed for each analysis area based on selection criteria that allowed annual analysis and provided sufficient spectral contrast to better distinguish land cover and use classes.

The themes of Pasture, Agriculture, Forest Planting, Urban Area, Coastal Area, and Mining processed the Landsat mosaics by individual scene, using a developed tool to assess images individually and enhance mosaic quality. In the Amazon biome, each Landsat image was classified using the Random Forest algorithm, and the results were reclassified to create the annual land cover and use map.

Classification: The classification scheme was hierarchical, divided into six levels, with each level containing subclasses representing different categories of land cover and use. Random forest and learning algorithms such as U-Net were used for classification, with specific parameters and training samples defined for each biome and cross-cutting theme.

At Level 1, there are six main classes: Forest, Non-Forest Formation, Agriculture, Non-Vegetated Area, Water, and Unobserved. Level 2 includes 16 classes that also combine LU/LC classes. The forest class at Level 1 is further divided into four subclasses: Forest Formation, Savanna Formation, Mangrove, and Shrubland Vegetation. The non-forest formation includes wetlands, pasture formations, salt flats, rocky outcrops, other non-forest formations, and the new Sand Bench Herbaceous Vegetation Class. The agriculture class is subdivided into Pasture, Agriculture, Forest Planting, and Mosaic Uses. The water class is divided into River, Lake, Ocean, and Aquaculture.

Post-processing: Filters are applied to the classified maps to reduce noise and improve accuracy. These filters include gap-filling to replace no-data values, spatial filters to remove isolated pixels, temporal filters to correct disallowed transitions, frequency filters for classes with low occurrence, incident filters to stabilize noisy pixel trajectories, and integration of maps from different biomes and cross-cutting themes.

Map Accuracy Assessment: Two validation approaches were utilized: a comparative analysis with reference maps of specific biomes/regions for determined years, and an accuracy assessment based on statistical techniques using independent sampling points with visual interpretation across Brazil and over the time series. For spatial agreement analysis with reference maps, each biome and cross-cutting theme conducted analyses based on available reference maps. Validation with independent points involved visual interpretation of Landsat data, MODIS-NDVI time series, and high-resolution Google Earth images (when available).

For the main objectives of this project, the original MapBiomas classes were grouped according to REDD+ criteria established by the VMD0007 module. A reclassification based on the classified raster of deforestation year by year was performed. The table illustrates the reclassification and description of each class and the existing area before it.

Classification Criteria:

- **Forest Area:** Remaining forest areas, including all areas classified as forest since 2021 that have not changed class during the historical period.
- **Non-Forest Area:** Areas occupied by any anthropogenic use at any point during the historical period. MapBiomas defines deforestation as clear-cutting, involving complete removal of forest cover and conversion to other anthropogenic uses (agriculture, pasture, mining, etc.).
- **Other Natural Formations:** Natural formations that, when vegetated, present a different aspect and physiognomy (Savanna, Scrubland, Natural Grassland, etc.).
- **Water Bodies:** Any type of water bodies, rivers, and streams.
- **Mining:** Mining areas.
- **Urbanized Area:** Urbanized areas.

According to VMD0007, a 4-year period from the start of the project was selected as the verification period, spanning 2021 to 2024. Various GIS procedures were conducted to select and extract land use values over the years to generate a map containing information on accumulated deforestation and quantify the forest converted into non-forested anthropogenic areas. The map illustrating historical deforestation is represented in the figures below:

At the beginning of the verification period in 2021, a total of 549,273.46 hectares were covered by forests, while 6,986.44 hectares were occupied by non-forested anthropogenic uses in the RRD. Figure 39 illustrates the reference map of forest cover for RRD at the start of the verification period (2021), and Figure 42 at the end of the historical verification period (2024).

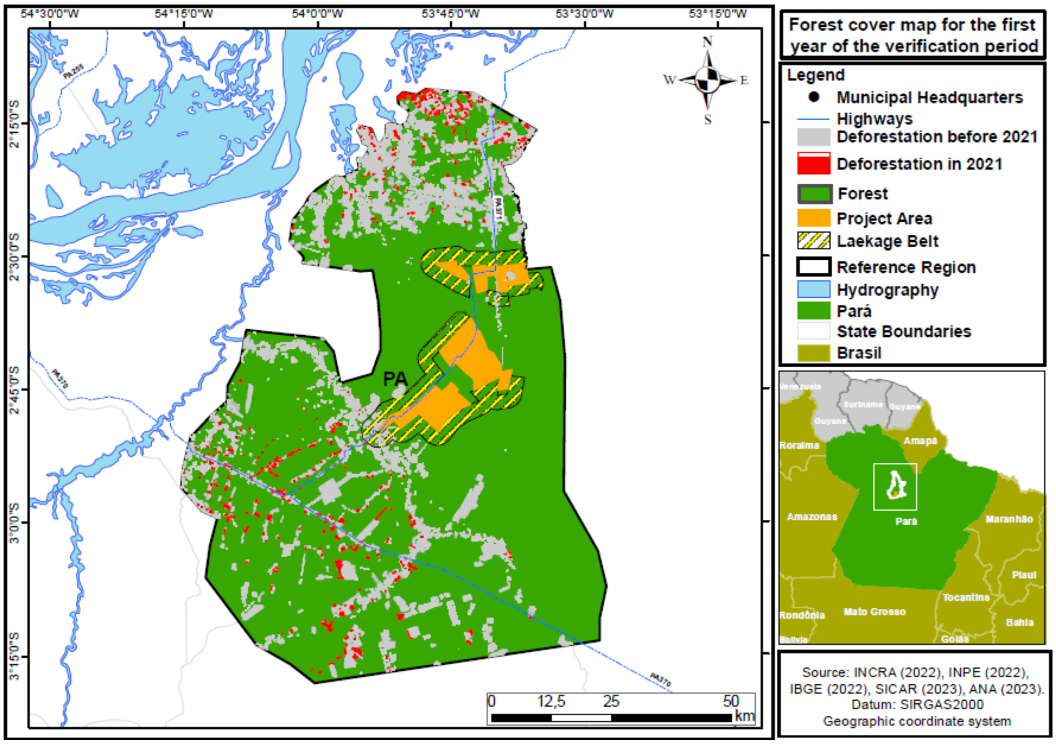

Figure 39. Mapa de referência de cobertura florestal para o primeiro ano do período de verificação em PA, LB e RRD (2021).

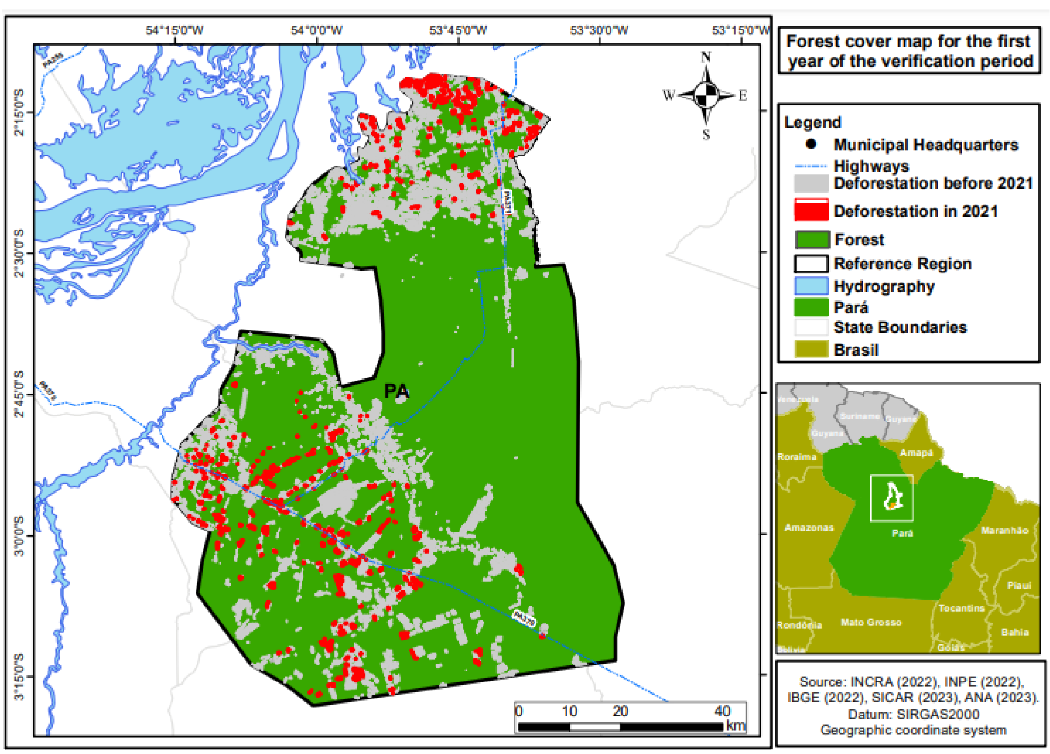

Figure 40. Reference map of forest cover for the first year of the verification period in RRD (2021).

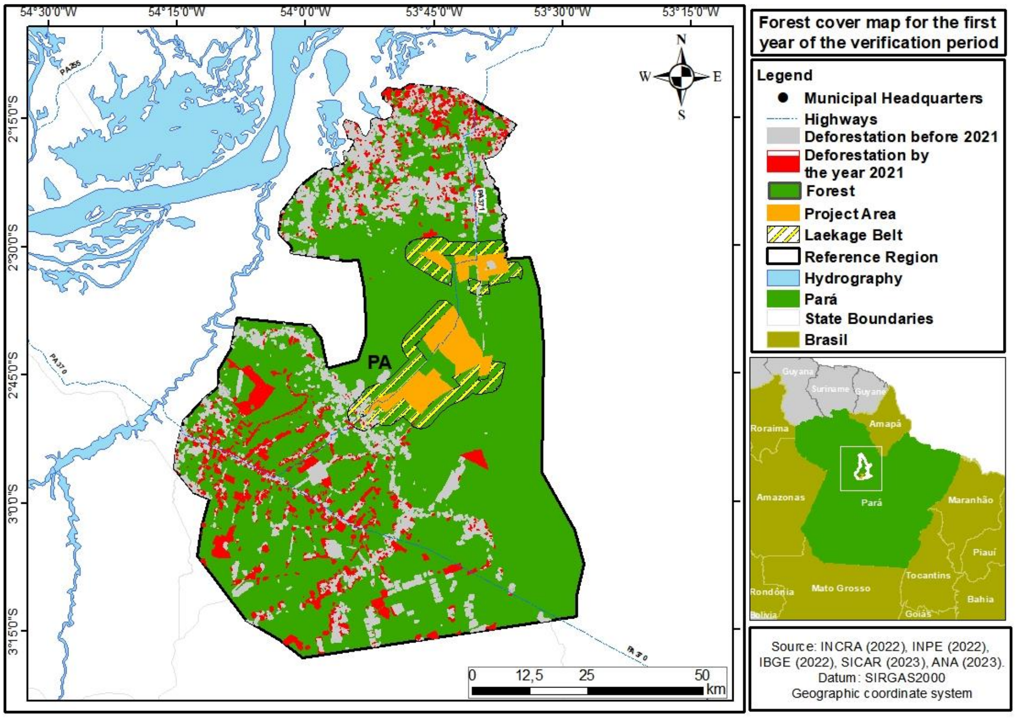

Figure 41. Reference map of forest cover for the last year of the verification period in PA, LB, and RRD (2024).

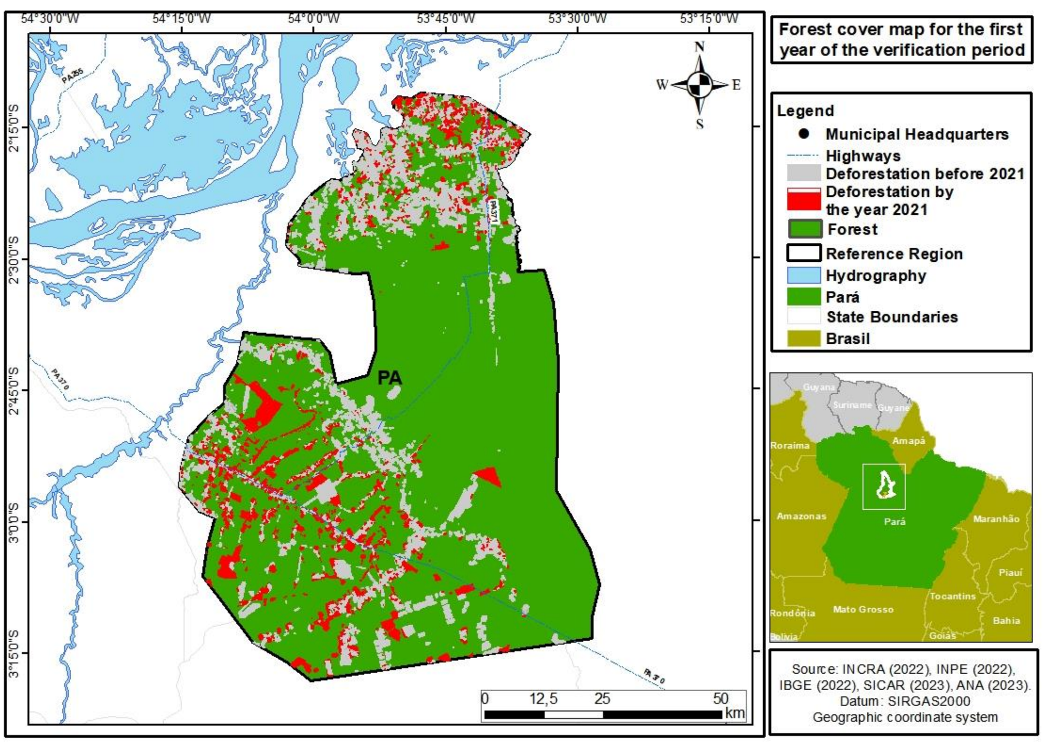

Figure 42. (b) Mapa de referência de cobertura florestal para o último ano do período de verificação em RRD (2024).

Estimation of Annual Unplanned Deforestation Areas in RRD

The methodology provides three approaches to calculate the deforestation area in RRD, which are:

1. Average annual historical deforestation during the historical reference period
2. A linear regression of the deforested area as a function of time
3. A non-linear regression of the deforested area as a function of time

Since the linear relationship does not show a significant trend, the historical average of the historical period was adopted, equal to 0.60% (approximation 1. / Step 2.2 of VMD0007). Table X corresponds to the deforestation areas throughout the historical period, both for the RRD. The deforestation rate was derived from an analysis of deforestation that occurred within the RRD during the verification period, 2021 to 2024. Deforestation was analyzed during the historical reference period within the respective PA, LB, and RRD identified based on the following data:

- **PRODES**: The PRODES (Satellite Deforestation Monitoring Project) monitors deforestation by satellite— a forestry practice where all or most trees in an area are uniformly felled— in the Amazon and Cerrado since 2000.
- **DETER**: This data is generated from a system that provides alerts for areas of degradation and deforestation from 2008 for the Amazon and from 2018 for the Cerrado. The mapping utilizes Landsat satellite images or similar to record and quantify the alert areas produced in the DETER (Real-Time Deforestation Detection) project.
- **SAD**: The SAD (Deforestation Alert System) is a monthly and automatic monitoring system that issues alerts for the suppression of native vegetation throughout the Cerrado and Amazon biomes, starting in 2022. The system uses advanced processing techniques of Sentinel-2 satellite images (10 m resolution) to detect, validate, and refine alerts.
- **MapBiomas Alert**: It is a validation and refinement system for deforestation alerts (like SAD and DETER) using high-resolution images, where each alert is validated, refined, and defined in a time window of occurrence. For each validated alert, a report is generated identifying before and after images of the deforestation.

The annual deforestation estimates for the RRD were obtained by calculating the amount of area (ha) deforested within the limits of the RRD from the deforestation layer from 2021 to 2024 for the Agroflorestal project. By aggregating the described data, we ensured that there were no overlapping deforestation polygons, and the result was the amount of deforestation within the RRD during the verification period year by year.

Table 70. Quantity in hectares of deforestation year by year during the verification period (2021-2024)

Year	Accumulated Deforestation (ha)	Annual Deforestation (ha)	Average annual rate (%)
2.021	6.986,44	5.895,35	1,08
2.022	21.441,64	15.546,29	2,86
2.023	26.006,14	10.459,85	1,92
2.024	15.682,00	5.222,15	0,96
Average	17.529,05	9.280,91	1,71
Total	-	37.123,63	-

Calculation of baseline carbon stock changes

The resulting changes in carbon stock for initial forest classes for the reference region, project area and leakage belt are shown in tables below.

Table 71.Baseline carbon stock change in the reference region

Carbon Stock Changes in Aboveground Biomass per Initial Forest Class icl		Total Carbon Stock Change in Aboveground Biomass of Initial Forest Classes in the Reference Region	
ID icl	1	ΔCab BSLRRicl,t Annual	ΔCab BSLRR icl Cumulative
Name	Managed Ombrophilous Dense Forest		
Project year	t CO2e	t CO2e	t CO2e
2021	-2030644,63	-2030644,63	-2030644,63
2022	-2018665,77	-2018665,77	-4049310,40
2023	-2006751,38	-2006751,38	-6056061,78
2024	-1994913,18	-1994913,18	-8050974,96

Carbon Stock Changes in belowground Biomass per Initial Forest Class icl		Total Carbon Stock Change in Belowground Biomass of Initial Forest Classes in the Reference Region	
ID icl	1	ΔCab BSLRRicl,t Annual	ΔCab BSLRR icl Cumulative
Name	Managed Ombrophilous Dense Forest		
Project year	t CO2e	t CO2e	t CO2e
2021	-62949,98	-62949,98	-62949,98
2022	-125528,62	-125528,62	-188478,61
2023	-187737,92	-187737,92	-376216,52
2024	-249580,22	-249580,22	-625796,75

Carbon Stock Changes in Aboveground Biomass Per Post-deforestation Zone z		Total Carbon Stock Change in Aboveground Biomass of Post-deforestation Zones in the Refence Region	
ID z	1	ΔCab BSLRRz,t Annual	ΔCabBSLRRz Cumulative
Name			
Project year	t CO2e	t CO2e	t CO2e
2021	0,0	0	0
2022	0,0	0	0
2023	0,0	0	0
2024	0,0	0	0

Total Net Carbon Stock Change in ALL POOLS of Post-deforestation Zones in the Reference Region	
ΔC BSLRRt Annual	ΔC BSLRR Cumulative
t CO2e	t CO2e
-2107407,16	-2107407,16
-2157925,46	-4265332,62
-2208139,32	-6473471,95
-2258062,90	-8731534,85

Table 72. Baseline carbon stock change in the project area

Carbon Stock Changes in Aboveground Biomass Per Post-deforestation Zone z		Total Carbon Stock Change in Aboveground Biomass of Initial Forest Classes in the Reference Region	
ID z	1	ΔCab BSLRRicl,t Annual	ΔCab BSLRR icl Cumulative
Name			
Project year	t CO2e	t CO2e	t CO2e
2021	0,0	0,00	0,00
2022	0,0	0,00	0,00
2023	0,0	0,00	0,00
2024	0,0	0,00	0,00

Carbon Stock Changes in Below ground Biomass Per Post-deforestation Zone z		Total Carbon Stock Change in Below ground Biomass of Post-deforestation Zones in the Project Area	
ID z	1	ΔCbb BSLPAz,t Annual	ΔCbbBSLPAz Cumulative
Name			
Project year	t CO2e	t CO2e	t CO2e
2021	0,0	0,0	0,0
2022	0,0	0,0	0,0
2023	0,0	0,0	0,0
2024	0,0	0,0	0,0

Project year	Total Net Carbon Stock Change in ALL POOLS of Post-deforestation Zones in the Project Area	
	ΔCBSLPAt Annual	ΔCBSLPA Cumulative
	t CO2e	t CO2e
2020	0,0	0,0
2021	0,0	0,0
2022	0,0	0,0
2023	0,0	0,0

Table 73. Baseline carbon stock change in the leakage belt

Carbon Stock Changes in Aboveground Biomass per Initial Forest Class icl	Total Carbon Stock Change in Aboveground Biomass of Initial Forest Classes in the Leakage Belt	Carbon Stock Changes in Below ground Biomass per Initial Forest Class icl	Total Carbon Stock Change in belowground Biomass of Initial Forest Classes in the Leakage Belt	Carbon Stock Changes in Below ground Biomass Per Post-deforestation Zone z	Total Carbon Stock Change in Aboveground Biomass of Post-deforestation Zones in the Leakage Belt	Total Net Carbon Stock Change in ALL POOLS of Post-deforestation Zones in the Lekage Belt

ID icl	1	ΔCab BSLLKicl,t Annual	ΔCab BSLLK icl Cumulative
Name	Managed Ombrophilous Dense Forest		
Project year	t CO2e	t CO2e	t CO2e
2021	0,00	0,00	0,00
2022	-24174,55	-24174,55	-24174,55
2023	-5819,47	-5819,47	-29994,02
2024	-18378,52	-18378,52	-48372,54

ID icl	1	ΔCbb BSLLKicl,t Annual	ΔCbb BSLLK icl Cumulative
Name	Managed Ombrophilous Dense Forest		
Project year	t CO2e	t CO2e	t CO2e
2021	0,00	0,00	0,00
2022	-749,41	-749,41	-749,41
2023	-929,81	-929,81	-1679,23
2024	-1499,55	-1499,55	-3178,77

ID z	1	ΔCbb BSLLKz,t Annual	ΔCbbBSLLKz Cumulative
Name			
Project year	t CO2e	t CO2e	t CO2e
2021	0,00	0,00	0,00
2022	0,00	0,00	0,00
2023	0,00	0,00	0,00
2024	0,00	0,00	0,00

ΔC BSLLKt Annual	ΔC BSLLK Cumulative
t CO2e	t CO2e
0,00	0,00
-24923,96	-24923,96
-6749,29	-31673,25
-19878,07	-51551,32

7.3 Project Emissions

During the monitoring period from 2021 to 2024, no deforestation was recorded within the project area. This positive outcome indicates that the protection and conservation activities implemented by the project are effectively preventing deforestation and preserving the forests. The absence of deforestation underscores the success of the management and control strategies adopted, aligned with the goals of emissions reduction and carbon conservation.

Despite this favorable result, it is essential to continue monitoring critical parameters such as above-ground biomass, soil carbon, occurrence of fires, and land-use changes to ensure ongoing compliance with the VM015 methodology of Verra. This continuous monitoring ensures that emission reductions are verified, maintained, and accurately accounted for, thereby strengthening the project's integrity and the generation of carbon credits.

Fossil fuel emissions from activities are likely to be less than 5% of the total GHG emissions reductions benefits generated by the present project. Considering that emissions from deforestation and forest degradation would be much higher than those associated with operations, the emissions from fossil fuel during transport and machinery use can be considered de minimis . In addition, according to VCS AFOLU Requirements, fossil fuel emissions from transport and machinery use in REDD project activities can be considered de minimis

No production of fuel wood or charcoal occurred within the project area during the monitoring period.

Table 74. Total ex post carbon stock decrease due to planned activities in the project area

	Total carbon stock decrease due to planned deforestation		Total carbon stock decrease due to planned logging activities		Total carbon stock decrease due to planned activities	
Project year t	annual ΔCPDdPAt tCO_2-e	cumulative ΔCPDdPAt tCO_2-e	annual ΔCPLdPAt tCO_2-e	cumulative ΔCPLdPAt tCO_2-e	annual ΔCPAdPAt tCO_2-e	cumulative ΔCPAdPA tCO_2-e
2021	0,00	0,00	0,00	0,00	0,00	0,00
2022	0,00	0,00	0,00	0,00	0,00	0,00
2023	0,00	0,00	0,00	0,00	0,00	0,00
2024	0,00	0,00	0,00	0,00	0,00	0,00

No forest fires or catastrophic events occurred within the project area

Project year	Total Carbon Stock Decrease due to Forest Fires		Total Carbon Stock Decrease due to Catastrophic Events		Total Carbon Stock Decrease due to Forest Fires and Catastrophic Events	
	annual ΔCUFdPAt tCO_2-e	cumulative ΔCUFdPA tCO_2-e	annual ΔCUCdPAt tCO_2-e	cumulative ΔCUCdPAt tCO_2-e	annual ΔCFCdPAt tCO_2-e	cumulative ΔCFCdPA tCO_2-e
2021	0	0	0	0	0	0
2022	0	0	0	0	0	0
2023	0	0	0	0	0	0
2024	0	0	0	0	0	0

Ex post carbon stock change in the project area under the project scenario is shown below.

Table 75. Ex post net carbon stock change in the project area under the project scenario

Project year	Total carbon stock decrease due to planned logging activities		Total carbon stock increase due to growth without harvest		Total carbon stock decrease unavoided unplanned deforestation		Total carbon stock change in the project case	
	annual $\Delta CPLdPA_t$ tCO_2-e	cumulative ΔCPLdPA tCO_2-e	annual $\Delta CPLiPA_t$ tCO_2-e	cumulative ΔCPLiPA tCO_2-e	annual $\Delta CUDdPA_t$ tCO_2-e	cumulative ΔCUDdPA tCO_2-e	annual $\Delta CPSPA_t$ tCO_2-e	cumulative ΔCPSPA tCO_2-e
2021	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0
2022	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0
2023	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0
2024	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0

Table 76. Total ex post estimated actual net changes in carbon stocks and emissions of GHG gases in the project area

Total carbon stock increase due to growth without harvest		Total Carbon Stock Decrease due to Forest Fires and Catastrophic Events		Total carbon stock decrease due to unavoided unplanned deforestation		Total carbon stock change in the project case		Total Estimated Actual Non-CO2 Emissions from Forest Fires in the Project Area	
annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative

$\Delta CPLiPA_t$ tCO2-e	$\Delta CPLiPA$ tCO2-e	$\Delta CFCdPA_t$ tCO2-e	$\Delta CFCdPA$ tCO2-e	$\Delta CUDdPA_t$ tCO2-e	$\Delta CUDdPA$ tCO2-e	$\Delta CPSPA_t$ tCO2-e	$\Delta CPSPA$ tCO2-e	$EBBPSPA_t$ tCO2-e	EBBPSPA tCO2-e
0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0
0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0
0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0
0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0

7.4 Leakage Emissions

According to the applied methodology, two primary sources of leakage are considered: (a) decrease in carbon stocks and increase in GHG emissions associated with leakage prevention measures, and (b) decrease in carbon stocks and increase in GHG emissions associated with activity displacement leakage.

To mitigate the risk of activity displacement leakage, agents responsible for baseline deforestation may engage in alternative activities within the project and leakage management areas. These activities aim to replace baseline income, product generation, and livelihoods, thereby reducing deforestation and minimizing the risk of displacement. However, implementing these measures may lead to a decrease in carbon stocks or an increase in GHG emissions compared to the baseline scenario.

If these impacts are significant, they must be quantified, and ex post monitoring is required to ensure accurate accounting of any changes in carbon stocks and GHG emissions. Specifically, if the net sum of carbon stock changes within a monitoring period is greater than zero, it indicates that leakage prevention measures are not causing a decrease in carbon stocks, and the net increase shall be conservatively excluded from the calculation of net GHG emission reductions of the project activity. Conversely, if the net sum is negative, indicating a significant decrease in carbon stocks, it must be accounted for in the project's emission reductions calculations.

Table 77. Ex post estimated net carbon stock change in leakage management areas

Project year	Total Net Carbon Stock Change of Post-deforestation Zones in the Lekage Belt		Total carbon stock change in the project case		Carbon Stock Decrease due to Leakage Prevention Measures	
	ΔC_{BSLLKt} Annual t CO2e	ΔC_{BSLLK} Cumulative t CO2e	annual $\Delta CPSPA_t$ tCO2-e	cumulative $\Delta CPSPA$ tCO2-e	annual $\Delta CLPMLK_t$ tCO2-e	cumulative $\Delta CLPMLK$ tCO2-e
2021	0,00	0,00	0,0	0,0	0,0	0,0

2022	-24174,55	-24174,55	0,0	0,0	0,0	0,0
2023	-5819,47	-29994,02	0,0	0,0	0,0	0,0
2024	-18378,52	-48372,54	0,0	0,0	0,0	0,0

The implementation of the REDD AUD project could result in the displacement of activities that cause deforestation within the project boundary to areas outside it. A greater-than-expected decrease in carbon stocks within the leakage belt compared to the baseline scenario indicates the displacement of deforestation activities due to the project. As defined in VCS Methodology VM0015 v1.1, if carbon stocks in the leakage belt area decrease more than projected in the baseline scenario, it suggests that leakage from the displacement of baseline activities has occurred. However, if strong evidence demonstrates that deforestation in the leakage belt is caused by agents unrelated to the project area, such deforestation may not be attributed to the project and should not be considered leakage.

Leakage emissions due to activity displacement are calculated as the difference between ex ante (predicted) and ex post (actual) assessments, as detailed in the table below.

Table 78. Table 63: Total Net Baseline Carbon Stock Change in the Leakage Belt (Method 1)

Project year	Total Ex ante Net Increase in Emissions due to Leakage		Total Ex post Net Increase in Emissions due to Leakage		Total Ex Post Emissions Leakage	
	annual ELKt tCO2-e	cumulative ELK tCO2-e	annual ELKt tCO2-e	cumulative ELK tCO2-e	annual ΔCLK_t tCO2-e	cumulative ΔCLK tCO2-e
2021	3122,55	3122,55	0,00	0,00	0,00	0,00
2022	2970,97	6093,52	552,34	552,34	0,00	0,00
2023	2953,43	9046,95	132,96	685,30	0,00	0,00
2024	2936,03	11982,98	419,91	1105,21	0,00	0,00

The results of all ex post estimations of leakage are summarized in the table below.

Table 62: Ex Post Estimated Total Leakage

Project year	Total Ex Ante GHG Emissions from Increased Grazing Activities		Total Ex Ante Increase in GHG Emissions due to Displaced Forest Fires		Total Ex Ante Decrease in Carbon Stocks due to Displaced Deforestation		Carbon Stock Decrease due to Leakage Prevention Measures		Total Net Carbon Stock Change due to Leakage		Total Net Increase in Emissions due to Leakage	
	annual EgLKt tCO2-e	cumulative EgLK tCO2-e	annual EADLKt tCO2-e	cumulative EADLK tCO2-e	annual ΔCADLKt tCO2-e	cumulative ΔCADLK tCO2-e	annual ΔCLPMLKt tCO2-e	cumulative ΔCLPMLK tCO2-e	annual ΔCLK_t tCO2-e	cumulative ΔCLK tCO2-e	annual ELKt tCO2-e	cumulative ELK tCO2-e

2021	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
2022	0,00	0,00	552,34	552,34	24923,96	24923,96	0,00	0,00	24923,96	24923,96	552,34	552,34
2023	0,00	0,00	132,96	685,30	6749,29	31673,25	0,00	0,00	6749,29	31673,25	132,96	685,30
2024	0,00	0,00	419,91	1105,21	19878,07	51551,32	0,00	0,00	19878,07	51551,32	419,91	1105,21

7.5 GHG Emission Reductions and Carbon Dioxide Removals

The net anthropogenic GHG emission reduction of the proposed AUD project activity is calculated as follows:

$\Delta REDDt$=($\Delta CBSLPAt$+$EBBBSLPAt$)−($\Delta CPSPAt$+$EBBPSPAt$)−($\Delta CLKt$ + $ELKt$)

Where:

ΔREDDt = Ex ante estimated net anthropogenic GHG emission reduction attributable to the AUD project activity at year t (t CO2e)

ΔCBSLPAt = Sum of baseline carbon stock changes in the project area at year t (t CO2e)

Note: If ΔCPSPAtrepresents a net increase in carbon stocks, a negative sign before theabsolute value of ΔCPSPAt shall be used. If ΔCPSPAtrepresents a net decrease, the positive sign shall be used.

EBBBSLPAt = Sum of baseline emissions from biomass burning in the project area at

year t (t CO2e)

ΔCPSPAt = Sum of ex ante estimated actual carbon stock changes in the project area at year t (t CO2e)

EBBPSPAt = Sum of (ex ante estimated) actual emissions from biomass burning in the project area at year t (t CO2e)

ΔCLKt = Sum of ex ante estimated leakage net carbon stock changes at year t (t CO2e)

Note – where the cumulative sum of ΔCLKt within a baseline validity period is greater than zero, ΔCLKt must be set to zero.

ELKt = Sum of ex ante estimated leakage emissions at year t (t CO2e) t = 1, 2, 3, ..., T, a year of the proposed project crediting period (dimensionless)

The number of Verified Carbon Units (VCUs) to be generated through the proposed AUD project activity at year *t* is calculated as follows:

VCUt= *ΔREDDt*–*VBCt*

VBCt= (*ΔCBSLPAt*– *ΔCPSPAt*) x *RFt*

Where:

VCUt = Number of Verified Carbon Units that can be traded at time *t*; tCO2e

Note:If VCUt< 0 no credits (VCUs) will be awarded to the proponents of the AUD project activity.

Δ*REDDt* = *Ex post* estimated net anthropogenic greenhouse gas emission reduction attributable to the AUD project activity at year *t*; tCO2e

VBCt = Number of Buffer Credits deposited in the VCS Buffer at time *t*; tCO2e

Δ*CBSLPAt* = Sum of baseline carbon stock changes in the project area at year *t*; tCO2e

Δ*CPSPAt*Sum of *ex post*carbon stock changes in the project area at year *t*; tCO2e ha-1

*RFt*Risk factor used to calculate VCS buffer credits; %

*t*1, 2, 3 … *T,* a year of the proposed project crediting period; dimensionless.

The *RFt* was estimated using the most recent version of the *VCS-approved AFOLU Non-Permanence Risk Tool*and the resulting value of *RFt*for the present REDD project during the current monitoring period was 12%.

The specific summary of GHG reductions and removals in the Agroflorestal Novo Horizonte project during the current monitoring period is included in Table below.

Table 79. Ex post estimated net anthropogenic GHG emissions reductions and VCU

Project year	Baseline carbon stock changes		Baseline GHG emissions		Ex post project carbon stock changes		Ex post project GHG emissions		Ex post leakage carbon stock changes		Ex post leakage GHG emissions		Ex post net anthropogenic GHG emission reductions		Ex post VCUs tradable		Ex post buffer credits	
	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative
	ΔCBSLPAt	ΔCBSLPA	EBBBSLPAt	EBBBSLPA	ΔCPSPAt	ΔCPSPA	EBBPSPAt	EBBPSPA	ΔCLKt	ΔCLK	ELKt	ELK	ΔREDDt	ΔREDD	VCUt	VCU	VBCt	VBC
	tCO2-e	tCO2-e	tCO2-e	tCO2-e	tCO2-e	tCO2-e	tCO2-e	tCO2-e	tCO2-e	tCO2-e	tCO2-e	tCO2-e	tCO2-e	tCO2-e	tCO2-e	tCO2-e	tCO2-e	tCO2-e
2021	88.977,03	88.977,03	1.971,81	1.971,81	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	90948,84	90948,84	**80034,98**	80034,98	10913,86	10913,86
2022	91.126,71	180.103,74	1.960,16	3.931,97	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	93086,88	184035,72	**81916,45**	161951,43	11170,43	22084,29
2023	93.260,59	273.364,34	1.948,51	5.880,49	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	95209,10	279244,82	**83784,01**	245735,44	11425,09	33509,38
2024	95.390,75	368.755,08	1.937,13	7.817,62	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	97327,88	376572,70	**85648,53**	331383,98	11679,35	45188,72

The latter table includes estimates of GHG emissions reduction (*REDDt*), calculations of buffer and leakage, and the resulting calculation of tradable Verified Carbon Units(*VCUt*).

Table 80. Net GHG emissions reductions and removals in the Agroflorestal Novo Horizonte REDD AUD

Vintage period	Baseline emissions (tCO_2e)	Project emissions (tCO_2e)	Leakage emissions (tCO_2e)	Buffer pool allocation (tCO_2e)	Reductions VCUs (tCO_2e)	Removals VCUs (tCO_2e)	Total VCU issuance (tCO_2e)
2021	1.971,81	0	0	10913,86	90948,84	0,0	80034,98
2022	1.960,16	0	0	11170,43	93086,88	0,0	81916,45
2023	1.948,51	0	0	11425,09	95209,10	0,0	83784,01
2024	1.937,13	0	0	11679,35	97327,88	0,0	85648,53
Total	7.817,62	0	0	45.188,72	376.572,70	0	331.383,98

Table 81. Comparison between Estimated and Achieved Reductions and Removals.

Vintage period	Ex-ante estimated reductions/ removals	Achieved reductions/ removals	Percent difference	Explanation for the difference
2021	75260,17	90948,84	120,85	The difference between the achieved reductions and removals during the monitoring period is attributed to the increased deforestation rate in the reference region compared to the baseline scenario.
2022	77029,39	93086,88	120,85	
2023	78785,53	95209,10	120,85	
2024	80538,82	97327,88	120,85	
Total	311613,91	376.572,70		

i) Provide the requested information using the table below:

State the non-permanence risk rating (%)	12
Has the non-permanence risk report been attached as either an appendix or a separate document?	☒ Yes ☐ No
For ARR and IFM projects with harvesting, state, in $tCO_2e$the Long-term Average (LTA).	N/A
Has the LTA been updated based on monitored data, if applicable?	☐ Yes ☒ No If no, provide justification. Not Applicable
State, in tCO_2e, the expected total GHG benefit to date	486,892.1
If a loss occurred (including a loss event or reversal), state the amount of tCO_2e lost:	N/A

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Use the table below to describe the commercially sensitive information included in the project description to be excluded in the public version.

Section	Information	Justification

APPENDIX X: <TITLE OF APPENDIX>

Use appendices for supporting information. Delete this appendix (title and instructions) where no appendix is required.